

GATE EC: Electromagnetics & Transmission Lines

Previous Year Solved Questions Archive

Athena Project Archive

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Question 1: Transmission Lines (GATE EC 2026)

MCQ

2 Marks

A complex load (in Ω) is represented as $\Gamma_L = 0.5\angle 30^\circ$ on the Smith chart. A co-axial cable with a characteristic impedance of 50Ω is connected to the load. The new input impedance of the load now moves to a diametrically opposite point on the same Γ circle on the Smith chart.

Which option is the nearest input impedance of the cable connected load (in Ω)?

- (A) $20.7 - j5.1$
- (B) **(Correct)** $17.7 - j11.8$
- (C) $97.5 - j65.0$
- (D) $97.5 + j65.0$

Correct Answer: (B)

Detailed Solution – Question 1

Given:

Reflection coefficient of load: $\Gamma_L = 0.5\angle 30^\circ$

Characteristic impedance: $Z_0 = 50\Omega$

The input impedance moves to the **diametrically opposite point** on the same Smith chart circle.

On a Smith chart, diametrically opposite point means:

$$\Gamma_{in} = 0.5\angle(30^\circ + 180^\circ) = 0.5\angle 210^\circ$$

Step 1: Convert reflection coefficient into rectangular form

$$\Gamma_{in} = 0.5(\cos 210^\circ + j \sin 210^\circ)$$

$$= 0.5(-0.866 - j0.5)$$

$$\Gamma_{in} = -0.433 - j0.25$$

Step 2: Use impedance-reflection coefficient relation

Substituting values,

$$\begin{aligned} Z_{in} &= 50 \cdot \frac{1 + (-0.433 - j0.25)}{1 - (-0.433 - j0.25)} \\ &= 50 \cdot \frac{0.567 - j0.25}{1.433 + j0.25} \end{aligned}$$

Multiplying numerator and denominator by conjugate:

$$Z_{in} = 50 \cdot \frac{(0.567 - j0.25)(1.433 - j0.25)}{(1.433)^2 + (0.25)^2}$$

After simplification,

$$Z_{in} \approx 17.7 - j11.8\Omega$$

Question 2: Waveguides & Optical Fibers (GATE EC 2026)

NAT

1 Mark

The cutoff frequency (in GHz) for the dominant TE_{10} mode of an air-filled rectangular waveguide of inner dimension $0.28 \text{ inch} \times 0.14 \text{ inch}$ is _____. (rounded off to two decimal places)

Detailed Solution – Question 2

Given:

Inner dimensions: $a = 0.28 \text{ inch}$, $b = 0.14 \text{ inch}$

Air-filled waveguide, dominant mode: TE_{10}

Step 1: Convert dimensions to meters

$1 \text{ inch} = 0.0254 \text{ m}$

$a = 0.28 \times 0.0254 = 7.112 \times 10^{-3} \text{ m}$

Step 2: Apply cutoff frequency formula for TE_{10}

For TE_{mn} mode, cutoff frequency is:

$$f_c = \frac{c}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}$$

For TE_{10} : $m = 1, n = 0$

$$f_c = \frac{c}{2a} = \frac{3 \times 10^8}{2 \times 7.112 \times 10^{-3}}$$

Step 3: Calculate

$$f_c = \frac{3 \times 10^8}{1.4224 \times 10^{-2}} = 21.09 \times 10^9 \text{ Hz}$$

Question 3: Basics of Electromagnetics & Maxwell's Eqns (GATE EC 2026)

MSQ

1 Mark

Which option(s) represents/represent the dielectric loss tangent of a substrate?

- (A) **(Correct)** Ratio of the real to imaginary parts of the total displacement current
- (B) **(Correct)** $(\omega\epsilon'' + \sigma)/(\omega\epsilon')$
- (C) Ratio of the electric susceptibility to permittivity
- (D) Ratio of the polarization vector $\vec{P}$ to the displacement vector $\vec{D}$

Correct Answer: (A, B)

Detailed Solution – Question 3

The loss tangent $\tan \delta$ is defined as the ratio of the total loss current (conduction + displacement loss) to the lossless displacement current:

$$\tan \delta = \frac{\omega\epsilon'' + \sigma}{\omega\epsilon'}$$

- (A) "Ratio of real to imaginary parts of displacement current", this is exactly $\tan \delta$
- (B) $\frac{\omega\epsilon'' + \sigma}{\omega\epsilon'}$, this is the standard formula
- (C) "Ratio of electric susceptibility to permittivity", χ/ϵ , this is NOT the loss tangent
- (D) "Ratio of $\vec{P}$ to $\vec{D}$ ", $P/D = (\epsilon - \epsilon_0)/\epsilon$, this is NOT the loss tangent

Answer:

Options (A) and (B) correctly represent the dielectric loss tangent.

Question 4: Uniform Plane Waves (GATE EC 2026)

MCQ

1 Mark

The electric field of a monochromatic plane wave travelling in a lossless isotropic and homogenous medium is given by

$$\vec{E}(z, t) = E_0[\hat{x} \cos(\omega t - kz) + \hat{y} \sin(\omega t - kz)]$$

in a right-handed orthogonal co-ordinate system.

Which of the following is the correct polarization of the electromagnetic wave?

- (A) **(Correct)** Right-handed circularly polarized
- (B) Left-handed circularly polarized
- (C) Linearly polarized
- (D) Linearly polarized with -45° angle to $\hat{x}$

Correct Answer: (A)

Detailed Solution – Question 4

Given,

$$\vec{E}(z, t) = E_0 [\cos(\omega t - kz)\hat{a}_x + \sin(\omega t - kz)\hat{a}_y]$$

At $z = 0$,

$$\vec{E}(t) = E_0 \cos \omega t, \hat{a}_x + E_0 \sin \omega t, \hat{a}_y$$

Here,

$$|E_{x0}| = |E_{y0}| = E_0$$

and the phase difference is,

$$|\phi| = 90^\circ$$

Therefore, the wave is **circularly polarized**.

Now, at $t = 0$, $\omega t = 0$,

$$\vec{E} = E_0 \hat{a}_x$$

$$\text{At } t = \frac{T}{4}, \omega t = \frac{\pi}{2},$$

$$\vec{E} = E_0 \hat{a}_y$$

Thus, the electric field rotates from $\hat{a}_x$ toward $\hat{a}_y$ with time.

Hence, it is a **Right-handed circularly polarized** wave.

Question 5: Miscellaneous Topics (GATE EC 2026)

MCQ

1 Mark

Consider the Friis' transmission equation $P_R = \frac{P_T G_T G_R \lambda^2}{(4\pi D)^2}$, where P_R and P_T are the received and the transmitted powers, respectively.

G_T and G_R are the gain of transmitting and receiving antennas, respectively, D is the distance between the transmitting and receiving antennas, and λ is the wavelength in free space.

Given: $G_T = G_R = 1.0$, $\lambda = 0.30$ m and $P_T = +10$ dBm.

Choose the distance (D), in km, from the following options at which the received power, $P_R = -90$ dBm?

- (A) $\frac{15}{4\pi}$
- (B) **(Correct)** $\frac{15}{2\pi}$

(C) $\frac{75}{2\pi}$

(D) $\frac{3}{4\pi}$

Correct Answer: (B)

Detailed Solution – Question 5

Given:

$$G_T = G_R = 1, \lambda = 0.30 \text{ m}, P_T = +10 \text{ dBm}, P_R = -90 \text{ dBm}$$

Step 1: Power ratio

$$\frac{P_R}{P_T} = \frac{-90-10}{10} \text{ dB} = -100 \text{ dB} = 10^{-10}$$

Step 2: Friis equation

$$\frac{P_R}{P_T} = G_T G_R \left(\frac{\lambda}{4\pi D} \right)^2 = 1 \times 1 \times \left(\frac{0.3}{4\pi D} \right)^2$$

Step 3: Solve for D

$$10^{-10} = \left(\frac{0.3}{4\pi D} \right)^2$$

$$10^{-5} = \frac{0.3}{4\pi D}$$

$$D = \frac{0.3}{4\pi \times 10^{-5}} = \frac{0.3}{4\pi} \times 10^5 = \frac{30000}{4\pi} \text{ m} = \frac{30}{4\pi} \text{ km} = \frac{15}{2\pi} \text{ km}$$

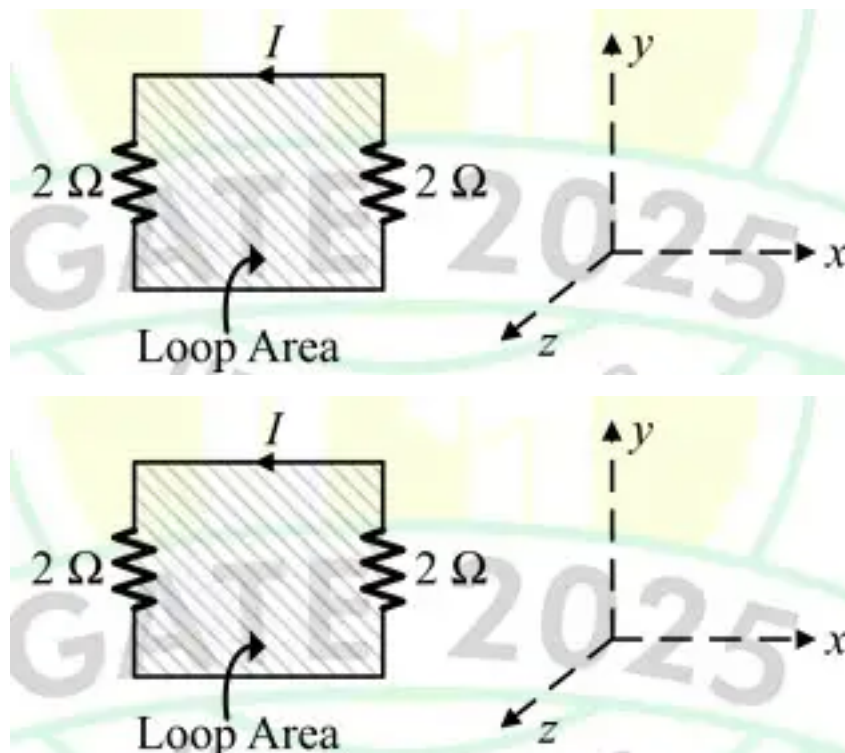
Question 6: Miscellaneous Topics (GATE EC 2025)

NAT

2 Marks

Two resistors are connected in a circuit loop of area 5 m^2 , as shown in the figure below. The circuit loop is placed on the $x - y$ plane.

When a time-varying magnetic flux, with fluxdensity $B(t) = 0.5t$ (in Tesla), is applied along the positive z -axis, the magnitude of current I (in Amperes, rounded off to two decimal places) in the loop is _____ .



Detailed Solution – Question 6

Given:

Loop area, $A = 5 \text{ m}^2$

Magnetic field, $B(t) = 0.5t \text{ T}$

Total resistance in the loop $= 2 \Omega + 2 \Omega = 4 \Omega$

We use Faraday's Law of electromagnetic induction: $\mathcal{E} = -\frac{d\Phi}{dt}$ where $\Phi = B(t) \cdot A$ is the magnetic flux.

So, $\Phi(t) = B(t) \cdot A = 0.5t \cdot 5 = 2.5t$

$\frac{d\Phi}{dt} = \frac{d}{dt}(2.5t) = 2.5 \text{ V}$

So, the induced EMF is: $\mathcal{E} = -2.5 \text{ V}$

(The negative sign indicates direction, but we are interested in magnitude of current)

Now, using Ohm's Law: $I = \frac{|\mathcal{E}|}{R} = \frac{2.5}{4} = 0.625 \text{ A}$

Question 7: Transmission Lines (GATE EC 2025)

NAT

2 Marks

A 50Ω lossless transmission line is terminated with a load Z_L of $(50 - j75)\Omega$.

If the average incident power on the line is 10 mW , then the average power delivered to the load (in mW , rounded off to one decimal place) is _____.

Detailed Solution – Question 7

Step 1: Characteristic Impedance Given: $Z_0 = 50 \Omega$ (lossless line)

$Z_L = 50 - j75 \Omega$

Incident power $= P_{\text{inc}} = 10 \text{ mW}$

Step 2: Reflection Coefficient Reflection coefficient $\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}$ First compute: $Z_L + Z_0 = (50 - j75) + 50 = 100 - j75$

$Z_L - Z_0 = (50 - j75) - 50 = -j75$

So: $\Gamma = \frac{-j75}{100 - j75}$ Multiply numerator and denominator by the complex conjugate of the denominator: $\Gamma = \frac{-j75 \cdot (100 + j75)}{(100 - j75)(100 + j75)}$ Numerator: $-j75(100 + j75) = -j7500 - j^2 75 \cdot 75 = -j7500 + 5625$

Denominator: $100^2 + 75^2 = 10000 + 5625 = 15625$ So: $\Gamma = \frac{5625 - j7500}{15625} = 0.36 - j0.48$

Step 3: Reflected Power $|\Gamma|^2 = (0.36)^2 + (0.48)^2 = 0.1296 + 0.2304 = 0.36$

So, reflected power =

$$P_{\text{ref}} = |\Gamma|^2 \cdot P_{\text{inc}} = 0.36 \cdot 10 = 3.6 \text{ mW}$$

Step 4: Delivered Power

$$P_{\text{delivered}} = P_{\text{inc}} - P_{\text{ref}} = 10 - 3.6 = \boxed{6.4 \text{ mW}}$$

Question 8: Basics of Electromagnetics & Maxwell's Eqns (GATE EC 2025)

MCQ

2 Marks

An electric field of 0.01 V/m is applied along the length of a copper wire of circular cross-section with diameter 1 mm . Copper has a conductivity of $5.8 \times 10^7 \text{ S/m}$.

The current (in Amperes, rounded off to two decimal places) flowing through the wire is _____ .

- (A) **(Correct)** 0.46
(B) 1.82
(C) 0.58
(D) 1.12

Correct Answer: (A)

Detailed Solution – Question 8

$J = \sigma E = 0.01 \times 5.8 \times 10^7$
Current,

$$\begin{aligned} I &= J\pi r^2 \\ &= 0.01 \times 5.8 \times 10^7 \times \pi \times (0.5 \times 10^{-3})^2 \\ &= 0.045 \times 10^7 \times 10^{-6} = 0.46 \text{ A} \end{aligned}$$

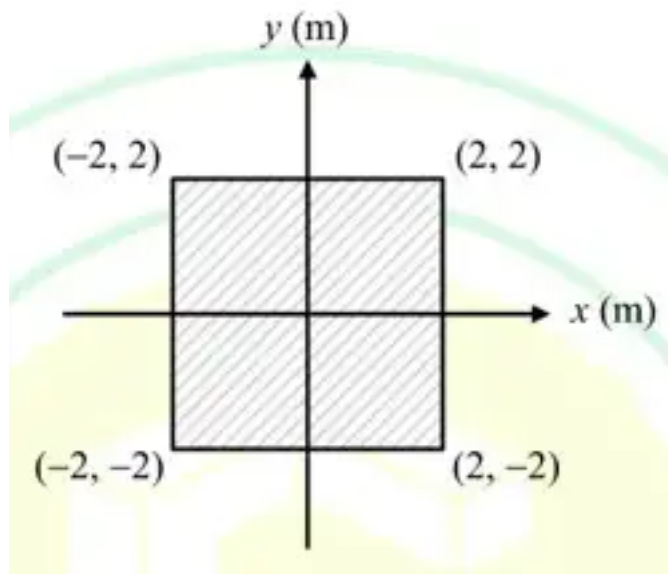
Question 9: Basics of Electromagnetics & Maxwell's Eqns (GATE EC 2025)

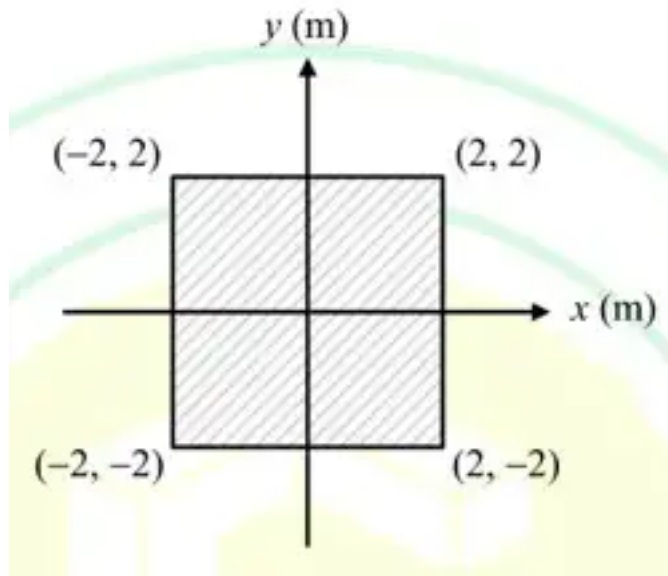
MCQ

2 Marks

A square metal sheet of $4 \text{ m} \times 4 \text{ m}$ is placed on the $x - y$ plane as shown in the figure below.

If the surface charge density (in $\mu\text{C}/\text{m}^2$) on the sheet is $\rho_s(x, y) = 4|y|$, then the total charge (in μC , rounded off to the nearest integer) on the sheet is _____ .





- (A) 16
- (B) 85
- (C) **(Correct)** 64
- (D) 256

Correct Answer: (C)

Detailed Solution – Question 9

$$\begin{aligned}
 Q &= \iint 4|y|dx dy = 4 \iint 4y dx dy \\
 &= 16 \int dx \int y dy = 16[x]_0^2 \left[\frac{y^2}{2} \right]_0^2 \\
 &= 16 \times 2 \times \frac{2^2}{2} = 16 \times 4 = 64 \mu\text{C}
 \end{aligned}$$

Question 10: Transmission Lines (GATE EC 2024)

NAT

2 Marks

A lossless transmission line with characteristic impedance $Z_0 = 50\Omega$ is terminated with an unknown load. The magnitude of the reflection co-efficient is $|\Gamma| = 0.6$. As one moves towards the generator from the load, the maximum value of the input impedance magnitude looking towards the load (in Ω) is _____

Detailed Solution – Question 10

As we know that

$$Z_{\max} = SZ_0$$

Now,

$$S = \frac{1 + |\Gamma|}{1 - |\Gamma|} = \frac{1 + 0.6}{1 - 0.6} = \frac{1.6}{0.4} = 4$$

$$\therefore Z_{\max} = 4 \times 50 = 200\Omega.$$

Question 11: Uniform Plane Waves (GATE EC 2024)

MCQ

2 Marks

A uniform plane wave with electric field $\vec{E}(x) = A_y \hat{a}_y e^{-j\frac{2\pi x}{3}}$ V/m is travelling in the air (relative permittivity, $\epsilon_r = 1$ and relative permeability, $\mu_r = 1$) in the $+x$ direction (A_y is a positive constant, $\hat{a}_y$ is the unit vector along the y axis). It is incident normally on an ideal electric conductor (conductivity, $\sigma = \infty$) at $x = 0$. The position of the first null of the total magnetic field in the air (measured from $x = 0$, in metres) is _____

(A) (Correct) $-\frac{3}{4}$

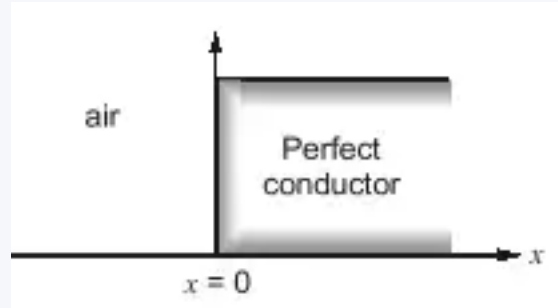
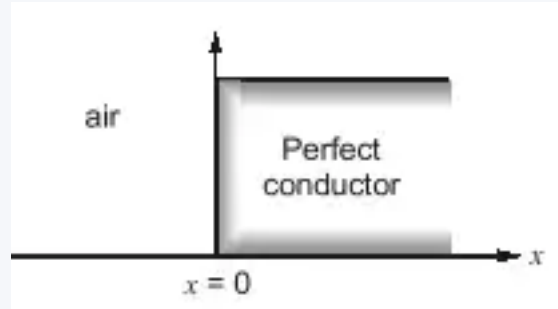
(B) $-\frac{3}{2}$

(C) -6

(D) -3

Correct Answer: (A)

Detailed Solution – Question 11



At perfect conductor,

$$\begin{aligned}\Gamma &= -1 \\ \Rightarrow \Gamma &= 1 \mid \pi\end{aligned}$$

As we know that $H_{\min}$ occurs at $E_{\max}$. Hence,

$$\begin{aligned}E_{\max} &= E_0[1 + |\Gamma|] \\ H_{\min} &= \frac{E_0}{\eta}[1 - |\Gamma|] \text{ at } 2\beta x_{\max} = 2n\pi + \theta_{\Gamma}\end{aligned}$$

So,

$$2\beta x_{\max} = 2n\pi + \theta_{\Gamma}$$

$$\Rightarrow \frac{4\pi}{\lambda} x_{\max} = 2n\pi + \pi$$

$$\Rightarrow x_{\max} = (2n + 1)\frac{\lambda}{4}; n = 0, 1, 2, 3 \dots$$

So, for 1st value of x for H-field to be zero is

$$x_{\max} = \frac{\lambda}{4} \quad [\text{at } n = 0]$$

Now, from the equation given,

$$\begin{aligned}\beta &= \frac{2\pi}{3} \\ \Rightarrow \frac{2\pi}{\lambda} &= \frac{2\pi}{3} \\ \Rightarrow \lambda &= 3\end{aligned}$$

Hence,

$$x_{\max} = \frac{3}{4}$$

If the reference is not at interference then $x = -\frac{3}{4}$.

Question 12: Basics of Electromagnetics & Maxwell's Eqns (GATE EC 2024)

MCQ

1 Mark

Let $\hat{i}$ and $\hat{j}$ be the unit vectors along x and y axes, respectively and let A be a positive constant. Which one of the following statements is true for the vector fields $\vec{F}_1 = A(\hat{i}y + \hat{j}x)$ and $\vec{F}_2 = A(\hat{i}y - \hat{j}x)$?

- (A) Both $\vec{F}_1$ and $\vec{F}_2$ are electrostatic fields.
- (B) **(Correct)** Only $\vec{F}_1$ is an electrostatic field.
- (C) Only $\vec{F}_2$ is an electrostatic field.
- (D) Neither $\vec{F}_1$ nor $\vec{F}_2$ is an electrostatic field.

Correct Answer: (B)

Detailed Solution – Question 12

For an electrostatic field, $\nabla \times \vec{F} = 0$

$$\begin{aligned}
 \vec{F}_1 &= A[y\hat{i} + x\hat{j}] \\
 \Rightarrow \\
 \Delta \times \vec{F}_1 &= A \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ y & x & 0 \end{vmatrix} \\
 &= A[0\hat{i} - 0\hat{j} + (1 - 1)\hat{k}] \\
 &= 0 \\
 \vec{F}_2 &= A[y\hat{i} - x\hat{j}] \\
 \Rightarrow \\
 \Delta \times \vec{F}_2 &= A \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ y & -x & 0 \end{vmatrix} \\
 &= A[0\hat{i} - 0\hat{j} + (-1 - 1)\hat{k}] \\
 &= -2A\hat{k}
 \end{aligned}$$

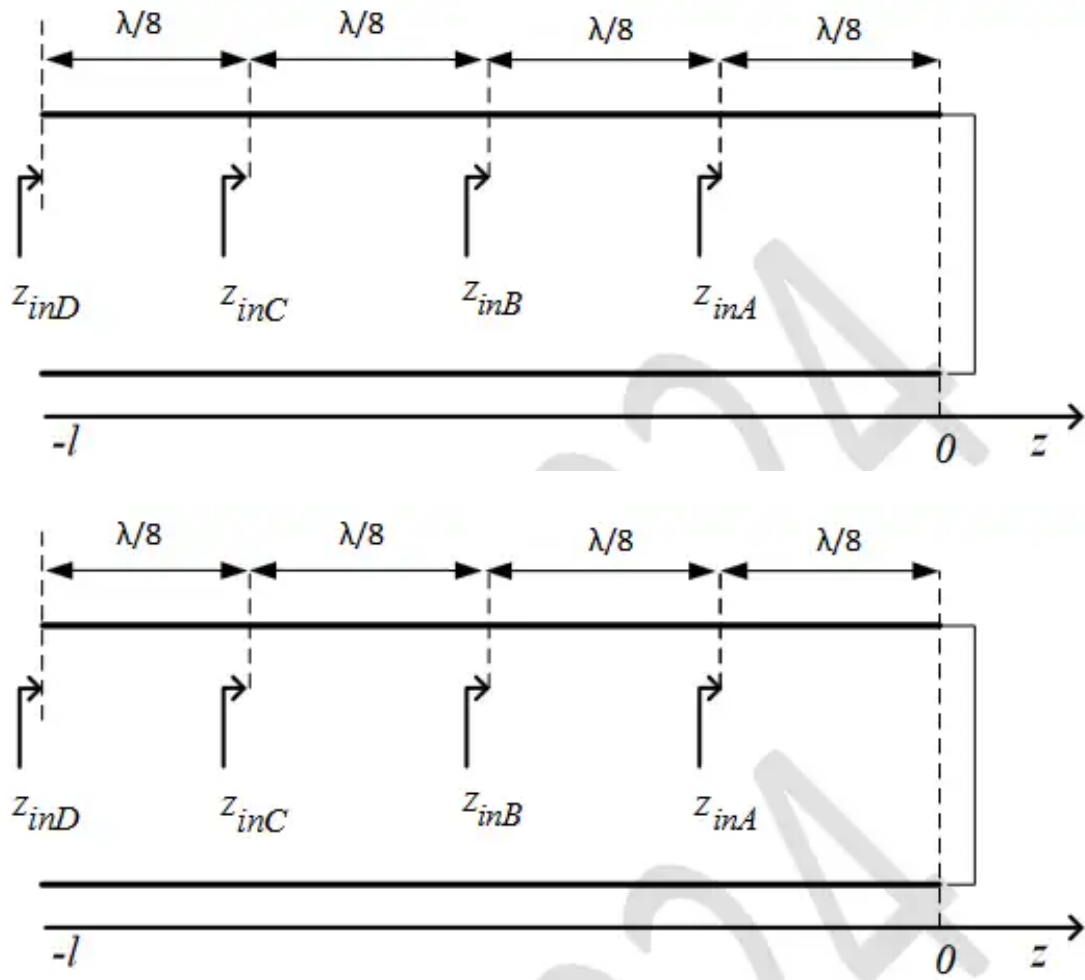
Hence, $\vec{F}_1$ is electrostatic, $\vec{F}_2$ is not electrostatic.

Question 13: Transmission Lines (GATE EC 2024)

MCQ

1 Mark

Consider a lossless transmission line terminated with a short circuit as shown in the figure below. As one moves towards the generator from the load, the normalized impedances $z_{in A}$, $z_{in B}$, $z_{in C}$ and $z_{in D}$ (indicated in the figure) are _____



(A)

$$z_{in A} = +0.4j\Omega, z_{in B} = \infty, z_{in C} = -0.4j\Omega, z_{in D} = 0$$

(B)

$$z_{in A} = \infty, z_{in B} = +0.4j\Omega, z_{in C} = 0, z_{in D} = +0.4j\Omega$$

(C)

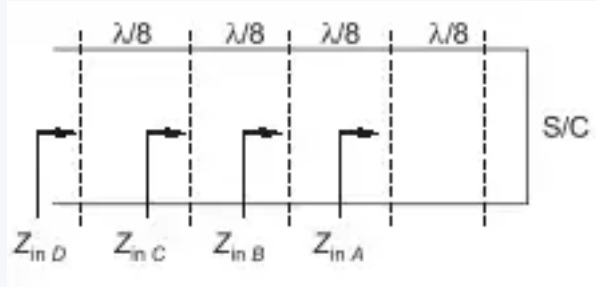
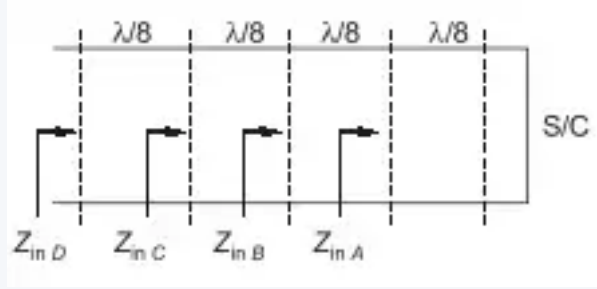
$$z_{in A} = -1j\Omega, z_{in B} = 0, z_{in C} = +1j\Omega, z_{in D} = \infty$$

(D) (Correct)

$$z_{in A} = +1j\Omega, z_{in B} = \infty, z_{in C} = -1j\Omega, z_{in D} = 0$$

Correct Answer: (D)

Detailed Solution – Question 13



$$Z_{s/c} = jZ_0 \tan \beta l.$$

$\Rightarrow$ Normalized impedance,

$$\bar{Z}_{S/C} = \frac{Z_{S/C}}{Z_0} = j \tan \beta l$$

$$\bar{Z}_{in A} = j \tan \left(\frac{2\pi}{\lambda} \cdot \frac{\lambda}{8} \right) = j \tan(\pi/4) = j1\Omega$$

$$\bar{Z}_{in B} = j \tan \left(\frac{2\pi}{\lambda} \cdot \frac{\lambda}{4} \right) = \infty$$

$$\bar{Z}_{in C} = j \tan \left(\frac{2\pi}{\lambda} \cdot \frac{3\lambda}{8} \right) = -j1\Omega$$

$$\bar{Z}_{in D} = j \tan \left(\frac{2\pi}{\lambda} \cdot \frac{\lambda}{2} \right) = 0$$

Question 14: Basics of Electromagnetics & Maxwell's Eqns (GATE EC 2023)

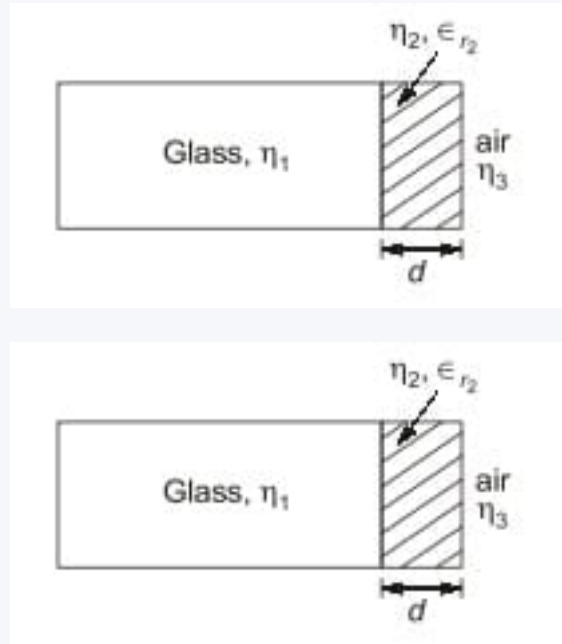
NAT

2 Marks

A transparent dielectric coating is applied to glass ($\epsilon_r = 4, \mu_r = 1$) to eliminate the reflection of red light ($\lambda_0 = 0.75\mu\text{m}$). The minimum thickness of the dielectric coating, in μm , that can be used is _____ (rounded off to two decimal places).

Detailed Solution – Question 14

For no reflection, impedance must be matched.
Hence, η_2 acts like a quarter wave impedance transformer.



So,

$$(i) \quad \eta_2 = \sqrt{\eta_1 \cdot \eta_3} \Rightarrow \epsilon_{r_2} = \sqrt{\epsilon_{r_1} \cdot \epsilon_{r_3}} \Rightarrow \epsilon_{r_2} = 2$$

(ii) For impedance matching,

$$d = (2n + 1) \frac{\lambda}{4}; n = 0, 1, 2, \dots$$

$$\lambda = \frac{\lambda_0}{\sqrt{\epsilon_r}} = \frac{\lambda_0}{\sqrt{\epsilon_{r_2}}}$$

$$\text{Here } \lambda = \frac{0.75 \times 10^{-6}}{\sqrt{2}} = 0.53 \times 10^{-6}$$

Hence, for minimum distance, $n = 0$

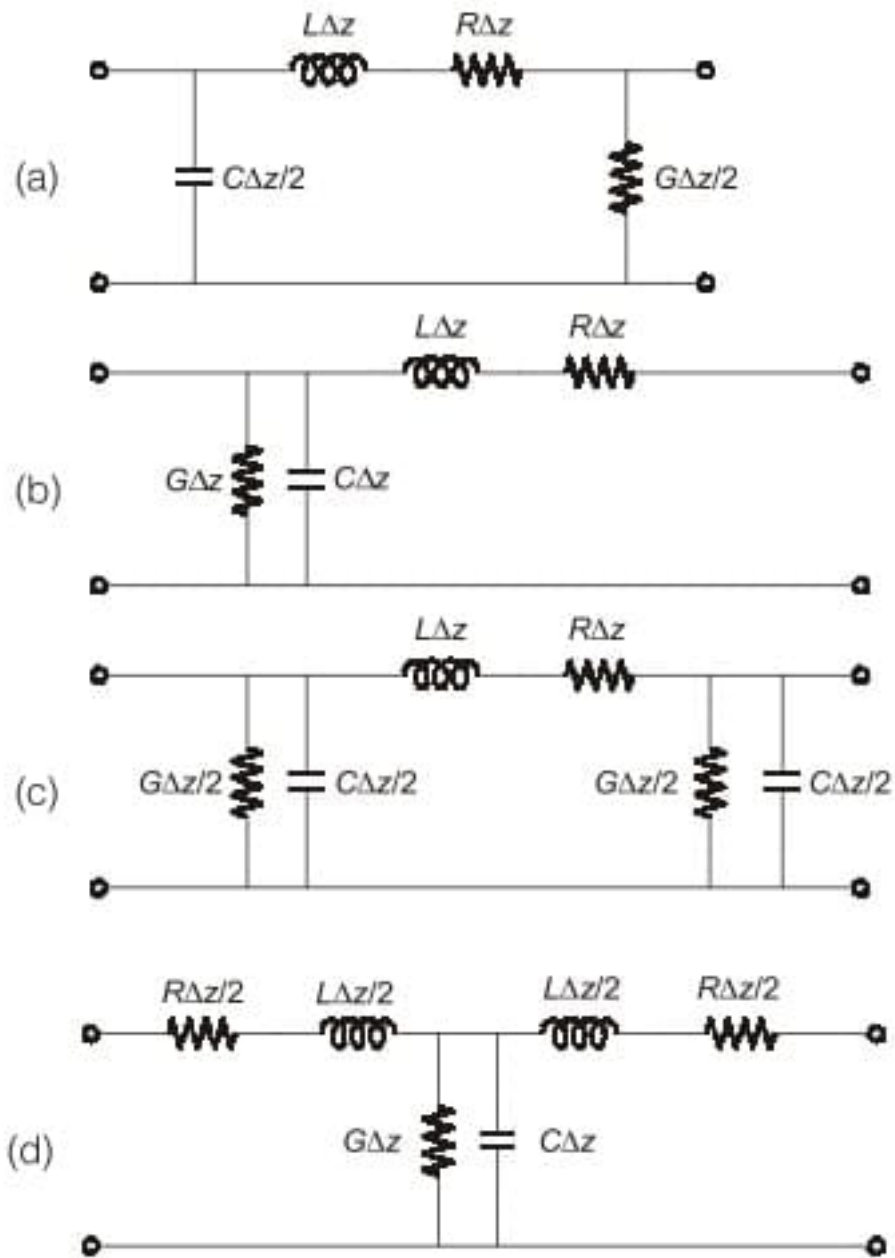
$$\text{So, } d = \frac{\lambda}{4} = \frac{0.53 \times 10^{-6}}{4} = 0.133 \mu\text{m}$$

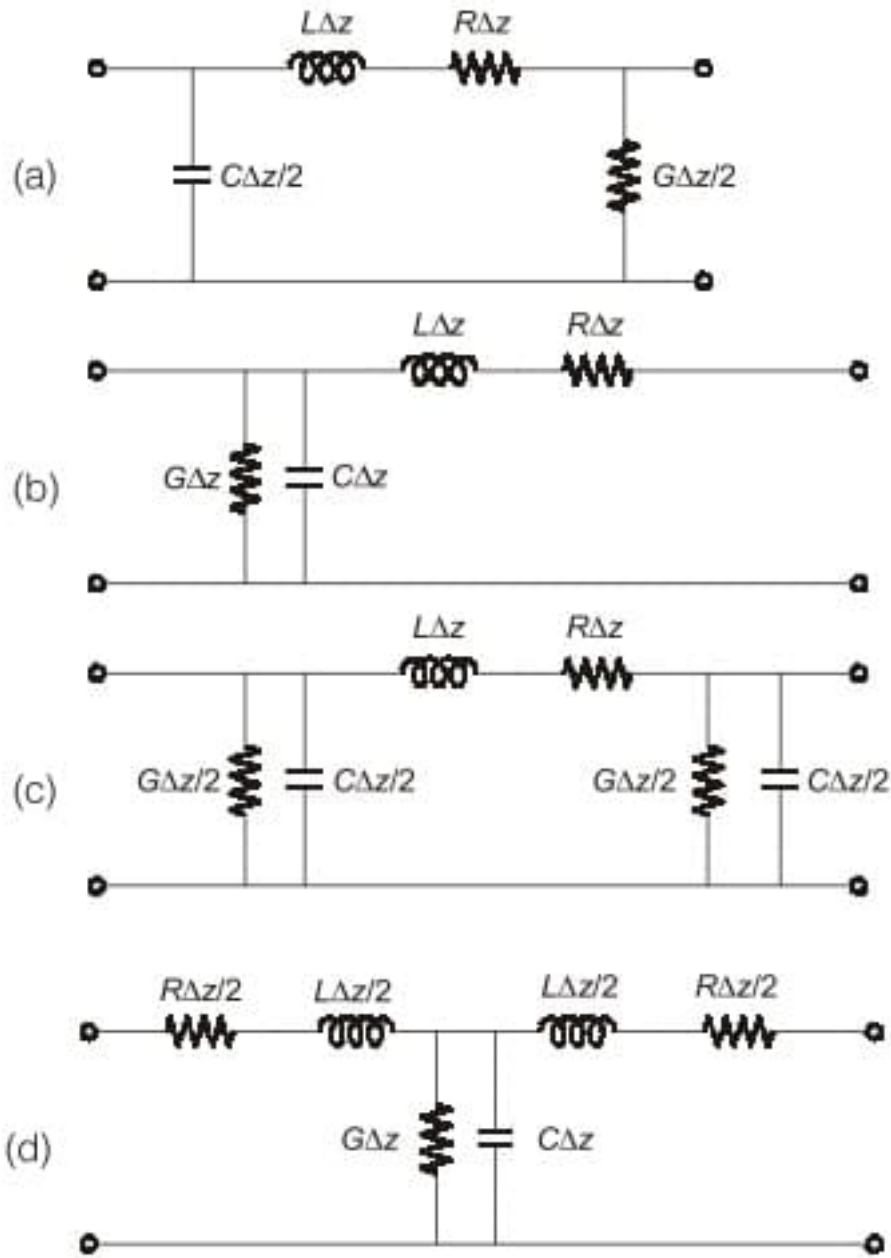
Question 15: Transmission Lines (GATE EC 2023)

MSQ

2 Marks

The following circuit(s) representing a lumped element equivalent of an infinitesimal section of a transmission line is/are





- (A) A
- (B) (Correct) B
- (C) (Correct) C
- (D) (Correct) D

Correct Answer: (B, C, D)

Question 16: Uniform Plane Waves (GATE EC 2023)

MSQ

2 Marks

The electric field of a plane electromagnetic wave is

$$E = a_x C_{1x} \cos(\omega t - \beta z) + a_y C_{1y} \cos(\omega t - \beta z + \theta) \text{ V/m}$$

Which of the following combination(s) will give rise to a left handed elliptically polarized (LHEP) wave?

- (A) **(Correct)** $C_{1x} = 1, C_{1y} = 1, \theta = \pi/4$
- (B) **(Correct)** $C_{1x} = 2, C_{1y} = 1, \theta = \pi/2$
- (C) $C_{1x} = 1, C_{1y} = 2, \theta = 3\pi/2$
- (D) **(Correct)** $C_{1x} = 2, C_{1y} = 1, \theta = 3\pi/4$

Correct Answer: (A, B, D)

Detailed Solution – Question 16

Given,

$$\vec{E} = \hat{a}_x C_{1x} \cos(\omega t - \beta z) + \hat{a}_y C_{1y} \cos(\omega t - \beta z + \theta)$$

at $z = 0$

$$\vec{E} = C_{1x} \cos \omega t \hat{a}_x + C_{1y} \cos(\omega t + \theta) \hat{a}_y$$

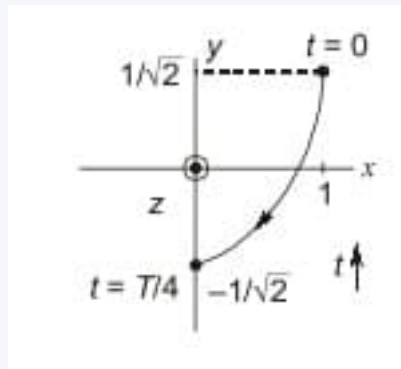
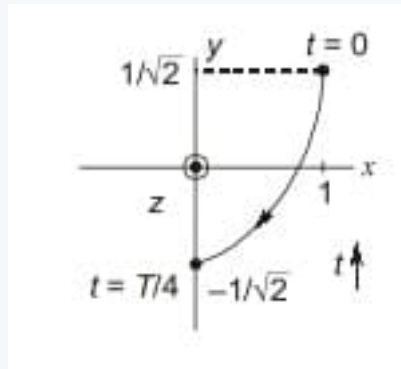
Going by options,

Option (A) $\vec{E} = \cos \omega t \hat{a}_x + \cos(\omega t + \pi/4) \hat{a}_y$

at $t = 0, \omega t = 0, \vec{E} = \hat{a}_x + \frac{1}{\sqrt{2}} \hat{a}_y$

at $t = T/4, \omega t = \pi/2, \vec{E} = 0 - \frac{1}{\sqrt{2}} \hat{a}_y$

$\Rightarrow$ Hence, it is LHEP.

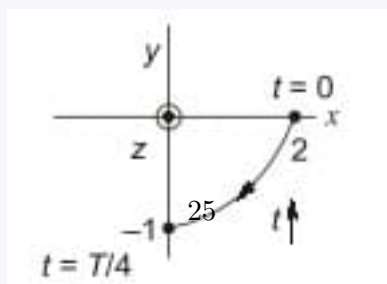
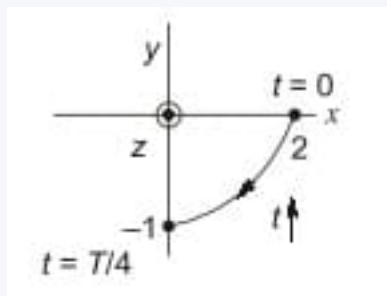


Option (B) $\vec{E} = 2 \cos \omega t \hat{a}_x + \cos(\omega t + \pi/2) \hat{a}_y$

at $t = 0, \omega t = 0, \vec{E} = 2 \hat{a}_x$

at $t = T/4, \omega t = \pi/2, \vec{E} = -1 \hat{a}_y$

$\Rightarrow$ Hence, it is LHEP.

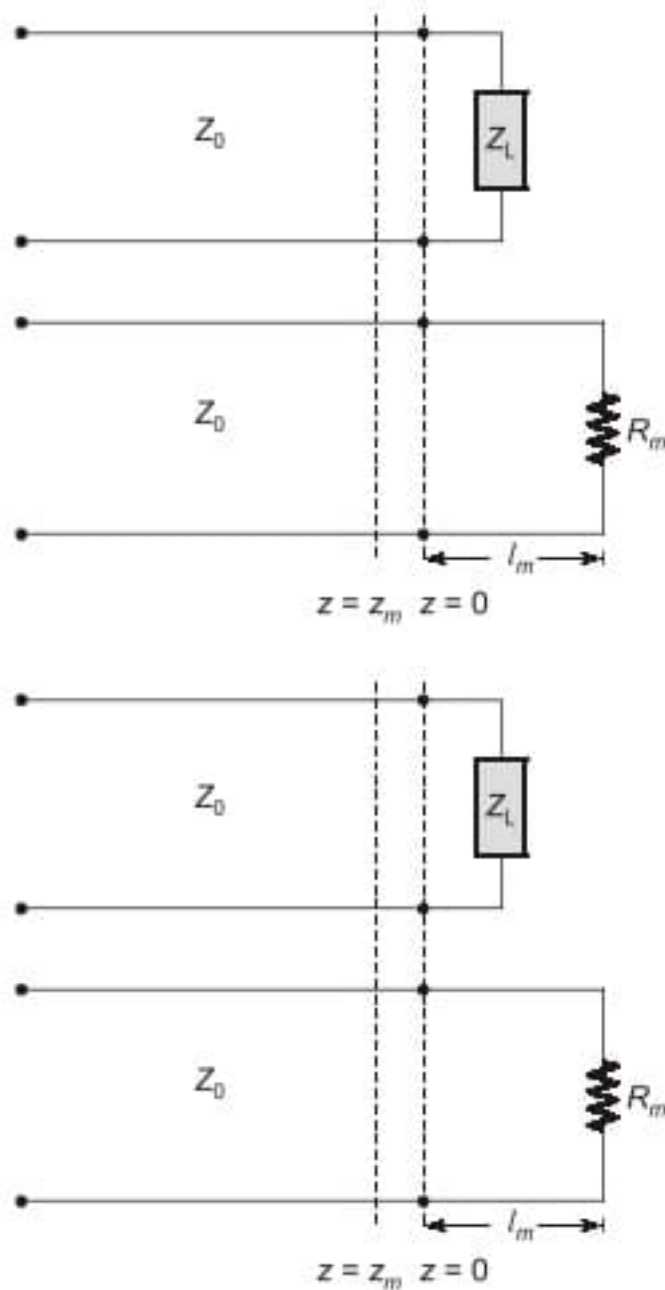


Question 17: Transmission Lines (GATE EC 2023)

MSQ

2 Marks

The standing wave ratio on a 50Ω lossless transmission line terminated in an unknown load impedance is found to be 2.0. The distance between successive voltage minima is 30 cm and the first minimum is located at 10 cm from the load. Z_L can be replaced by an equivalent length l_m and terminating resistance R_m of the same line. The value of R_m and l_m , respectively, are



- (A) $R_m = 100\Omega, l_m = 20 \text{ cm}$
- (B) **(Correct)** $R_m = 25\Omega, l_m = 20 \text{ cm}$
- (C) **(Correct)** $R_m = 100\Omega, l_m = 5 \text{ cm}$
- (D) $R_m = 25\Omega, l_m = 5 \text{ cm}$

Correct Answer: (B, C)

Detailed Solution – Question 17

Given $S = 2$, $Z_{\min} = 10 \text{ cm}$, $Z_0 = 50\Omega$

As we know that, $|\Gamma| = \frac{S-1}{S+1} = \frac{2-1}{2+1} = \frac{1}{3}$

Now, distance between successive voltage minima = 30 cm

$$\Rightarrow \frac{\lambda}{2} = 30 \text{ cm}$$

$$\Rightarrow \lambda = 60 \text{ cm}$$

Also, for minima,

$$2\beta Z_{\min} = (2n+1)\pi + \theta_\Gamma$$

At $n = 0$, 1st minima, $Z_{\min} = 10 \text{ cm}$

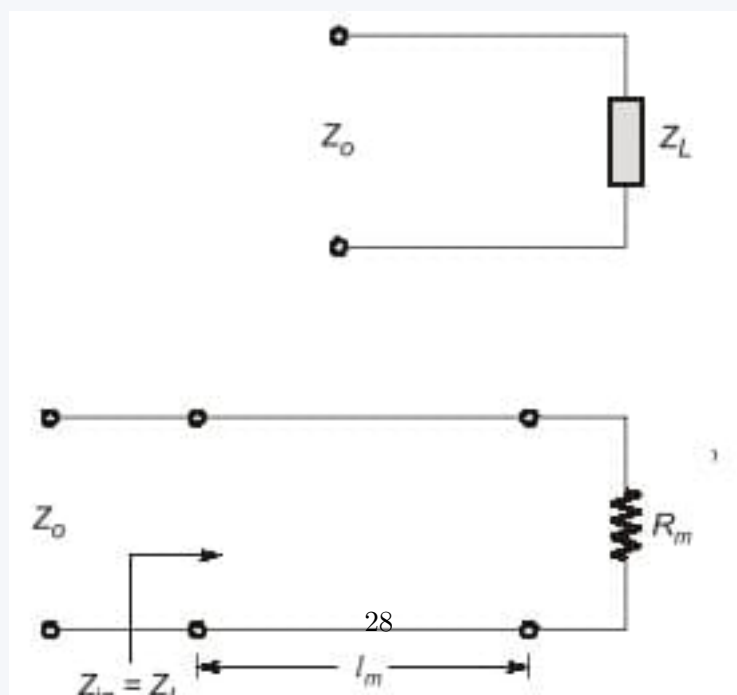
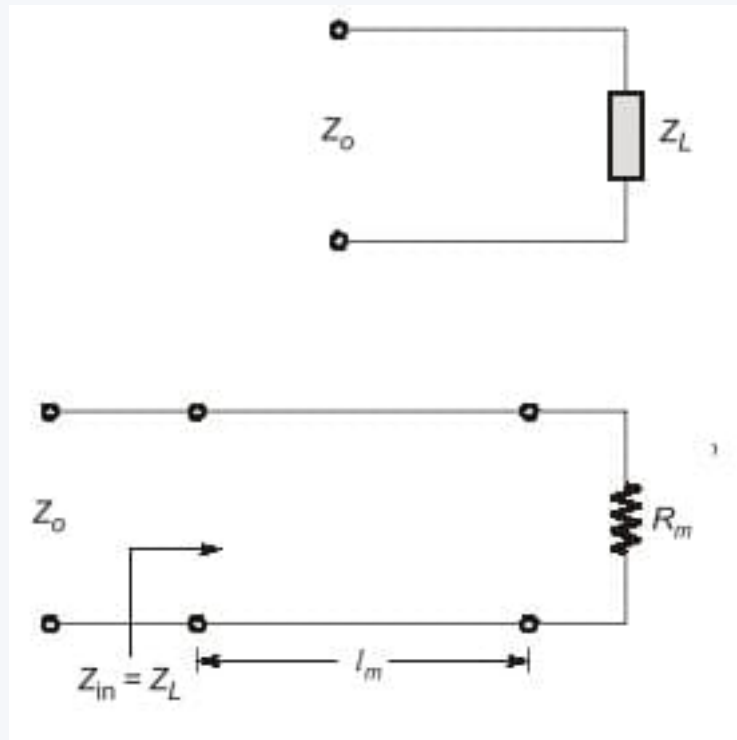
$$\frac{4\pi}{\lambda} Z_{\min} = \pi + \theta_\Gamma$$

$$\Rightarrow \frac{4\pi}{60} * 10 = \pi + \theta_\Gamma$$

$$\Rightarrow \frac{2\pi}{3} - \pi = \theta_\Gamma$$

$$\Rightarrow \theta_\Gamma = \frac{-\pi}{3} \quad \therefore \Gamma = \frac{1}{3} \angle -60^\circ$$

Now, $\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}$



Question 18: Transmission Lines (GATE EC 2023)

MCQ

1 Mark

A cascade of common-source amplifiers in a unity gain feedback configuration oscillates when

- (A) the closed loop gain is less than 1 and the phase shift is less than 180° .
- (B) the closed loop gain is greater than 1 and the phase shift is less than 180° .
- (C) the closed loop gain is less than 1 and the phase shift is greater than 180° .
- (D) **(Correct)** the closed loop gain is greater than 1 and the phase shift is greater than 180° .

Correct Answer: (D)

Detailed Solution – Question 18

For oscillation,

Loop gain magnitude ≥ 1 .

Phase of loop gain = 360°

So, correct answer is option (D).

In options, closed loop gain is mentioned technically it should be loop gain.

Question 19: Miscellaneous Topics (GATE EC 2023)

MCQ

1 Mark

Consider a narrow band signal, propagating in a lossless dielectric medium ($\epsilon_r = 4, \mu_r = 1$), with phase velocity v_p and group velocity v_g . Which of the following statement is true? (c is the velocity of light in vacuum.)

- (A) $v_p > c, v_g > c$
- (B) $v_p < c, v_g > c$
- (C) $v_p > c, v_g < c$
- (D) **(Correct)** $v_p < c, v_g < c$

Correct Answer: (D)

Detailed Solution – Question 19

Phase velocity,

$$V_p = \frac{\omega}{\beta} = \frac{\omega}{\omega\sqrt{\mu\epsilon}} = \frac{1}{\sqrt{\mu_0\epsilon_0}}, \frac{1}{\sqrt{\mu_r\epsilon_r}} = \frac{C}{\sqrt{\mu_r\epsilon_r}}$$

$$\therefore V_p < C$$

Group velocity,

$$V_g = \frac{d\omega}{d\beta} = \frac{V_p}{1 - \frac{\omega}{V_p} \frac{dV_p}{d\omega}}$$

Here, $V_p \neq f(\omega)$

$$\therefore V_g = V_p < C$$

$$\text{Hence, } V_p < C \\ V_g < C$$

Question 20: Basics of Electromagnetics & Maxwell's Eqns (GATE EC 2022)

NAT

2 Marks

In an electrostatic field, the electric displacement density vector, $\vec{D}$, is given by $\vec{D}(x, y, z) = (x^3\vec{i} + y^3\vec{j} + xy^2\vec{k})C/m^2$, where $\vec{i}, \vec{j}, \vec{k}$ are the unit vectors along x-axis, y-axis, and z-axis, respectively. Consider a cubical region R centered at the origin with each side of length 1 m, and vertices at $(\pm 0.5m, \pm 0.5m, \pm 0.5m)$. The electric charge enclosed within R is _____ C (rounded off to two decimal places).

Detailed Solution – Question 20

$$\vec{D}(x, y, z) = (x^3\vec{i} + y^3\vec{j} + xy^2\vec{k})c/m^2$$

$$Q_{enc.} = \int_v \rho_v \cdot dV = \int (\nabla \cdot \vec{D}) dV$$

$$\nabla \cdot \vec{D} = 3x^2 + 3y^2$$

$$dV = dxdydz$$

$$\therefore Q_{enc.} = \int_v 3(x^2 + y^2)dxdydz = 3 \left[\int_{-0.5}^{0.5} x^2 dx \int_{-0.5}^{0.5} dy \int_{-0.5}^{0.5} dz + \int_{-0.5}^{0.5} dx \int_{-0.5}^{0.5} y^2 dy \int_{-0.5}^{0.5} dz \right]$$

$$= 3 \left[\frac{x^3}{3} \Big|_{-0.5}^{0.5} \times 1 \times 1 + \frac{y^3}{3} \Big|_{-0.5}^{0.5} \times 1 \times 1 \right] = 0.25 + 0.25 = 0.5C$$

Question 21: Waveguides & Optical Fibers (GATE EC 2022)

MSQ

2 Marks

A waveguide consists of two infinite parallel plates (perfect conductors) at a separation of 10^{-4} cm, with air as the dielectric. Assume the speed of light in air to be 3×10^8 m/s. The frequency/frequencies of TM waves which can propagate in this waveguide is/are _____.

(A) (Correct) $6 \times 10^{15} Hz$

- (B) $0.5 \times 10^{12} Hz$
 (C) **(Correct)** $8 \times 10^{14} Hz$
 (D) $1 \times 10^{13} Hz$

Correct Answer: (A, C)

Detailed Solution – Question 21

Cut-off frequency.

$$f_c = \frac{c}{2a} \quad (m=1) = \frac{3 \times 10^8}{2 \times 10^{-4} \times 10^{-2}} \\ = 1.5 \times 10^{14} Hz$$

$f > f_c$ will only propagate A and C will propagate.

Question 22: Uniform Plane Waves (GATE EC 2022)

MSQ

1 Mark

Consider the following wave equation,

$$\frac{\partial^2 f(x, t)}{\partial t^2} = 10000 \frac{\partial^2 f(x, t)}{\partial x^2}$$

Which of the given options is/are solution(s) to the given wave equation?

- (A) **(Correct)** $f(x, t) = e^{-(x-100t)^2} + e^{-(x+100t)^2}$
 (B) $f(x, t) = e^{-(x-100t)} + 0.5e^{-(x+1000t)}$
 (C) **(Correct)** $f(x, t) = e^{-(x-100t)} + \sin(x + 100t)$
 (D) $f(x, t) = e^{j100\pi(-100x+t)} + e^{j100\pi(100x+t)}$

Correct Answer: (A, C)

Detailed Solution – Question 22

As we know, wave equation is given by $\frac{\partial^2 f(x, t)}{\partial x^2} = \frac{C^2 d^2 f(x, y)}{dt^2}$ Here, option (A) and (C) are satisfying the above standard wave equation.

Question 23: Basics of Electromagnetics & Maxwell's Eqns (GATE EC 2022)

MCQ

1 Mark

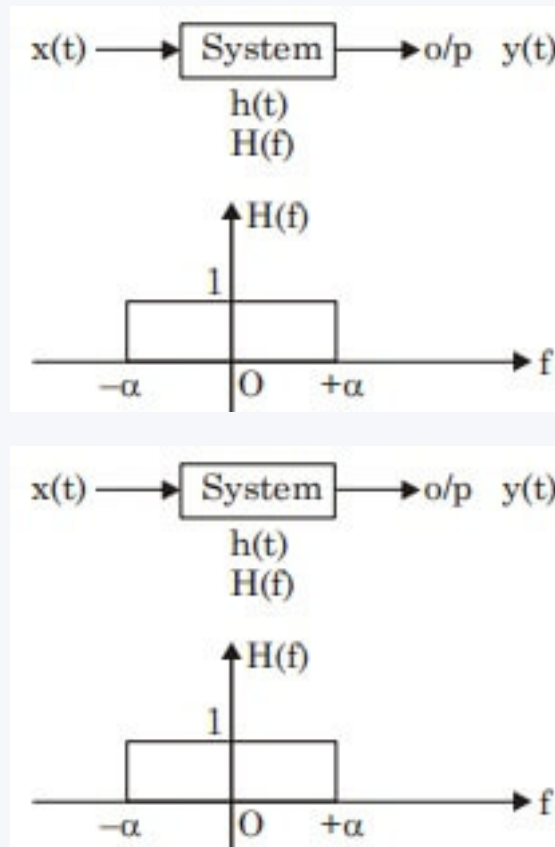
In a circuit, there is a series connection of an ideal resistor and an ideal capacitor. The conduction current (in Amperes) through the resistor is $2 \sin(t + \pi/2)$. The displacement current (in Amperes) through the capacitor is _____.

- (A) $2 \sin(t)$

- (B) $2 \sin(t + \pi)$
- (C) **(Correct)** $2 \sin(t + \pi/2)$
- (D) 0

Correct Answer: (C)

Detailed Solution – Question 23



In series connection, current pass through each element remain same. Hence, $i_c = i_d$ So,

$$i_d = 2 \sin(t + \pi/2)$$

Question 24: Antennas & Radiating Systems (GATE EC 2021)

NAT

2 Marks

An antenna with a directive gain of 6 dB is radiating a total power of 16 kW. The amplitude of the electric field in free space at a distance of 8 km from the antenna in the direction of 6 dB gain (rounded off to three decimal places) is _____ V/m .

Detailed Solution – Question 24

$$\begin{aligned}
G_d(\text{ dB}) &= 10 \log_{10} (G_d) \\
\Rightarrow 6 &= 10 \log_{10} G_d \\
\Rightarrow G_d &= 10^{0.6} \\
G_d &= 3.981 \\
E_m &= \frac{\sqrt{60 G_d P_{\text{rad}}}}{r} \\
E_m &= \frac{\sqrt{60 (16 \times 10^3) (3.981)}}{8 \times 10^3} = 0.244 \text{ (V/m)}
\end{aligned}$$

Question 25: Waveguides & Optical Fibers (GATE EC 2021)**NAT****2 Marks**

A standard air-filled rectangular waveguide with dimensions $a = 8\text{ cm}$, $b = 4\text{ cm}$, operates at 3.4 GHz . For the dominant mode of wave propagation, the phase velocity of the signal in v_p . The value (rounded off to two decimal places) of v_p/c , where c denotes the velocity of light, is _____

Detailed Solution – Question 25

$$\begin{aligned}
f_{c10} &= \frac{c}{2a} = \frac{3 \times 10^8}{2(8 \times 10^{-2})} = 1.875\text{GHz} \\
\text{Guide phase velocity, } V_p &= \frac{c}{\sqrt{1 - \left(\frac{f_{c10}}{f}\right)^2}} \\
\frac{V_p}{c} &= \frac{1}{\sqrt{1 - \left(\frac{f_{c10}}{f}\right)^2}} = \frac{1}{\sqrt{1 - \left(\frac{1.875}{3.4}\right)^2}} = 1.198
\end{aligned}$$

Question 26: Basics of Electromagnetics & Maxwell's Eqns (GATE EC 2021)**NAT****2 Marks**

For a vector field $D = \rho \cos^2 \varphi a_\rho + z^2 \sin^2 \varphi a_\varphi$ in a cylindrical coordinate system (ρ, φ, z) with unit vectors a_ρ, a_φ and a_z , the net flux of D leaving the closed surface of the cylinder ($\rho = 3, 0 \leq z \leq 2$) (rounded off to two decimal places) is _____

Detailed Solution – Question 26

Method 1: $\vec{D} = \rho \cos^2 \phi \hat{a}_\rho + z^2 \sin^2 \phi \hat{a}_\phi$ Electric flux crossing the closed surface is

$$\begin{aligned}
 \psi &= \oiint \vec{D} \cdot \vec{dS} = \iiint (\vec{\nabla} \cdot \vec{D}) dv \\
 \vec{\nabla} \cdot \vec{D} &= \frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho D_\rho) + \frac{1}{\rho} \frac{\partial D_\phi}{\partial \phi} + \frac{\partial D_z}{\partial z} \\
 &= \frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho \cos^2 \phi) + \frac{1}{\rho} \frac{\partial}{\partial \phi} (z^2 \sin^2 \phi) + 0 \\
 &= \frac{1}{\rho} (2 \cos^2 \phi) + \frac{1}{\rho} z^2 2 \sin \phi \cos \phi = 2 \cos^2 \phi + \frac{z^2}{\rho} \sin 2\phi \\
 \iiint (\vec{\nabla} \cdot \vec{D}) dv &= \iiint 2 \cos^2 \phi (\rho d\rho d\phi dz) + \iiint \left(\frac{z^2}{\rho} \sin 2\phi \right) \rho d\rho d\phi dz \\
 &= 2 \int_{\rho=0}^3 \rho d\rho \int_{\phi=0}^{2\pi} \left(\frac{1 + \cos 2\phi}{2} \right) d\phi \int_{z=0}^2 dz + \int_{\rho=0}^2 d\rho \int_{\phi=0}^{2\pi} \sin 2\phi d\phi \int_{z=0}^2 z^2 dz \\
 &= 2 \left(\frac{\rho^2}{-2} \right)_{\rho=0}^3 \frac{1}{2} (2\pi) (z)_{z=0}^2 + 0 \\
 &= 2 \left(\frac{3^2}{2} \right) \pi (2) = 18\pi \text{ (Coulomb)} = 56.55 \text{ (Coulomb)}
 \end{aligned}$$

Method 2: Electric flux crossing the closed surface is $\psi = \oiint \vec{D} \cdot \vec{dS}$ Electric flux crossing $\rho=3$ cylindrical surface is

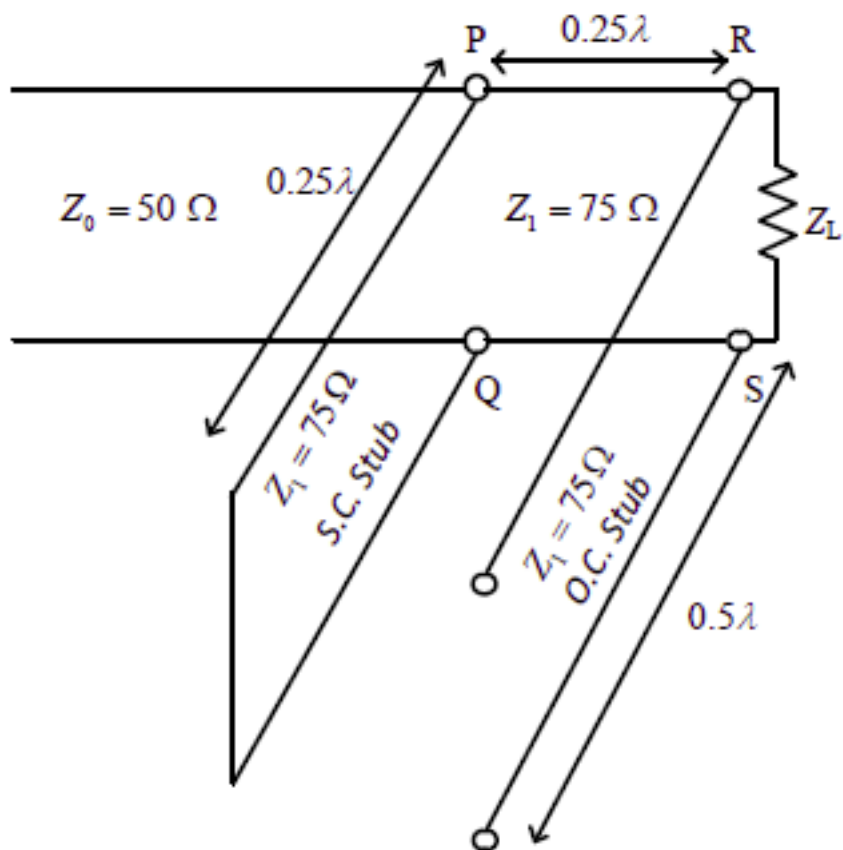
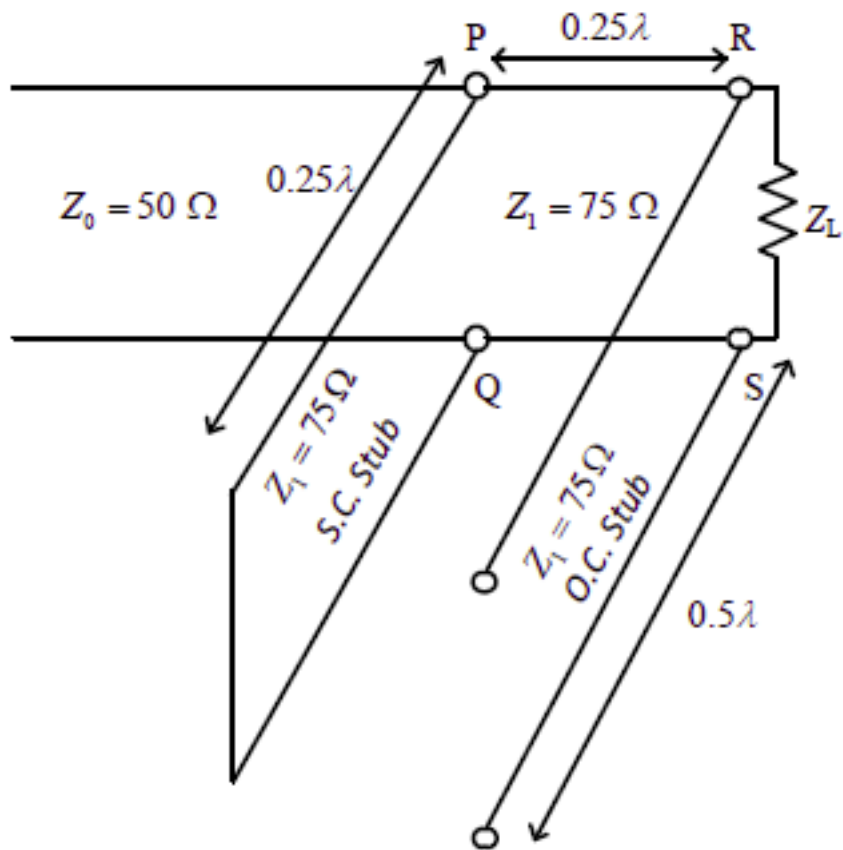
$$\begin{aligned}
 \psi|_{\rho=3} &= \oiint (\rho \cos^2 \phi \hat{a}_\rho) \cdot (\rho d\phi dz) \hat{a}_\rho \\
 &= 3^2 \int_{\phi=0}^{2\pi} \cos^2 \phi d\phi \int_{z=0}^2 dz \\
 &= 9 \frac{1}{2} (2\pi) (2) = 18\pi \text{ (coulomb)} = 56.55 \text{ (coulomb)}
 \end{aligned}$$

Question 27: Transmission Lines (GATE EC 2021)

MCQ

2 Marks

The impedance matching network shown in the figure is to match a lossless line having characteristic impedance $Z_0 = 50 \Omega$ with a load impedance Z_L . A quarter-wave line having a characteristic impedance $Z_1 = 75 \Omega$ is connected to Z_L . Two stubs having characteristic impedance of 75Ω each are connected to this quarter-wave line. One is a short-circuited (S.C) stub of length 0.25λ connected across PQ and the other one is an open-circuited (O.C) stub of length 0.5λ connected across RS.



The impedance matching is achieved when the real part of Z_L is

- (A) (Correct) 112.5Ω

- (B) 75.0Ω
 (C) 50.0Ω
 (D) 33.3Ω

Correct Answer: (A)

Detailed Solution – Question 27

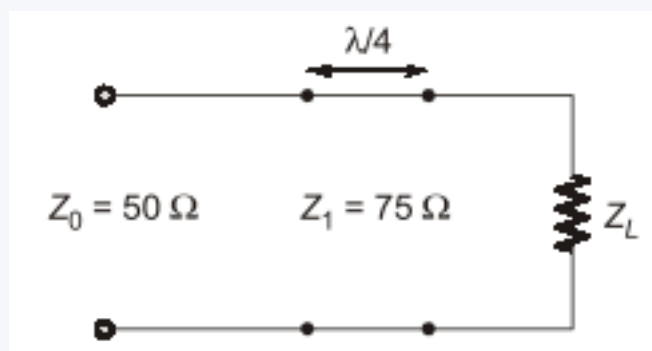
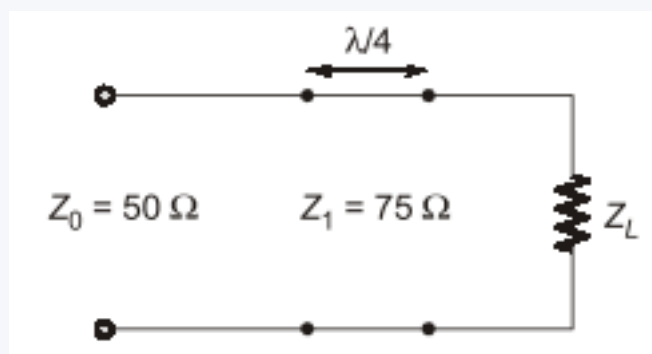
$$Z_{in \lambda/4} = \frac{Z_0^2}{Z_L} = \frac{(75)^2}{0} = \infty \quad \left[\text{for } \frac{\lambda}{4} T_X \text{ line} \right]$$

Given Z_L of $\frac{\lambda}{4}$ line is $0(\text{SC})$

$$Z_{in \lambda/2} = Z_L = \infty \quad \left[\text{for } \frac{\lambda}{2} T_X \text{ line} \right]$$

Given Z_L of $\frac{\lambda}{2}$ line is $\infty(O.C)$

The input impedance of $\frac{\lambda}{4}$ transmission line, as well as $\frac{\lambda}{2}$ transmission line is ∞ , and they are in parallel with main transmission line, so they are not effective for main. Transmission line. Final configuration of given line is



For impedance matching, $Z_L = \frac{Z_1^2}{Z_0} = \frac{(75)^2}{50} = 112.5$

Question 28: Waveguides & Optical Fibers (GATE EC 2021)

NAT

1 Mark

The refractive indices of the core and cladding of an optical fiber are 1.50 and 1.48, respectively. The critical propagation angle, which is defined as the maximum angle that the light beam makes with the

axis of the optical fiber to achieve the total internal reflection, (rounded off to two decimal places) is _____ degree.

Detailed Solution – Question 28

Given that Refractive index of core $\eta_1 = 1.50$ Refractive index of clad $\eta_2 = 1.48$ Critical propagation angle (θ_P)

$$\theta_P = \sin^{-1} \left[\frac{\sqrt{\eta_1^2 - \eta_2^2}}{\eta_1} \right] = \sin^{-1} \left[\frac{\sqrt{1.5^2 - 1.48^2}}{1.5} \right] = 9.37$$

Question 29: Basics of Electromagnetics & Maxwell's Eqns (GATE EC 2021)

NAT

1 Mark

Consider the vector field

$$\vec{F} = a_x (4y - c_1 z) + \hat{a}_y (4x + 2z) + a_z (2y + z)$$

in a rectangular coordinate system (x,y,z) with unit vectors a_x , a_y and a_z . If the field $\vec{F}$ is irrotational (conservative), then the constant c_1 (in integer) is _____

Detailed Solution – Question 29

$$\begin{aligned} \nabla \times \vec{F} &= 0 \\ \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 4y - c_1 z & 4x + 2z & 2y + z \end{vmatrix} &= 0 \\ &= i(2 - 2) - j(0 + c_1) + k(4 - 4) = 0 \\ c_1 &= 0 \end{aligned}$$

Question 30: Waveguides & Optical Fibers (GATE EC 2021)

MCQ

1 Mark

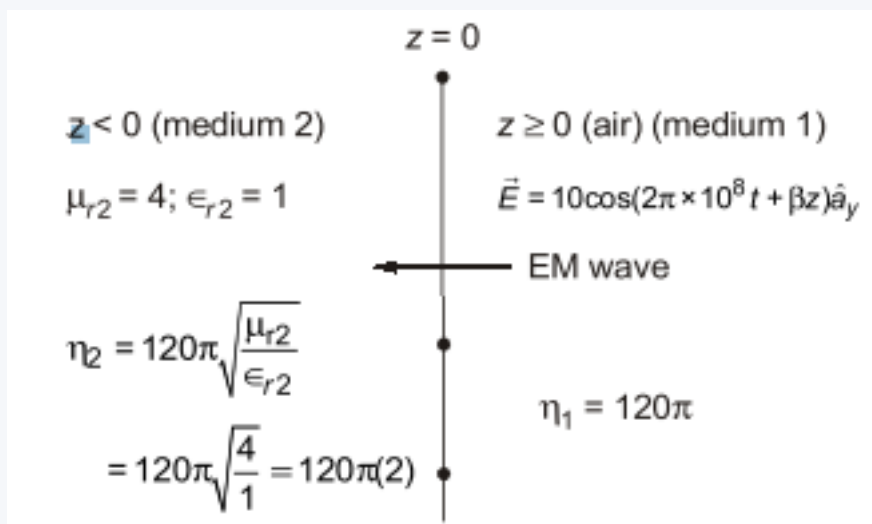
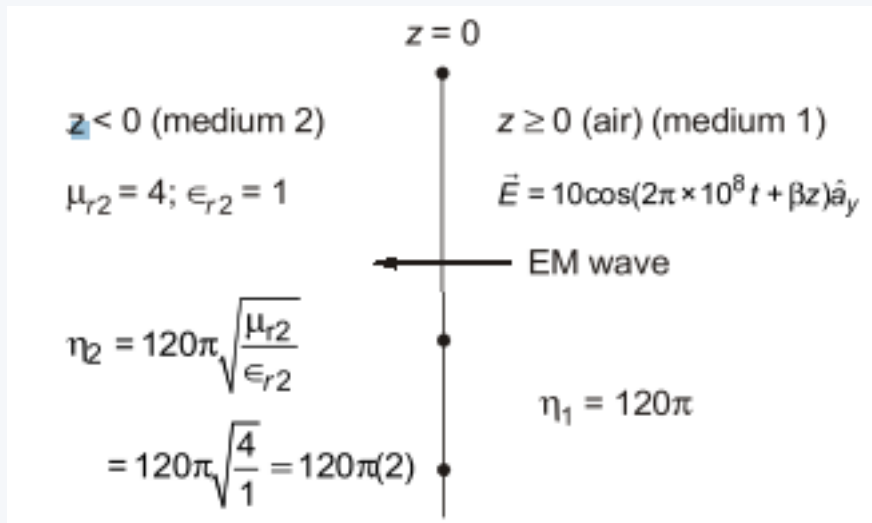
Consider a rectangular coordinate system (x,y,z) with unit vectors a_x , a_y and a_z . A plane wave traveling in the region $z \geq 0$ with electric field vector $\vec{E} = 10 \cos(2 \times 10^8 t + \beta z) a_y$ is incident normally on the plane at $z = 0$, where β is the phase constant. The region $z \geq 0$ is in free space and the region $z < 0$ is filled with a lossless medium (permittivity $\epsilon = \epsilon_0$, permeability $\mu = 4\mu_0$, where $\epsilon_0 = 8.85 \times 10^{-12} \text{ F/m}$ and $\mu_0 = 4\pi \times 10^{-7} \text{ H/m}$). The value of the reflection coefficient is

- (A) (Correct) $\frac{1}{3}$
- (B) $\frac{3}{5}$
- (C) $\frac{2}{5}$
- (D) $\frac{2}{3}$

Correct Answer: (A)

Detailed Solution – Question 30

Given $\vec{E} = 10 \cos(2\pi \times 10^8 t + \beta z) \hat{a}_y$ for $z \geq 0$ having free space. For $z < 0$ medium has $\epsilon_{r2} = 1$; $\mu_{r2} = 4$



$$\Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} = \frac{120\pi(2) - 120\pi}{120\pi(2) + 120\pi} = \frac{1}{3}$$

Question 31: Uniform Plane Waves (GATE EC 2020)

NAT

2 Marks

The magnetic field of a uniform plane wave in vacuum is given by

$$\vec{H}(x, y, z, t) = (\hat{a}_x + 2\hat{a}_y + b\hat{a}_z) \cos(\omega t + 3x - y - z)$$

The value of b is _____

Detailed Solution – Question 31

For uniform plane wave $\hat{a}_H \cdot \hat{a}_\rho = 0$

$\hat{a}_H$ is unit vector in magnetic field direction $\hat{a}_\rho$ is unit vector in power flow direction

$$\hat{a}_H = \frac{1\hat{a}_x + 2\hat{a}_y + b\hat{a}_z}{\sqrt{1^2 + 2^2 + b^2}}$$

$$\hat{a}_\rho = \frac{-3\hat{a}_x + \hat{a}_y + \hat{a}_z}{\sqrt{3^2 + 1^2 + 1^2}}$$

$$\hat{a}_H \cdot \hat{a}_\rho = 0(\hat{a}_x + 2\hat{a}_y + b\hat{a}_z) \cdot (-3\hat{a}_x + \hat{a}_y + \hat{a}_z) = 0$$

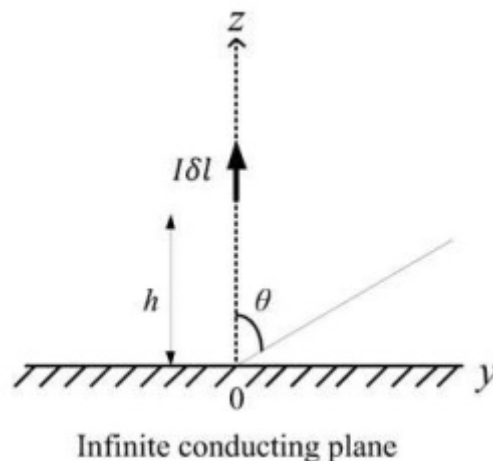
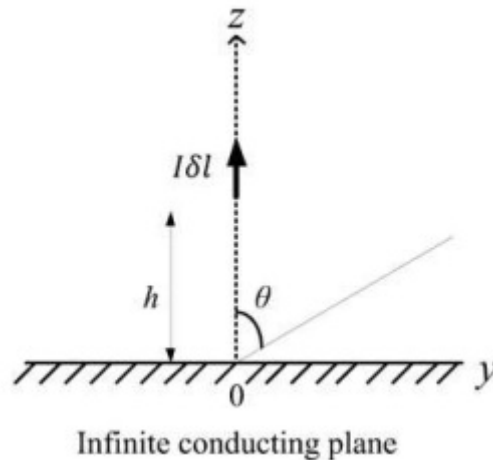
$$-3 + 2 + b = 0 \quad b = 1$$

Question 32: Antennas & Radiating Systems (GATE EC 2020)

MCQ

2 Marks

For an infinitesimally small dipole in free space, the electric field E_θ in the far field is proportional to $(e^{-jkr}/r) \sin \theta$, where $k = 2\pi/\lambda$. A vertical infinitesimally small electric dipole ($\delta l \ll \lambda$) is placed at a distance h ($h > 0$) above an infinite ideal conducting plane, as shown in the figure. The minimum value of h , for which one of the maxima in the far field radiation pattern occurs at $\theta = 60^\circ$, is



(A) (Correct) λ

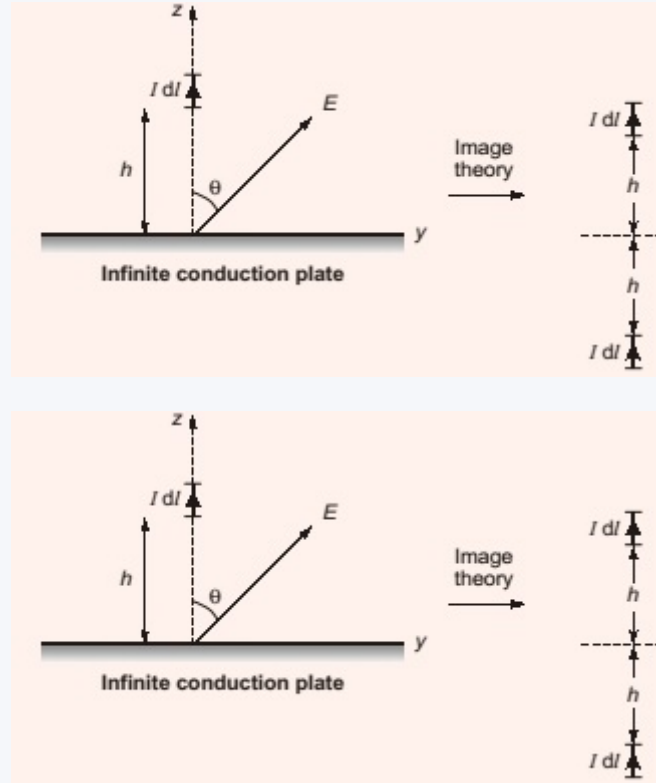
(B) 0.5λ

(C) 0.25λ

(D) 0.75λ

Correct Answer: (A)

Detailed Solution – Question 32



$$|\text{Total } E| = |(E_{\text{single element}})| |(A.F.)|$$

$$|(A.F.)| = \frac{\sin\left(N\frac{\psi}{2}\right)}{\sin\left(\frac{\psi}{2}\right)} = \frac{\sin\left(2\frac{\psi}{2}\right)}{\sin\left(\frac{\psi}{2}\right)}$$

$$= \frac{2 \sin\left(\frac{\psi}{2}\right) \cos\left(\frac{\psi}{2}\right)}{\sin\frac{\psi}{2}}$$

$$= 2 \cos\left(\frac{\psi}{2}\right)$$

$$|A.F_N| = \frac{A.F}{A.F_{\max}} = \frac{2 \cos\left(\frac{\psi}{2}\right)}{2}$$

$$= \left| \cos\left(\frac{\psi}{2}\right) \right|$$

$$\text{where, } \psi = \beta d \cos \theta = \frac{2\pi}{\lambda} (2h) \cos \theta$$

$$|A.F_N|_{\theta=60^\circ} = \left| \cos\left(\frac{2\pi}{\lambda} h \cos 60^\circ\right) \right|$$

$$= \left| \cos\left(\frac{\pi h}{\lambda}\right) \right|$$

$\cos \theta$ is maximum, where $\theta = n\pi$ $n = 0, 1, 2, \dots$

$$\frac{\pi h}{\lambda} = n\pi \Rightarrow h = n\lambda$$

$$\Rightarrow \text{For } n = 1, h_{\min} = \lambda$$

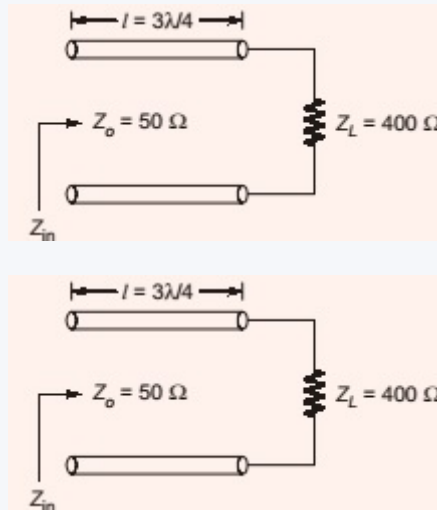
Question 33: Transmission Lines (GATE EC 2020)

NAT

1 Mark

A transmission line of length $3\lambda/4$ and having a characteristic impedance of 50Ω is terminated with a load of 400Ω . The impedance (rounded off to two decimal places) seen at the input end of the transmission line is _____ Ω .

Detailed Solution – Question 33



$$Z_{in} \text{ for } (l = \lambda/4) = \frac{Z_0^2}{Z_L} = \frac{50^2}{400} = \frac{25}{4} = 6.25\Omega$$

Question 34: Transmission Lines (GATE EC 2020)

MCQ

1 Mark

The impedances $Z=jX$, for all X in the range $(-\infty, \infty)$, map to the Smith chart as

- (A) **(Correct)** a circle of radius 1 with centre at $(0, 0)$.
- (B) a point at the centre of the chart.
- (C) a line passing through the centre of the chart
- (D) a circle of radius 0.5 with centre at $(0.5, 0)$.

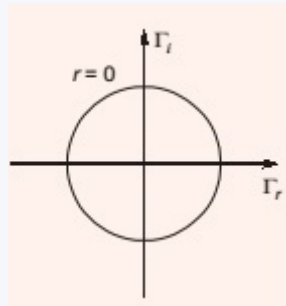
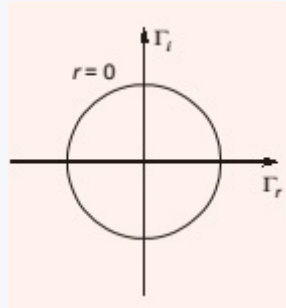
Correct Answer: (A)

Detailed Solution – Question 34

For given impedance Normalized impedance is

$$\begin{aligned}\frac{Z}{Z_0} &= \frac{jX}{Z_0} \\ Z &= jX \\ \Rightarrow Z &= 0 + jX\end{aligned}$$

Normalized Resistance = 0 $\Rightarrow r = 0$ $X = -\infty$ to ∞ $r = 0$ and X from $-\infty$ to ∞ is a unit circle (radius 1) and centre (0,0) on a complex reflection coefficient plane:



Question 35: Waveguides & Optical Fibers (GATE EC 2019)

NAT

2 Marks

A rectangular waveguide of width w and height h has cut-off frequencies for TE_{10} and TE_{11} modes in the ratio 1:2. The aspect ratio w/h , rounded off to two decimal places, is _____

Detailed Solution – Question 35

$$f_{cmn} = \frac{c}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}$$

For TE mode $f_{c10} = \frac{c}{2w} \dots (i)$ and For TE_{11} mode.

$$f_{c11} = \frac{c}{2} \sqrt{\left(\frac{1}{w}\right)^2 + \left(\frac{1}{h}\right)^2} \dots (i)$$

$$= \frac{c}{2w} \sqrt{1 + \left(\frac{w}{h}\right)^2} \dots (ii)$$

$$\text{given } \frac{f_{c10}}{f_{11}} = \frac{1}{2} \dots (iii)$$

put (i). (in) in (ii)

$$\Rightarrow \frac{\frac{c}{2w}}{\frac{c}{2w} \sqrt{1 + \left(\frac{w}{h}\right)^2}} = \frac{1}{2} \Rightarrow \sqrt{1 + \left(\frac{w}{h}\right)^2} = 2$$

On solving above equation, we get, $\frac{w}{h} = \sqrt{3} = 1.732$

Question 36: Waveguides & Optical Fibers (GATE EC 2019)

MCQ

2 Marks

The dispersion equation of a waveguide, which relates the wavenumber k to the frequency ω , is

$$k(\omega) = (1/c) \sqrt{\omega^2 - \omega_0^2}$$

where the speed of light $c = 3 \times 10^8$ m/s, and ω_0 is a constant. If the group velocity is 2×10^8 m/s, then the phase velocity is

- (A) 1.5×10^8 m/s
- (B) 2×10^8 m/s
- (C) 3×10^8 m/s
- (D) **(Correct)** 4.5×10^8 m/s

Correct Answer: (D)

Detailed Solution – Question 36

By definition $v_p = \frac{\omega}{\beta} = \frac{\omega}{k}$ where, $k(\omega) = \left(\frac{1}{c}\right) \sqrt{\omega^2 - \omega_0^2}$ (given) $\therefore v_p = \frac{c}{\sqrt{1 - \left(\frac{\omega_0}{\omega}\right)^2}}$ by definition,

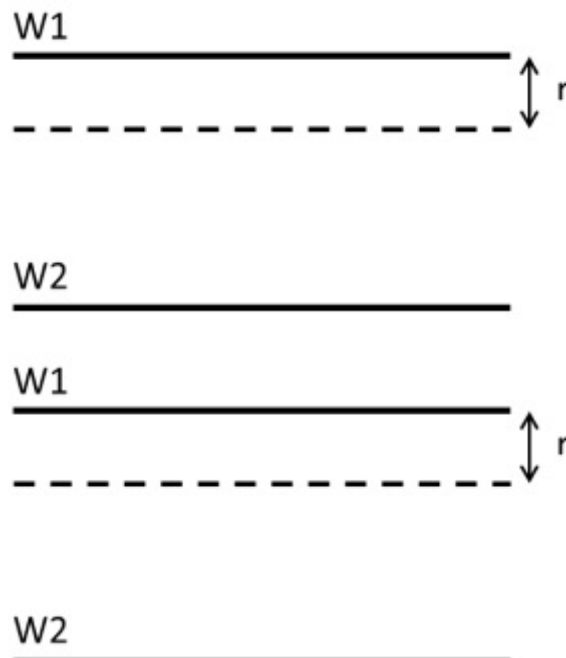
$$\begin{aligned} v_g &= \frac{d\omega}{d\beta} = \frac{d\omega}{dk} \\ &= \frac{dk}{d\omega} = \frac{1}{c} \frac{1}{2\sqrt{\omega^2 - \omega_0^2}} \times 2\omega \\ \text{or } v_g &= c\sqrt{1 - \left(\frac{\omega_0}{\omega}\right)^2} \\ \therefore v_p \cdot v_g &= c^2 \\ \therefore v_p &= \frac{c^2}{v_g} = \frac{(3 \times 10^8)^2}{2 \times 10^8} \\ &= 4.5 \times 10^8 \text{ m/sec} \end{aligned}$$

Question 37: Basics of Electromagnetics & Maxwell's Eqns (GATE EC 2019)

MCQ

2 Marks

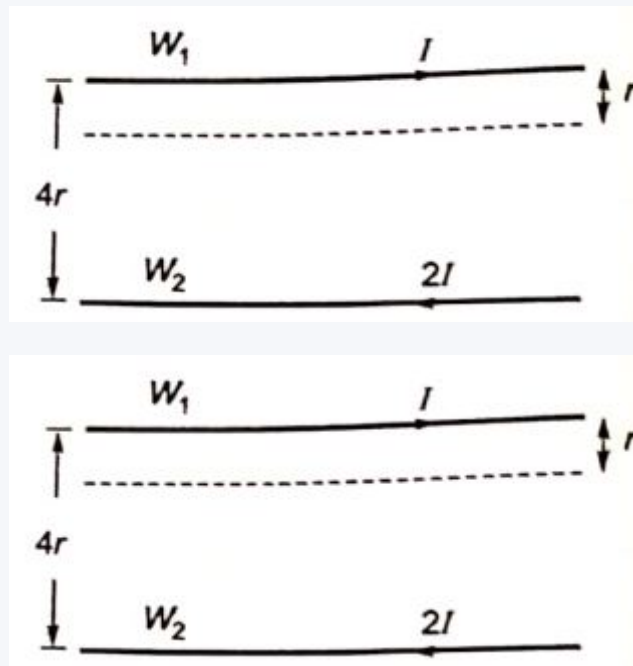
Two identical copper wires W1 and W2, placed in parallel as shown in the figure, carry currents I and $2I$, respectively, in opposite directions. If the two wires are separated by a distance of $4r$, then the magnitude of the magnetic field $\vec{B}$ between the wires at a distance r from W1 is



- (A) $\frac{\mu_0 I}{6\pi r}$
- (B) $\frac{6\mu_0 I}{5\pi r}$
- (C) **(Correct)** $\frac{5\mu_0 I}{6\pi r}$
- (D) $\frac{\mu_0^2 I^2}{2\pi r^2}$

Correct Answer: (C)

Detailed Solution – Question 37



Magnetic flux density ($\vec{B}$) at r distance due to infinite line carrying current I is $|\vec{B}| = \frac{\mu_0 I}{2\pi r}$. $\vec{B}$ at r distance due to W_1 wire = $|\vec{B}_1| = \frac{\mu_0 I}{2\pi r}$... (i) $\vec{B}$ at $3r$ distance due to W_2 wire = $|\vec{B}_2| = \frac{\mu_0 (2I)}{2\pi (3r)}$... (ii) From right hand thumb rule, $\vec{B}$ due to both lines add in between conductors.

$$\text{So, } |\vec{B}| = |\vec{B}_1| + |\vec{B}_2|$$

$$\therefore |\vec{B}| = \frac{\mu_0 I}{2\pi r} + \frac{2\mu_0 I}{6\pi r} = \frac{5\mu_0 I}{6\pi r}$$

Question 38: Antennas & Radiating Systems (GATE EC 2019)

NAT

1 Mark

Radiation resistance of a small dipole current element of length l at a frequency of 3 GHz is 3 ohms. If the length is changed by 1

Detailed Solution – Question 38

Radiation resistance of a small dipole current element of length 'l' is

$$R_{\text{rad}} = 80\pi^2 \left(\frac{l}{\lambda}\right)^2 \Rightarrow R \propto l^2$$

$$\frac{R_2}{R_1} = \left(\frac{l_2}{l_1}\right)^2$$

If length is changed by 1 $\frac{R_2}{R_1} = \left(\frac{1.01l}{l}\right)^2 = 1.0201$ Percentage change in radiation resistance

$$= \frac{R_2 - R_1}{R_1} \times 100$$

$$= 0.0201 \times 100 = 2.01\%$$

Question 39: Basics of Electromagnetism & Maxwell's Eqns (GATE EC 2019)

MCQ

1 Mark

In the table shown, List I and List II, respectively, contain terms appearing on the left-hand side and the right-hand side of Maxwell's equations (in their standard form). Match the left-hand side with the corresponding right-hand side.

List I		List II	
1	$\nabla \cdot \mathbf{D}$	P	0
2	$\nabla \times \mathbf{E}$	Q	ρ
3	$\nabla \cdot \mathbf{B}$	R	$-\frac{\partial \mathbf{B}}{\partial t}$
4	$\nabla \times \mathbf{H}$	S	$\mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$

List I		List II	
1	$\nabla \cdot \mathbf{D}$	P	0
2	$\nabla \times \mathbf{E}$	Q	ρ
3	$\nabla \cdot \mathbf{B}$	R	$-\frac{\partial \mathbf{B}}{\partial t}$
4	$\nabla \times \mathbf{H}$	S	$\mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$

- (A) 1-P, 2-R, 3-Q, 4-S
 (B) **(Correct)** 1-Q, 2-R, 3-P, 4-S
 (C) 1-Q, 2-S, 3-P, 4-R
 (D) 1-R, 2-Q, 3-S, 4-P

Correct Answer: (B)

Detailed Solution – Question 39

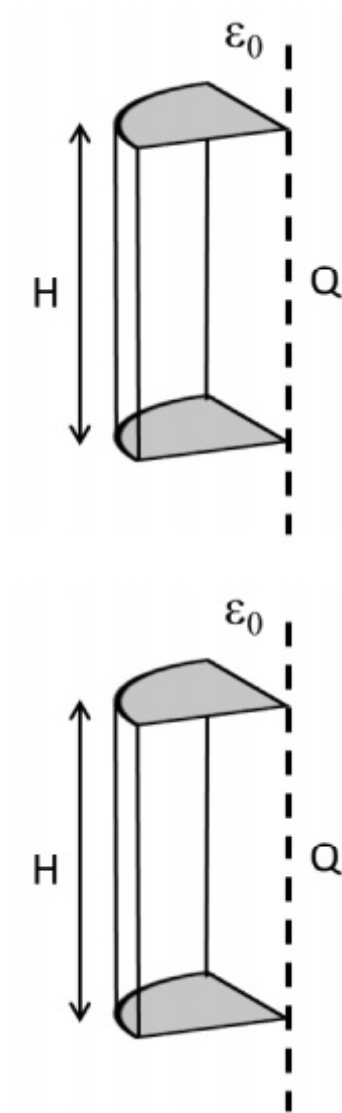
$$\begin{aligned}\nabla \cdot \vec{D} &= \rho_v \\ \nabla \times \vec{E} &= -\frac{\partial \vec{B}}{\partial t} \\ \nabla \cdot \vec{B} &= 0 \\ \nabla \times \vec{H} &= \vec{J} + \frac{\partial \vec{D}}{\partial t}\end{aligned}$$

Question 40: Basics of Electromagnetics & Maxwell's Eqns (GATE EC 2019)

MCQ

1 Mark

What is the electric flux ($\int \vec{E} \cdot d\hat{a}$) through a quarter-cylinder of height H (as shown in the figure) due to an infinitely long line charge along the axis of the cylinder with a charge density of Q ?



- (A) $\frac{HQ}{\epsilon_0}$
- (B) **(Correct)** $\frac{HQ}{4\epsilon_0}$
- (C) $\frac{H\epsilon_0}{4Q}$
- (D) $\frac{4H}{Q\epsilon_0}$

Correct Answer: (B)

Detailed Solution – Question 40

Electric field intensity ($\vec{E}$) at ' ρ ' distance due to infinite long line having line charge density Q is

$$\begin{aligned}\vec{E} &= \frac{Q}{2\pi\epsilon_0\rho}\hat{a}_\rho \\ \int \vec{E} \cdot d\vec{a} &= \iint \frac{Q}{2\pi\epsilon_0\rho}\hat{a}_\rho \cdot \rho d\phi dz \hat{a}_\rho \\ &= \frac{Q}{2\pi\epsilon_0} \int_{<\pi/2>} d\phi \int_{z=0}^H dz \\ &= \frac{Q}{2\pi\epsilon_0} \left(\frac{\pi}{2}\right) H = \frac{HQ}{4\epsilon_0}\end{aligned}$$

Question 41: Uniform Plane Waves (GATE EC 2018)

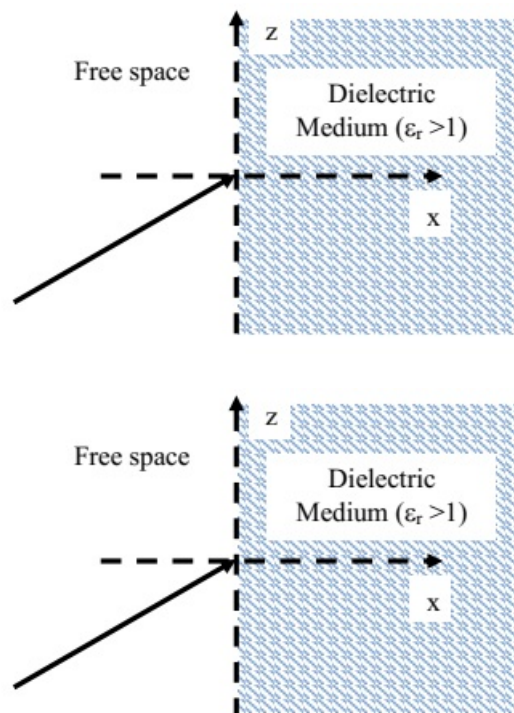
NAT

2 Marks

A uniform plane wave traveling in free space and having the electric field

$$\vec{E} = (\sqrt{2}\hat{a}_x - \hat{a}_z)\cos[6\sqrt{3}\pi \times 10^8 t - 2\pi(x + \sqrt{2}z)]V/m$$

is incident on a dielectric medium (relative permittivity >1 , relative permeability $= 1$) as shown in the figure and there is no reflected wave.



The relative permittivity (correct to two decimal places) of the dielectric medium is_____.

Detailed Solution – Question 41

The wave has E_i direction and propagation direction both in same plane (ZX). The wave is plane of incidence (P) polarized.

$$\beta_x = 1, \quad \beta_z = \sqrt{2}$$
$$\tan \theta_i = \frac{\beta_z}{\beta_x} = \sqrt{2}$$

No reflection in P polarized wave under Brewster's angle of incidence.

$$\theta_i = \theta_B$$
$$\tan \theta_B = \sqrt{2} = \sqrt{\epsilon_r}$$
$$\epsilon_r = 2$$

Question 42: Waveguides & Optical Fibers (GATE EC 2018)

NAT

2 Marks

The cutoff frequency of TE_{01} mode of an air filled rectangular waveguide having inner dimensions $a \text{ cm} \times b \text{ cm} (a > b)$ is twice that of the dominant TE_{10} mode. When the waveguide is operated at a frequency which is 25% dominant mode, the guide wavelength is found to be 4 cm. The value of b (in cm, correct to two decimal places) is _____.

Detailed Solution – Question 42

$$f_{c(01)} = 2f_{c(10)} = \frac{2c}{2a} = \frac{c}{a}$$
$$\frac{c}{2b} = \frac{c}{a} \Rightarrow a = 2b \Rightarrow b = \frac{a}{2}$$

Operating frequency, $f = 1.25f_{c(10)}$ $f_{c(10)} < 1.25f_{c(10)} < [f_{c(01)} = 2f_{c(10)}]$ So, for the given frequency, the waveguide will work in TE_{10} mode.

$$\text{So, } \lambda_g = \frac{\lambda_0}{\sqrt{1 - \left(\frac{f_{c(10)}}{f}\right)^2}}$$
$$= \frac{c/f}{\sqrt{1 - \left(\frac{1}{1.25}\right)^2}} = \frac{c/f}{0.6}$$
$$\frac{c}{(1.25)f_{c(10)}(0.6)} = \lambda_g = 4 \text{ cm}$$
$$\frac{c}{f_{c(10)}} = 3 \times 10^{-2} = 2a$$
$$a = 1.5 \text{ cm}$$
$$b = \frac{a}{2} = 0.75 \text{ cm}$$

Question 43: Uniform Plane Waves (GATE EC 2018)

MCQ

2 Marks

The distance (in meters) a wave has to propagate in a medium having a skin depth of 0.1 m so that the amplitude of the wave attenuates by 20 dB, is

- (A) 0.12
- (B) **(Correct)** 0.23
- (C) 0.46
- (D) 2.3

Correct Answer: (B)

Detailed Solution – Question 43

Attenuation constant,

$$\begin{aligned}\alpha &= \frac{1}{\text{skin depth}} = 10 \text{ Np/m} \\ 20 \log_{10} \left(\frac{E_0}{E_x} \right) &= 20 \text{ dB} \\ \frac{E_0}{E_x} &= 10 \Rightarrow E_x = \frac{E_0}{10} \\ E_x &= E_0 e^{-\alpha x} = E_0 e^{-10x} = \frac{E_0}{10} \\ e^{-10x} &= \frac{1}{10} \\ x &= \frac{1}{10} \ln(10) = 0.23 \text{ m}\end{aligned}$$

Question 44: Transmission Lines (GATE EC 2018)

NAT

1 Mark

A lossy transmission line has resistance per unit length $R = 0.05 \, \Omega/\text{m}$. The line is distortionless and has characteristic impedance of $50 \, \Omega$. The attenuation constant (in Np/m, correct to three decimal places) of the line is _____.

Detailed Solution – Question 44

For a distortionless transmission line, $\frac{L}{R} = \frac{C}{G}$ Propagation constant,

$$\begin{aligned}\gamma &= \alpha + j\beta = \sqrt{(R + j\omega L)(G + j\omega C)} \\ &= \sqrt{RG} \left(1 + j\omega \frac{L}{R} \right)\end{aligned}$$

Attenuation constant, $\alpha = \sqrt{RG}$ Characteristic impedance,

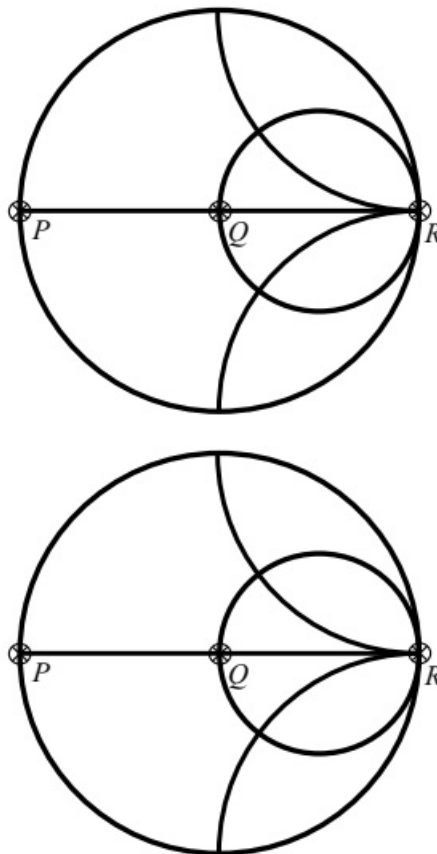
$$\begin{aligned}Z_0 &= \sqrt{\frac{(R + j\omega L)}{(G + j\omega C)}} = \sqrt{\frac{R}{G}} \\ \sqrt{G} &= \frac{\sqrt{R}}{Z_0} \\ \text{So, } \alpha &= \sqrt{R} \cdot \frac{\sqrt{R}}{Z_0} = \frac{R}{Z_0} \\ &= \frac{0.05}{50} = \frac{0.01}{10} = 0.001 \text{ Np/m}\end{aligned}$$

Question 45: Transmission Lines (GATE EC 2018)

MCQ

1 Mark

The points P, Q, and R shown on the Smith chart (normalized impedance chart) in the following figure represent:



(A) P: Open Circuit, Q: Short Circuit, R: Matched Load

- (B) P: Open Circuit, Q: Matched Load, R: Short Circuit
 (C) **(Correct)** P: Short Circuit, Q: Matched Load, R: Open Circuit
 (D) P: Short Circuit, Q: Open Circuit, R: Matched Load

Correct Answer: (C)

Detailed Solution – Question 45

For Short circuit, $r = x = 0 \Rightarrow$ Point "P" For Open circuit, $r = x = \infty \Rightarrow$ Point "R" For Matched load, $r = 1$ and $x = 0 \Rightarrow$ Point "Q" P: Short Circuit, Q: Matched Load R: Open circuit

Question 46: Waveguides & Optical Fibers (GATE EC 2017-SET-2)

MCQ

2 Marks

Standard air-filled rectangular waveguides of dimensions $a = 2.29$ cm and $b = 1.02$ cm are designed for radar applications. It is desired that these waveguides operate only in the dominant TE_{10} mode but not higher than 95

- (A) $8.19\text{GHz} \leq f \leq 13.1\text{GHz}$
 (B) **(Correct)** $8.19\text{GHz} \leq f \leq 12.45\text{GHz}$
 (C) $6.55\text{GHz} \leq f \leq 13.1\text{GHz}$
 (D) $1.64\text{GHz} \leq f \leq 10.24\text{GHz}$

Correct Answer: (B)

Detailed Solution – Question 46

$a = 2.29$ cm, $b = 1.02$ cm Waveguide is operating in (TE_{10}) dominant mode

$$f_c(TE_{10}) = \frac{c}{2a} = \frac{3 \times 10^{10}}{2 \times 2.29} = 6.55\text{GHz}$$

$\Rightarrow$ 25% above cut-off frequency $= 1.25 \times 6.55\text{GHz} = 8.1875\text{GHz}$ Next higher order mode is TE_{20} ($\because a_2 > 2b$)

$$f_c(TE_{20}) = \frac{c}{a} = \frac{3 \times 10^{10}}{2.29} = 13.1\text{GHz}_2$$

$$\Rightarrow 95\% \text{ of } f_c(TE_{20}) = 0.95 \times 13.1$$

$$= 12.445\text{GHz}$$

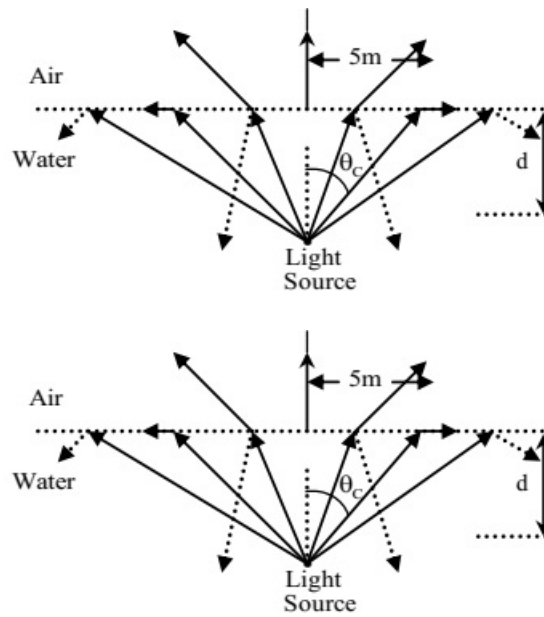
Range of allowable operating frequency $8.19\text{GHz} \leq f \leq 12.45\text{GHz}$

Question 47: Uniform Plane Waves (GATE EC 2017-SET-2)

NAT

2 Marks

The permittivity of water at optical frequencies is $1.75 \epsilon_0$. It is found that an isotropic light source at a distance d under water forms an illuminated circular area of radius 5m , as shown in the figure. The critical angle is θ_c .

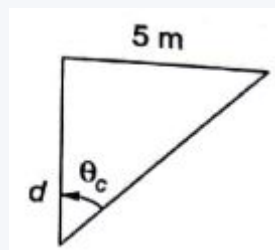
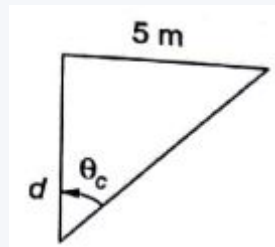


The value of d (in meter) is _____

Detailed Solution – Question 47

Critical angle,

$$\begin{aligned}\theta_c &= \sin^{-1} \sqrt{\frac{\epsilon_2}{\epsilon_1}} \\ &= \sin^{-1} \sqrt{\frac{\epsilon_0}{1.75\epsilon_0}} = 49.1^\circ\end{aligned}$$



$$\begin{aligned}\tan 49.1 &= 1.547 = \frac{5}{d} \\ \Rightarrow d &= \frac{5}{1.547} = 4.33\text{m}\end{aligned}$$

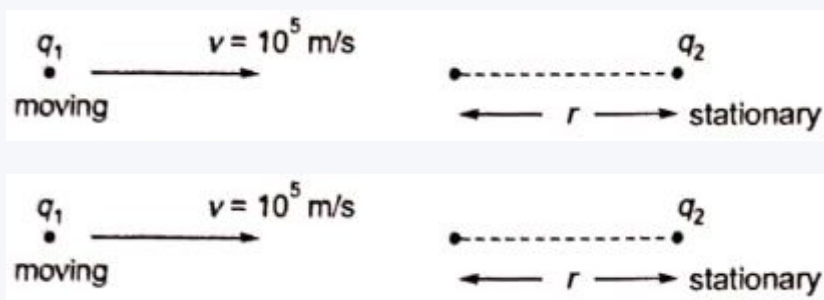
Question 48: Basics of Electromagnetics & Maxwell's Eqns (GATE EC 2017-SET-2)

NAT

2 Marks

An electron (q_1) is moving in free space with velocity 10^5 m/s towards a stationary electron (q_2) far away. The closest distance that this moving electron gets to the stationary electron before the repulsive force diverts its path is $\text{_____} \times 10^{-8}$ m. [Given, mass of electron $m = 9.11 \times 10^{-31}$ kg, charge of electron $e = -1.6 \times 10^{-19}$ C, and permittivity $\epsilon_0 = (1/36\pi) \times 10^{-9}$ F/m]

Detailed Solution – Question 48



r is the distance at which kinetic energy of q_1 becomes zero [(because kinetic energy (KE) is converted into potential energy (PE)]. When q_1 reaches ' r ', it starts diverting. Kinetic energy, $KE = \frac{1}{2}mv^2$ and work done in moving q_1 charge to distance ' r ' is

$$q_1 v_2 = q_1 \frac{q_2}{4\pi\epsilon_0 r}$$

$$(q_1 = q_2 = -1.6 \times 10^{-19} \text{ C})$$

$$\text{Now, } \frac{1}{2}mv^2 = \frac{q_1 q_2}{4\pi\epsilon_0 r}$$

$$\Rightarrow r = \frac{(2 \times -1.6 \times 10^{-19}) \times (-1.6 \times 10^{-19})}{4\pi \times \frac{10^{-9}}{36\pi} \times 9.11 \times 10^{-31} \times (10^5)^2}$$

$$\simeq 5.06 \times 10^{-8} \text{ m}$$

Question 49: Transmission Lines (GATE EC 2017-SET-2)

NAT

1 Mark

A two - wire transmission line terminates in a television set. The VSWR measured on the line is 5.8. The percentage of power that is reflected from the television set is _____

Detailed Solution – Question 49

Given that, VSWR (or) $s = 5.8$

$$s = \frac{1 + |\Gamma|}{1 - |\Gamma|} = 5.8$$

Γ = Reflection coefficient of television set

$$\Gamma = \frac{s - 1}{s + 1} = \frac{4.8}{6.8} = 0.70588$$

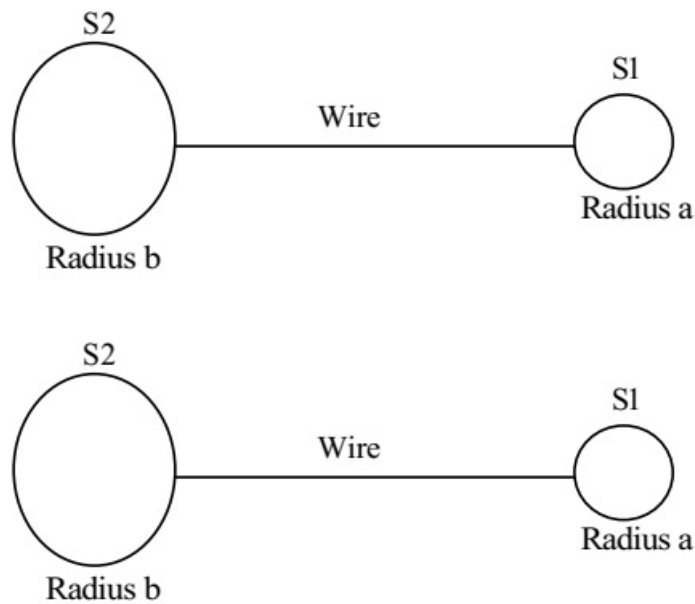
$$\frac{P_{\text{reflected}}}{P_{\text{incident}}} = \Gamma^2 = 0.4983 \text{ or } 49.83\%$$

Question 50: Basics of Electromagnetics & Maxwell's Eqns (GATE EC 2017-SET-2)

MCQ

1 Mark

Two conducting spheres S1 and S2 of radii a and b ($b > a$) respectively, are placed far apart and connected by a long, thin conducting wire, as shown in the figure.

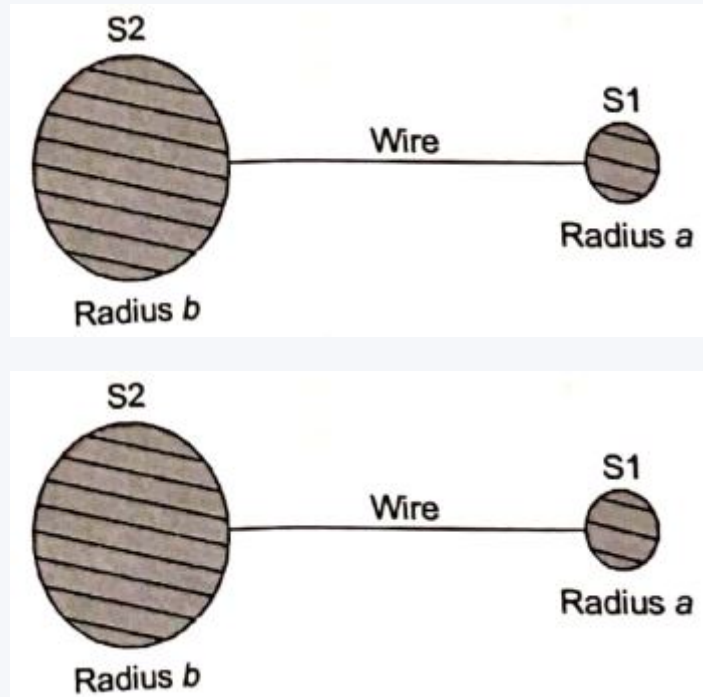


For some charge placed on this structure, the potential and surface electric field on S1 are V_a and E_a , and that on S2 are V_b and E_b , respectively, which of the following is CORRECT?

- (A) $V_a = V_b$ and $E_a < E_b$
- (B) $V_a > V_b$ and $E_a > E_b$
- (C) **(Correct)** $V_a = V_b$ and $E_a > E_b$
- (D) $V_a > V_b$ and $E_a = E_b$

Correct Answer: (C)

Detailed Solution – Question 50



When charge is placed on this structure equilibrium is established such that the spheres are at same potential i.e. $V_a = V_b$

$$\text{So, } \frac{V_a}{4\pi\epsilon_0 a} = \frac{V_b}{4\pi\epsilon_0 b}$$

$\frac{Q_b}{Q_a} = \frac{b}{a}$ Now, surface electric fields.

$$\frac{E_a}{E_b} = \left[\frac{Q_a/4\pi\epsilon_0 a^2}{Q_b/4\pi\epsilon_0 b^2} \right] = \frac{Q_a \times b^2}{Q_b \times a^2} = \frac{b}{a} > 1$$

So, $E_a > E_b$

Question 51: Antennas & Radiating Systems (GATE EC 2017-SET-1)

MCQ

2 Marks

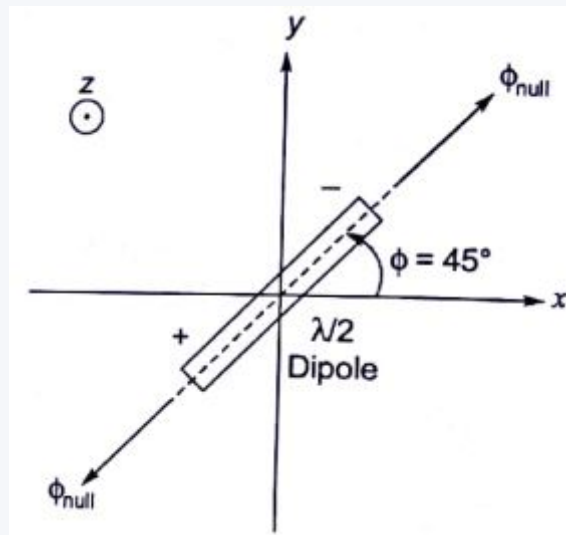
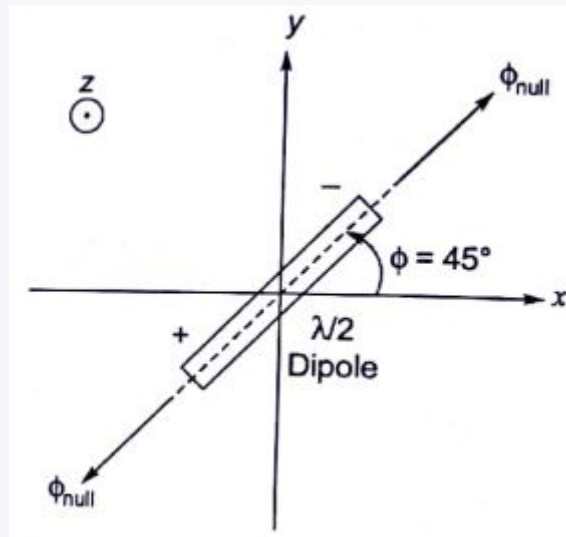
A half wavelength dipole is kept in the x-y plane and oriented along 45° from the x-axis. Determine the direction of null in the radiation pattern for $0 \leq \phi \leq \pi$. Here the angle θ ($0 \leq \theta \leq \pi$) is measured from the z-axis, and the angle ϕ ($0 \leq \phi \leq 2\pi$) is measured from the x-axis in the x-y plane.

- (A) (Correct) $\theta = 90^\circ$, $\phi = 45^\circ$
- (B) $\theta = 45^\circ$, $\phi = 90^\circ$
- (C) $\theta = 90^\circ$, $\phi = 135^\circ$
- (D) $\theta = 45^\circ$, $\phi = 135^\circ$

Correct Answer: (A)

Detailed Solution – Question 51

x-y plane means $Z=0$ with $Z = r \cos \theta$ (spherical coordinates) $\theta = 90^\circ$ With null points along the axis of the dipole it is $\phi = 45^\circ$ and 225°



Question 52: Uniform Plane Waves (GATE EC 2017-SET-1)

MCQ

2 Marks

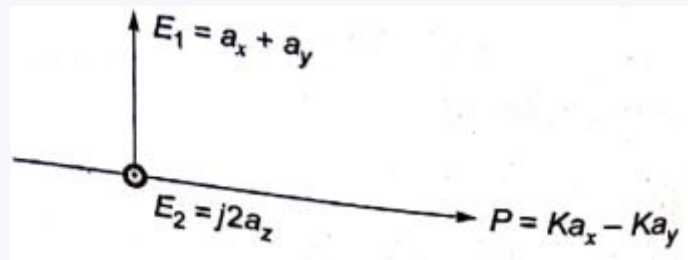
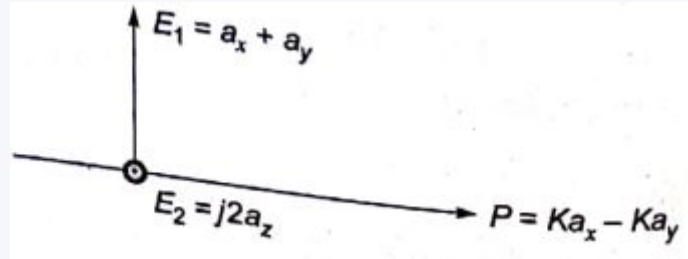
The expression for an electric field in free space is $E = E_0 = (\hat{x} + \hat{y} + j2\hat{z})e^{-j(\omega t - kx + ky)}$, where x, y, z represent the spatial coordinates, t represents time, and ω , k are constants. This electric field

- (A) does not represent a plane wave
- (B) represents a circular polarized plane wave propagating normal to the z-axis
- (C) **(Correct)** represents an elliptically polarized plane wave propagating along x-y plane.
- (D) represents a linearly polarized plane wave

Correct Answer: (C)

Detailed Solution – Question 52

$\vec{E}$ field direction $\Rightarrow (\hat{a}_x + \hat{a}_y + j2\hat{a}_z)$ Propagation direction $\Rightarrow k\hat{a}_x - k\hat{a}_y$ $\vec{E}$ is perpendicular to propagation $\vec{E} \cdot \vec{P} = 0$ Component in $\hat{a}_z$ has magnitude of 2. Component in X-Y plane has magnitude of $\sqrt{2}$. These two components are out of phase by 90° and have unequal amplitudes. So, it is elliptically polarized wave.



Question 53: Miscellaneous Topics (GATE EC 2017-SET-1)

NAT

2 Marks

An optical fiber is kept along the $\hat{z}$ direction. The refractive indices for the electric fields along $\hat{x}$ and $\hat{y}$ directions in the fiber are $n_x=1.5000$ and $n_y=1.5001$, respectively ($n_x \neq n_y$ due to the imperfection in the fiber cross-section). The free space wavelength of a light wave propagating in the fiber is $1.5\mu\text{m}$. If the light wave is circularly polarized at the input of the fiber, the minimum propagation distance after which it becomes linearly polarized, in centimeter, is _____.

Detailed Solution – Question 53

Initially the wave is circularly polarized. So, the initial phase difference between field components in $\hat{a}_x$ direction and $\hat{a}_y$ direction is $\frac{\pi}{2}$

To become linearly polarized, the wave must travel a minimum distance, such that, the phase difference at that point between the field components in $\hat{a}_x$ direction and $\hat{a}_y$ direction is π (i.e., the travel of this minimum distance should provide an additional phase difference of $\pi/2$ between $\hat{a}_x$ and $\hat{a}_y$ field components). So,

$$\begin{aligned} z_{min} k_z \sim z_{min} k_y &= \frac{\pi}{2} \\ z_{min} \left[\frac{\omega}{V_{px}} \sim \frac{\omega}{V_{py}} \right] &= \frac{\pi}{2} \\ 2\pi z_{min} \left[\frac{f}{c} \sqrt{\epsilon_{rx}} \sim \frac{f}{c} \sqrt{\epsilon_{ry}} \right] &= \frac{\pi}{2} \\ \frac{4z_{min}}{\lambda_0} [n_x \sim n_y] &= 1 \\ z_{min} &= \frac{\lambda_0}{4(n_x \sim n_y)} \\ &= \frac{1.5}{4(1.5 \sim 1.5001)} \mu m \\ &= \frac{1.5}{4(0.0001)} \mu m = \frac{1.5}{4} cm \\ z_{min} &= 0.375 cm \end{aligned}$$

Question 54: Antennas & Radiating Systems (GATE EC 2017-SET-1)

MCQ

1 Mark

Consider a wireless communication link between a transmitter and a receiver located in free space, with finite and strictly positive capacity. If the effective areas of the transmitter and the receiver antennas, and the distance between them are all doubled, and everything else remains unchanged, the maximum capacity of the wireless link

- (A) increases by a factor of 2
- (B) decreases by a factor of 2
- (C) **(Correct)** remains unchanged
- (D) decreases by a factor of $\sqrt{2}$

Correct Answer: (C)

Detailed Solution – Question 54

As per Friis free space propagation equation,

$$W_r = \frac{W_t A_{er} A_{et}}{(\lambda d)^2}$$

when A_{er} and A_{et} are doubled and d also doubled, W_r is same. Hence capacity is also same.

Question 55: Transmission Lines (GATE EC 2017-SET-1)

NAT

1 Mark

The voltage of an electromagnetic wave propagating in a coaxial cable with uniform characteristic impedance is $V(l) = e^{-\gamma l + j\omega t}$ volts, Where l is the distance along the length of the cable in meters. $\gamma = (0.1 + j40)m^{-1}$ is the complex propagation constant, and $\omega = 2\pi \times 10^9 rad/s$ is the angular frequency. The absolute value of the attenuation in the cable in dB/meter is _____.

Detailed Solution – Question 55

$$V(l) = V_0 e^{-\alpha l} e^{-j\beta l} e^{j\omega t}$$

$$\text{Attenuation} = \frac{|\text{Input}|}{|\text{Output}|} = \frac{|V_0(0)|}{|V_0(l)|}$$

$$\text{Attenuation per meter} = \frac{|V_0|}{|V_0(1m)|} = e^\alpha \quad \text{Attenuation in}$$

$$\text{dB/m} = (20 \log e^\alpha) \text{ dB/m}$$

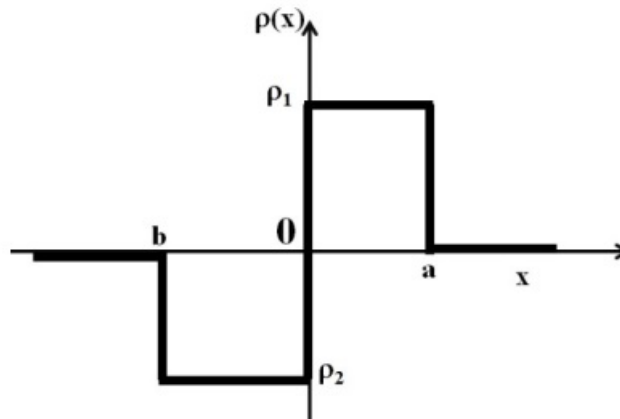
$$= 20(0.1) \log e = 0.868 \text{ dB/m}$$

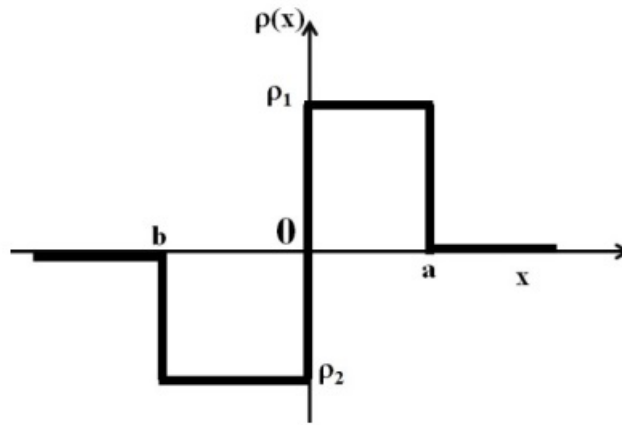
Question 56: Basics of Electromagnetics & Maxwell's Eqns (GATE EC 2016-SET-3)

MCQ

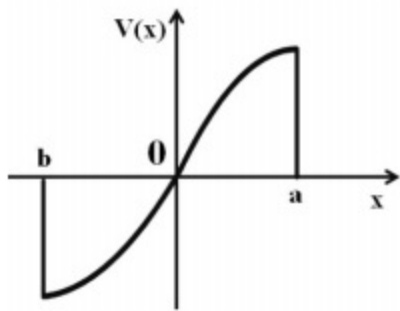
2 Marks

Consider the charge profile shown in the figure. The resultant potential distribution is best described by

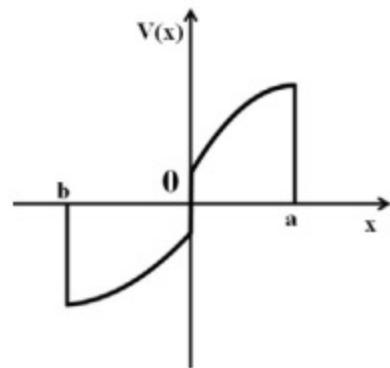




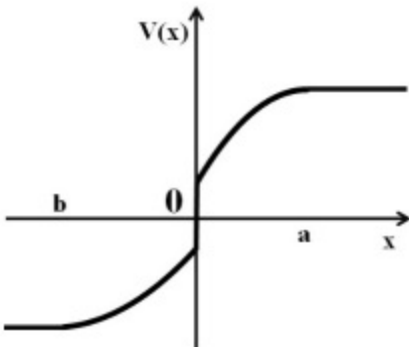
(A)



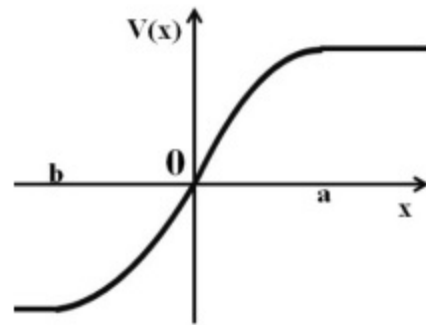
(B)

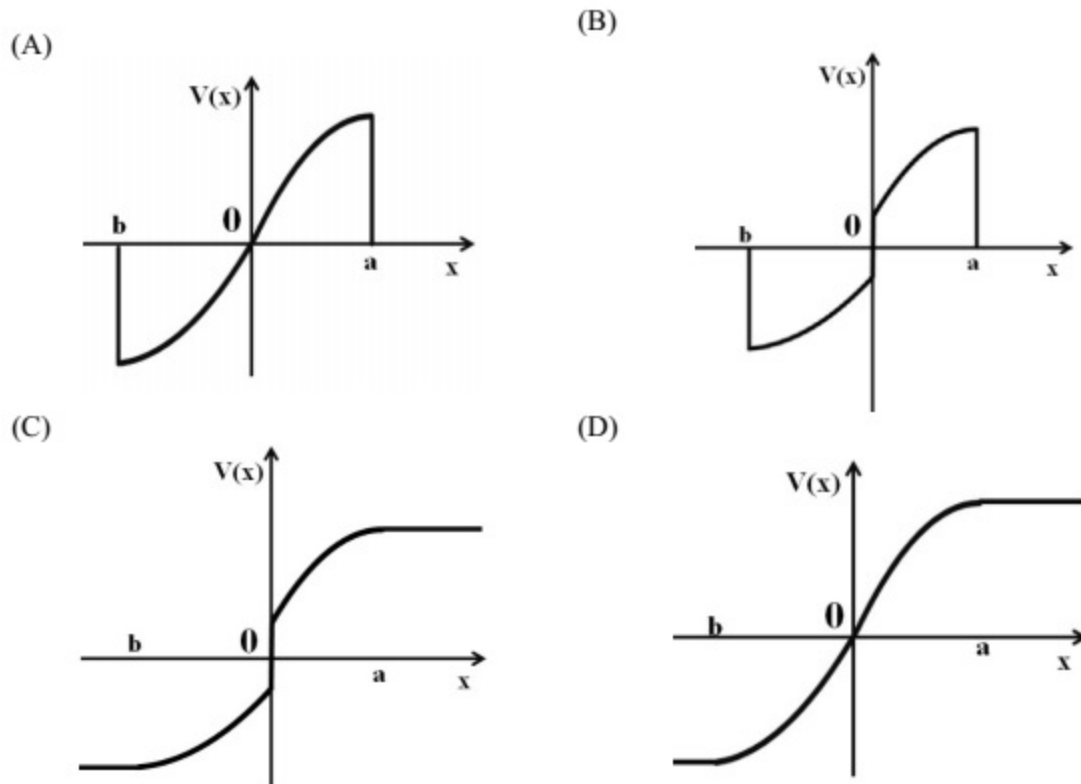


(C)



(D)





- (A) A
(B) B
(C) C
(D) **(Correct) D**

Correct Answer: (D)

Detailed Solution – Question 56

Applying Poisson's equations $\nabla^2 V = \frac{\partial^2 V}{\partial x^2} = -\frac{\rho_v}{\epsilon} = K$ Constant charge density

$$\frac{\partial V}{\partial x} = -Kx + K'$$

$$V = \frac{-Kx^2}{2} + K'x + K''$$

Towards positive x or negative side. It is a second order parabolic increase. Due to symmetry of + and - charges $K'' = 0$ is expected with $V = 0$ at centre and graph passing through origin. Beyond $x > 0$ or $x < b$, $E = 0$ due to capacitive nature of + and - charges $V = -\int 0 \cdot \vec{dl} = \text{Constant}$. This constant is same V at $x=a$ or $x=b$.

Question 57: Miscellaneous Topics (GATE EC 2016-SET-3)

NAT

2 Marks

A radar operating at 5 GHz uses a common antenna for transmission and reception. The antenna has a gain of 150 and is aligned for maximum directional radiation and reception to a target 1 km away having radar cross-section of 3 m^2 . If it transmits 100 kW, then the received power (in μW) is _____

Detailed Solution – Question 57

$$\begin{aligned}
f &= 5\text{GHz}, G = 150, R = 1\text{km} \\
P_t &= 100\text{kW}, \sigma = 3\text{m}^2 \\
P_r &= \frac{(G\lambda)^2 \cdot \sigma}{(4\pi)^3 R^4} \times P_t \\
&= \frac{\left(150 \times \frac{3 \times 10^8}{5 \times 10^9}\right)^2 \cdot 3}{(4\pi)^3 \cdot 10^{12}} \times 10^5 \\
&= \frac{81 \times 3 \times 10^5}{(4\pi)^3 \times 10^{12}} \\
&= 1.224 \times 10^{-8} = 0.012\mu\text{W}
\end{aligned}$$

Question 58: Waveguides & Optical Fibers (GATE EC 2016-SET-3)**MCQ****2 Marks**

Consider an air-filled rectangular waveguide with dimensions $a = 2.286\text{ cm}$ and $b = 1.016\text{ cm}$. The increasing order of the cut-off frequencies for different modes is

- (A) $TE_{01} < TE_{10} < TE_{11} < TE_{20}$
 (B) $TE_{20} < TE_{11} < TE_{10} < TE_{01}$
 (C) **(Correct)** $TE_{10} < TE_{20} < TE_{01} < TE_{11}$
 (D) $TE_{10} < TE_{11} < TE_{20} < TE_{01}$

Correct Answer: (C)**Detailed Solution – Question 58**

$$\begin{aligned}
a &= 2.286\text{cm} \\
b &= 1.016\text{cm} \\
f_{c_{TE_{10}}} &= \frac{c}{2a} \\
f_{c_{TE_{01}}} &= \frac{c}{2b} \\
f_{c_{TE_{20}}} &= \frac{c}{a} \\
f_{c_{TE_{11}}} &= \frac{c}{2} \sqrt{\left(\frac{1}{a}\right)^2 + \left(\frac{1}{b}\right)^2}
\end{aligned}$$

As $a > 2b \Rightarrow f_{c_{TE_{10}}} < f_{c_{TE_{20}}} < f_{c_{TE_{01}}} < f_{c_{TE_{11}}}$

Question 59: Waveguides & Optical Fibers (GATE EC 2016-SET-3)**NAT****2 Marks**

Consider an air-filled rectangular waveguide with dimensions $a = 2.286$ cm and $b = 1.016$ cm. At 10 GHz operating frequency, the value of the propagation constant (per meter) of the corresponding propagating mode is _____

Detailed Solution – Question 59

Operation frequency = 10GHz Cut off frequency for TE_{10} , $f_c = \frac{c}{2a}$

$$f_c = \frac{3 \times 10^8}{2 \times 2.286 \times 10^{-2}} \\ = 6.57\text{GHz}$$

Next mode TE_{20} has $f_c = 13\text{GHz}$ 10GHz operates only in dominant mode.

$$\begin{aligned} r &= \sqrt{\left(\frac{m\pi}{a}\right)^2 - \omega^2\mu\epsilon} \\ &= \sqrt{\left(\frac{\pi}{a}\right)^2 - \frac{(2\pi f)^2}{(3 \times 10^8)^2}} \\ &= \pi \sqrt{\left(\frac{1}{2.286 \times 10^{-2}}\right)^2 - \left(\frac{2 \times 10 \times 10^9}{3 \times 10^8}\right)^2} \\ &= j158 \text{ (m}^{-1}\text{)} \end{aligned}$$

Question 60: Uniform Plane Waves (GATE EC 2016-SET-3)

MCQ

1 Mark

Faraday's law of electromagnetic induction is mathematically described by which one of the following equations?

- (A) $\nabla \cdot \vec{B} = 0$
- (B) $\nabla \cdot \vec{D} = \rho_v$
- (C) **(Correct)** $\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$
- (D)

$$\nabla \times \vec{H} = \sigma \vec{E} + \frac{\partial \vec{D}}{\partial t}$$

Correct Answer: (C)

Detailed Solution – Question 60

Rate of change of magnetic field results in induced voltage, $\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$

Question 61: Uniform Plane Waves (GATE EC 2016-SET-3)

MCQ

1 Mark

If a right-handed circularly polarized wave is incident normally on a plane perfect conductor, then the reflected wave will be

- (A) right-handed circularly polarized
- (B) **(Correct)** left-handed circularly polarized
- (C) elliptically polarized with a tilt angle of 45°
- (D) horizontally polarized

Correct Answer: (B)

Detailed Solution – Question 61

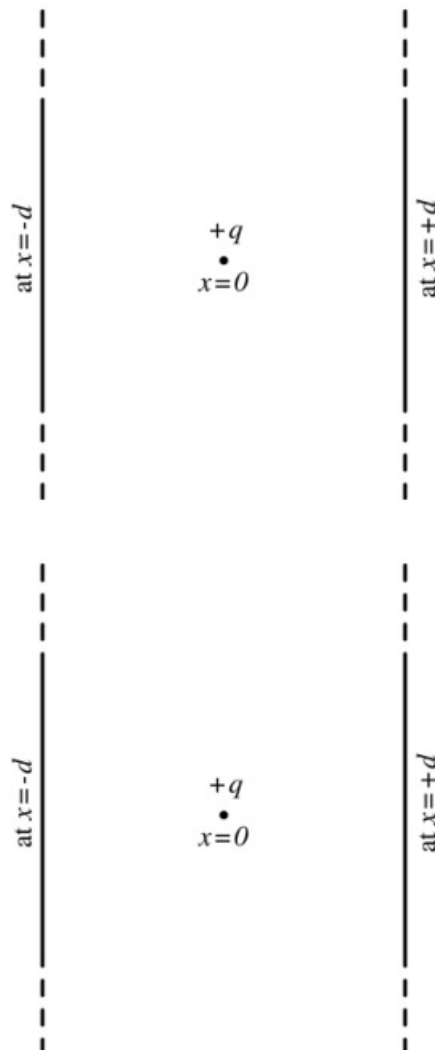
Left circularly polarized, due to direction change after the reflection from conductor.

Question 62: Basics of Electromagnetics & Maxwell's Eqns (GATE EC 2016-SET-2)

MCQ

2 Marks

A positive charge q is placed at $x = 0$ between two infinite metal plates placed at $x = -d$ and at $x = +d$ respectively. The metal plates lie in the yz plane.



The charge is at rest at $t = 0$, when a voltage $+V$ is applied to the plate at $-d$ and voltage $-V$ is applied to the plate at $x=+d$. Assume that the quantity of the charge q is small enough that it does not perturb the field set up by the metal plates. The time that the charge q takes to reach the right plate is proportional to

- (A) d/V
- (B) $\sqrt{d}/V$
- (C) **(Correct)** $d/\sqrt{V}$
- (D) $\sqrt{d/V}$

Correct Answer: (C)

Detailed Solution – Question 62

Velocity being free velocity,

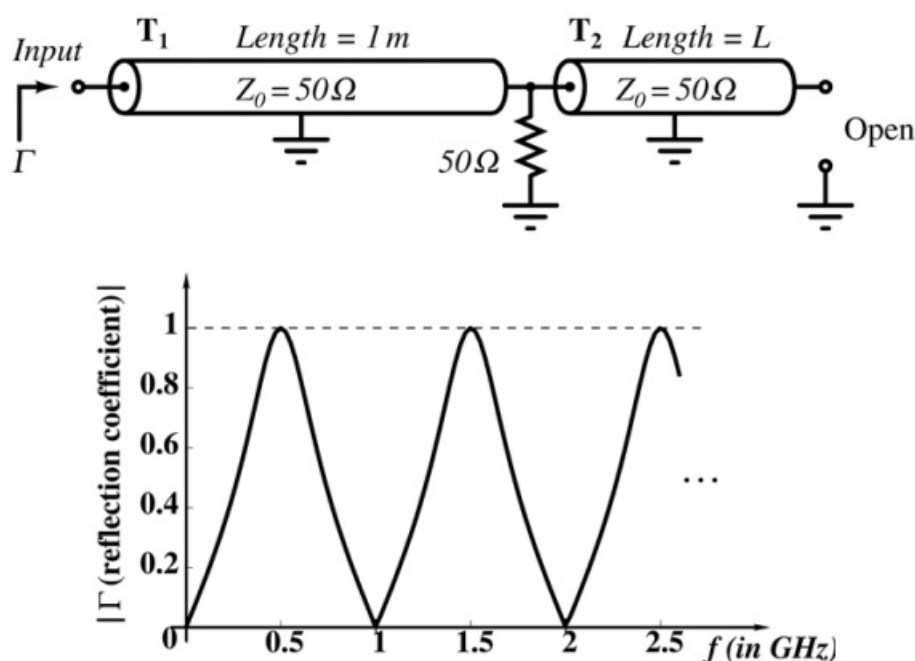
$$\begin{aligned}\frac{1}{2}mv^2 &= qV \\ v &= \frac{d}{t} = \sqrt{\frac{2qV}{m}} \\ \Rightarrow t &\propto \frac{d}{\sqrt{V}}\end{aligned}$$

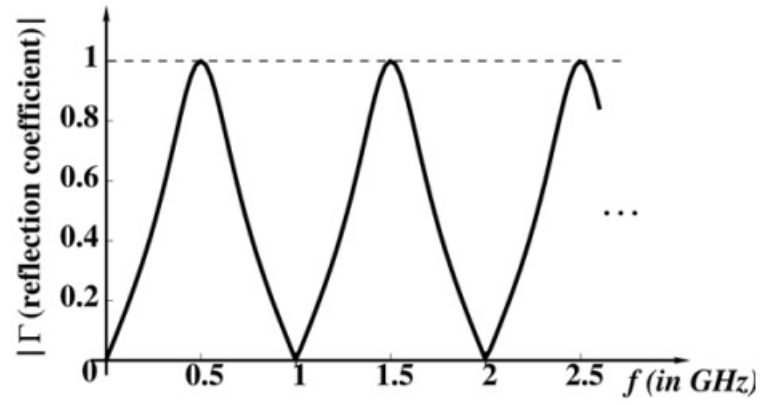
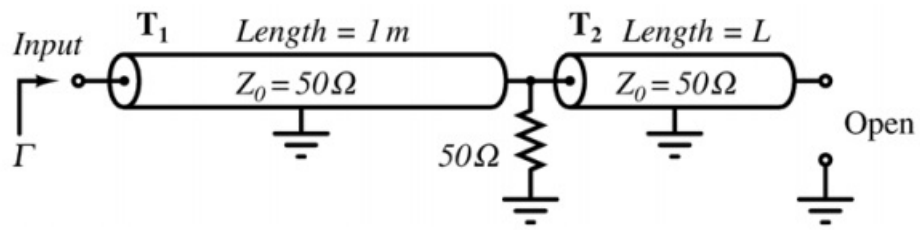
Question 63: Transmission Lines (GATE EC 2016-SET-2)

NAT

2 Marks

A microwave circuit consisting of lossless transmission lines T_1 and T_2 is shown in the figure. The plot shows the magnitude of the input reflection coefficient τ as a function of frequency f . The phase velocity of the signal in the transmission lines is $2 \times 10^8 \text{ m/s}$.

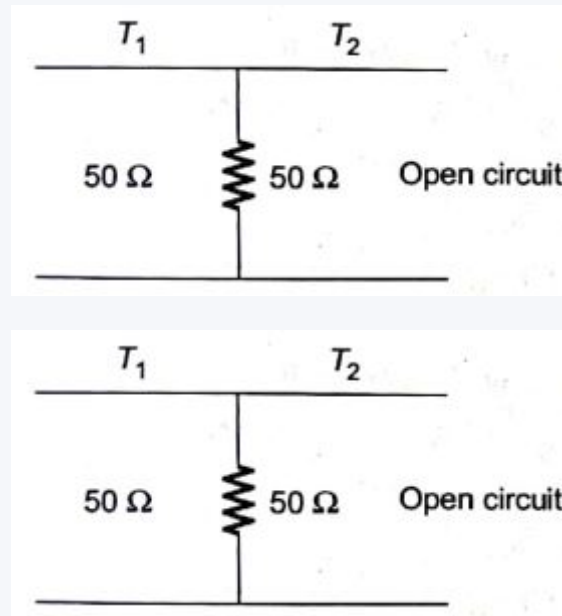




The length L (in meters) of T_2 is _____

Detailed Solution – Question 63

At frequency of 1GHz the line is matched and $|\Gamma| = 0$ as seen in the graph.



The line T_1 is having 50Ω load parallel to T_2 input impedance. This load is 50Ω when T_2 input impedance is infinite.

$$Z_{in\ 2} = \infty = Z_{L2} \text{ (open circuit)}$$

T_2 has to be a $\frac{\lambda}{2}$ line.

$$f = 1\text{GHz}$$

$$\lambda = \frac{2 \times 10^{10}}{1 \times 10^9} \text{cm}$$

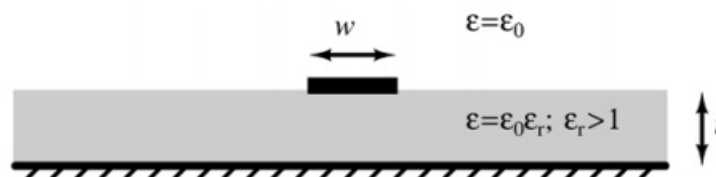
$$\frac{\lambda}{2} = L = 10\text{cm} = 0.1\text{m}$$

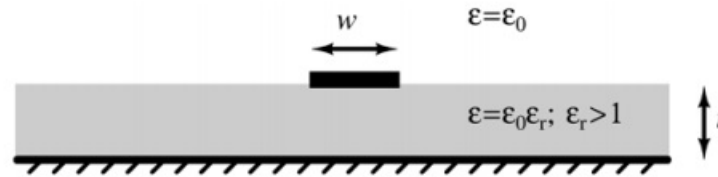
Question 64: Transmission Lines (GATE EC 2016-SET-2)

MCQ

2 Marks

A lossless microstrip transmission line consists of a trace of width w . It is drawn over a practically infinite ground plane and is separated by a dielectric slab of thickness t and relative permittivity $\epsilon_r > 1$. The inductance per unit length and the characteristic impedance of this line are L and Z_0 , respectively.





Which one of the following inequalities is always satisfied?

- (A) $Z_0 > \sqrt{\frac{Lt}{\epsilon_0 \epsilon_r w}}$
- (B) **(Correct)** $Z_0 < \sqrt{\frac{Lt}{\epsilon_0 \epsilon_r w}}$
- (C) $Z_0 > \sqrt{\frac{Lw}{\epsilon_0 \epsilon_r t}}$
- (D) $Z_0 < \sqrt{\frac{Lw}{\epsilon_0 \epsilon_r t}}$

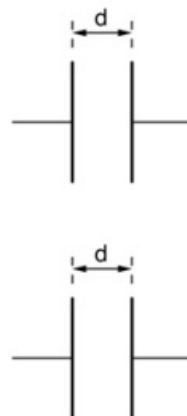
Correct Answer: (B)

Question 65: Basics of Electromagnetics & Maxwell's Eqns (GATE EC 2016-SET-2)

MCQ

2 Marks

The parallel-plate capacitor shown in the figure has movable plates. The capacitor is charged so that the energy stored in it is E when the plate separation is d . The capacitor is then isolated electrically and the plates are moved such that the plate separation becomes $2d$.



At this new plate separation, what is the energy stored in the capacitor, neglecting fringing effects?

- (A) **(Correct)** $2E$
- (B) $\sqrt{2}E$
- (C) E
- (D) $E/2$

Correct Answer: (A)

Detailed Solution – Question 65

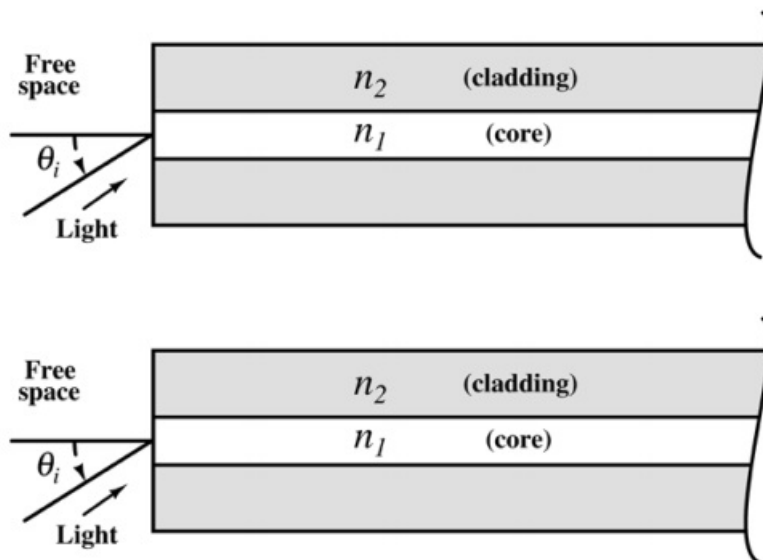
$$\begin{aligned}\text{Let, } E &= E_1, \text{ Energy } E_1 = \frac{Q_1^2}{2C_1} \\ \text{Electrically isolated } &\Rightarrow Q_2 = Q_1 \\ d_2 = 2d_1 &\Rightarrow C_2 = \frac{C_1}{2} \\ E_2 &= \frac{Q_2^2}{2C_2} = \frac{Q_1^2}{\frac{2C_1}{2}} = 2 \left(\frac{Q_1^2}{2C_1} \right) \\ &= 2E_1 = 2E\end{aligned}$$

Question 66: Miscellaneous Topics (GATE EC 2016-SET-2)

NAT

1 Mark

Light from free space is incident at an angle θ_i to the normal of the facet of a step-index large core optical fibre. The core and cladding refractive indices are $n_1 = 1.5$ and $n_2 = 1.4$, respectively.



The maximum value of θ_i (in degrees) for which the incident light will be guided in the core of the fibre is _____

Detailed Solution – Question 66

$$\begin{aligned}\sin \alpha_{\max} &= \sqrt{n_1^2 - n_2^2} = \sqrt{1.5^2 - 1.4^2} \\ \Rightarrow \alpha_{\max} &= \sin^{-1}(0.5385) = 32.58^\circ\end{aligned}$$

Question 67: Uniform Plane Waves (GATE EC 2016-SET-2)

MCQ

1 Mark

Let the electric field vector of a plane electromagnetic wave propagating in a homogenous medium be expressed as $E = \hat{x}E_x e^{-j(\omega t - \beta z)}$, where the propagation constant β is a function of the angular frequency ω . Assume that $\beta(\omega)$ and E_x are known and are real. From the information available, which one of the following CANNOT be determined?

- (A) The type of polarization of the wave.
- (B) The group velocity of the wave
- (C) The phase velocity of the wave.
- (D) **(Correct)** The power flux through the $z = 0$ plane.

Correct Answer: (D)

Detailed Solution – Question 67

$\beta(\omega)$ is known so V_g can be calculated. $v_p = \omega/\beta$ can be calculated. Polarization can be identified. μ_r and ϵ_r cannot be found, due to which power flux cannot be calculated as power flux

$$P = \frac{1}{2} \frac{|E|^2}{\eta}, \text{ where } \eta = 120\pi \times \sqrt{\frac{\mu_r}{\epsilon_r}}$$

Question 68: Basics of Electromagnetics & Maxwell's Eqns (GATE EC 2016-SET-2)

MCQ

1 Mark

A uniform and constant magnetic field $B = \hat{z}B$ exists in the $\hat{z}$ direction in vacuum. A particle of mass m with a small charge q is introduced into this region with an initial velocity $v = \hat{x}v_x + \hat{z}v_z$. Given that B , m , q , v_x and v_z are all non-zero, which one of the following describes the eventual trajectory of the particle?

- (A) **(Correct)** Helical motion in the $\hat{z}$ direction
- (B) Circular motion in the xy plane
- (C) Linear motion in the $\hat{z}$ direction
- (D) Linear motion in the $\hat{x}$ direction

Correct Answer: (A)

Detailed Solution – Question 68

Ba_z magnetic field $v_x a_x + v_z a_z$ velocity

$$\begin{aligned} F &= Q(v \times B) \text{ by Lorentz's law} \\ &= Q(v_x a_x + v_z a_z) \times Ba_z \\ F_y &= Qv_x \cdot B(-a_y) \end{aligned}$$

This results in a circular path in the XY plane with $v_z a_z$ component causing a linear path. Both result in a helical path along Z axis.

Question 69: Antennas & Radiating Systems (GATE EC 2016-SET-1)

MCQ

2 Marks

The far-zone power density radiated by a helical antenna is approximated as:

$$\vec{W}_{rad} = \vec{W}_{average} \approx \hat{a}_r C_o \frac{1}{r^2} \cos^4 \theta$$

The radiated power density is symmetrical with respect to ϕ and exists only in the upper hemisphere: $0 \leq \theta \leq \frac{\pi}{2}; 0 \leq \phi \leq 2\pi$; C_o is a constant. The power radiated by the antenna (in watts) and the maximum directivity of the antenna, respectively, are

- (A) $1.5C_o, 10dB$
- (B) **(Correct)** $1.256C_o, 10dB$
- (C) $1.256C_o, 12dB$
- (D) $1.5C_o, 12dB$

Correct Answer: (B)

Detailed Solution – Question 69

$$\begin{aligned} \text{Power radiated} &= \int P_{rad} \cdot ds \\ &= \int_{\theta=0}^{\pi/2} \int_{\phi=0}^{2\pi} C_o \frac{1}{r^2} \cos^4 \theta \cdot r^2 \sin \theta d\theta d\phi \\ &= 2\pi \cdot \frac{C_o}{r^2} \cdot r^2 \int_{\theta=0}^{\pi/2} \cos^4 \theta \cdot d(-\cos \theta) \\ &= 2\pi \cdot C_o \left(-\frac{\cos^5 \theta}{5} \right) \Big|_0^{\pi/2} \\ &= \frac{2\pi}{5} C_o = 1.256 C_o \end{aligned}$$

$$\begin{aligned} \text{Directivity} &= \frac{4\pi \cdot U(\theta, \phi)}{\int (\theta, \phi) \cdot d\Omega} \\ &= \frac{4\pi \cdot C_o \cdot \cos^4 \theta}{\int_{\theta=0}^{\pi/2} \int_{\phi=0}^{2\pi} C_o \cdot \cos^4 \theta \cdot \sin \theta d\theta d\phi} \\ &= \frac{2 \cos^5 \theta}{1/5} = 10 \cos^5 \theta \end{aligned}$$

$$\begin{aligned} \text{max} &= 10 \\ \text{value} & \end{aligned}$$

$$\Rightarrow \text{max value (in dB)} = 10 \log_{10} 10 = 10 \text{ dB}$$

Question 70: Antennas & Radiating Systems (GATE EC 2016-SET-1)

MSQ

2 Marks

The electric field of a uniform plane wave travelling along the negative z direction is given by the following equation:

$$\vec{E}_w^1 = (\hat{a}_x + j\hat{a}_y) E_0 e^{jkz}$$

This wave is incident upon a receiving antenna placed at the origin and whose radiated electric field towards the incident wave is given by the following equation:

$$\vec{E}_a = (\hat{a}_x + 2\hat{a}_y)E_I \frac{1}{r} e^{-jkr}$$

The polarization of the incident wave, the polarization of the antenna and losses due to the polarization mismatch are, respectively,

- (A) Linear, Circular (clockwise), -5dB
- (B) Circular (clockwise), Linear, -5dB
- (C) **(Correct)** Circular (clockwise), Linear, -3dB
- (D) **(Correct)** Circular (anti clockwise), Linear, -3dB

Correct Answer: (C, D)

Detailed Solution – Question 70

$\vec{E}_T = (\hat{a}_x + j\hat{a}_y) E_0 e^{jkz} \Rightarrow$ Wave contains two orthogonal components and Y component leads X component leads by 90° and also wave is travelling in negative z-direction. $\vec{E}_R = (\hat{a}_x + 2\hat{a}_y) E_1 \frac{1}{r} e^{-jkr} \Rightarrow$ Wave contains two orthogonal components with unequal amplitudes and both are in-phase. $\Rightarrow$ Linear polarization. Polarization Loss Factor, $PLF = \left| \hat{E}_T \cdot \hat{E}_R \right|^2$

$$\text{Where, } E_T = \frac{\hat{a}_x + j\hat{a}_y}{\sqrt{2}}$$

$$E_R = \frac{\hat{a}_x + 2\hat{a}_y}{\sqrt{5}}$$

$$PL = \left| \frac{1 + j2^2}{\sqrt{10}} \right| = \frac{5}{10} = \frac{1}{2}$$

$$\Rightarrow PLF(dB) = 10 \log \frac{1}{2} = -3dB$$

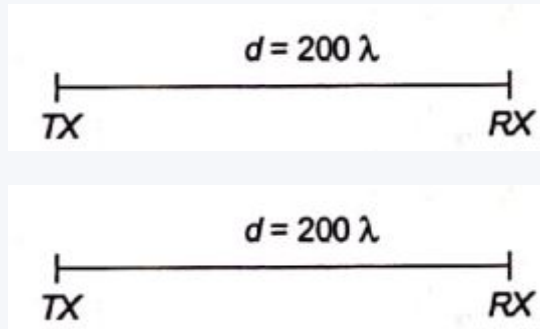
Question 71: Antennas & Radiating Systems (GATE EC 2016-SET-1)

NAT

2 Marks

Two lossless X-band horn antennas are separated by a distance of 200λ . The amplitude reflection coefficients at the terminals of the transmitting and receiving antennas are 0.15 and 0.18, respectively. The maximum directivities of the transmitting and receiving antennas (over the isotropic antenna) are 18 dB and 22 dB, respectively. Assuming that the input power in the lossless transmission line connected to the antenna is 2 W, and that the antennas are perfectly aligned and polarization matched, the power (in mW) delivered to the load at the receiver is _____

Detailed Solution – Question 71



$$\begin{aligned}
 G_t &= 10^{1.8}, G_r = 10^{2.2} \\
 P_r &= \frac{(1 - |\Gamma_t|^2)(1 - |\Gamma_r|^2) G_t G_r}{\left(\frac{4\pi d}{\lambda}\right)^2} \cdot P_t \\
 &= \frac{(1 - |0.15|^2)(1 - |0.18|^2) 10^{1.8} \cdot 10^{2.2}}{\left(\frac{4\pi 200\lambda}{\lambda}\right)^2} \times 2 \\
 &= 2.99476 \times 10^{-3} \text{ W} = 2.995 \text{ mW} \approx 3 \text{ mW}
 \end{aligned}$$

Question 72: Basics of Electromagnetics & Maxwell's Eqns (GATE EC 2016-SET-1)

MCQ

2 Marks

The current density in a medium is given by $\vec{j} = \frac{400 \sin \theta}{2\pi(r^2)+4} \hat{a}_r \text{ Am}^{-2}$. The total current and the average current density flowing through the portion of a spherical surface $r = 0.8 \text{ m}$, $\frac{\pi}{12} \leq \theta \leq \frac{\pi}{4}$, $0 \leq \phi \leq 2\pi$ are given, respectively, by

- (A) **(Correct)** 15.09 A, 12.86 Am^{-2}
- (B) 18.73 A, 13.65 Am^{-2}
- (C) 12.86 A, 9.23 Am^{-2}
- (D) 10.28 A, 7.56 Am^{-2}

Correct Answer: (A)

Detailed Solution – Question 72

$$\begin{aligned}
 I &= \int \vec{J} \cdot d\vec{s} \\
 &= \int_{\theta=\frac{\pi}{12}}^{\pi/4} \int_{\phi=0}^{2\pi} \frac{400 \sin \theta}{2\pi (r^2 + 4)} r^2 \sin \theta d\theta d\phi \\
 &= \frac{400}{2\pi (r^2 + 4)} \cdot r^2 \cdot \phi \Big|_0^{2\pi} \int_{\pi/12}^{\pi/4} \sin^2 \theta d\theta \\
 &= \frac{400r^2}{(r^2 + 4)} \int_{\pi/12}^{\pi/4} \left(\frac{1 - \cos 2\theta}{2} \right) d\theta \\
 &= \frac{400 \cdot r^2}{(r^2 + 4)} \left(\frac{\left(\frac{\pi}{4} - \frac{\pi}{12} \right)}{2} - \left(\frac{\sin 2\theta}{4} \right) \Big|_{\pi/12}^{\pi/4} \right) \\
 &= \frac{400 \cdot r^2}{(r^2 + 4)} \left(\frac{\pi}{12} - \left(\frac{1 - 1/2}{4} \right) \right) \Big|_{r=0.8} \\
 &= \frac{400 \times 0.8 \times 0.8}{4.64} \times 0.13 = 7.17 \text{ Amp} \\
 \text{Total area} &= \int ds = \iint r^2 \sin \theta d\theta d\phi \\
 &= r^2 \int_{\theta=\frac{\pi}{12}}^{\pi/4} \sin \theta d\theta \cdot 2\pi \\
 &= r^2 \cdot 2\pi \cdot 0.259 \Big|_{r=0.8} \\
 &= 0.8^2 \times 0.5 \times 2\pi \times \frac{1}{4} \\
 &= 1.041 \text{ m}^2 \\
 \text{Average current} &= \frac{7.17}{1.041} = 6.88 \text{ A/m}^2
 \end{aligned}$$

Note : GATE key mentioned (MTA - marks to all)

Question 73: Transmission Lines (GATE EC 2016-SET-1)

MCQ

1 Mark

The propagation constant of a lossy transmission line is $(2 + j5)m^{-1}$ and its characteristic impedance is $(50 + j0)\Omega$ at $\omega = 10^6 \text{ rad s}^{-1}$. The values of the line constants L, C, R, G are, respectively,

- (A) $L = 200\mu\text{H}/\text{m}, C = 0.1\mu\text{F}/\text{m}, R = 50\Omega/\text{m}, G = 0.02\text{S}/\text{m}$
- (B) **(Correct)** $L = 250\mu\text{H}/\text{m}, C = 0.1\mu\text{F}/\text{m}, R = 100\Omega/\text{m}, G = 0.04\text{S}/\text{m}$
- (C) $L = 200\mu\text{H}/\text{m}, C = 0.2\mu\text{F}/\text{m}, R = 100\Omega/\text{m}, G = 0.04\text{S}/\text{m}$
- (D) $L = 250\mu\text{H}/\text{m}, C = 0.2\mu\text{F}/\text{m}, R = 50\Omega/\text{m}, G = 0.04\text{S}/\text{m}$

Correct Answer: (B)

Detailed Solution – Question 73

$$\gamma = \sqrt{(R + j\omega L)(G + j\omega C)}$$

$$Z_0 = \sqrt{\frac{R + j\omega L}{G + j\omega C}}$$

$$\gamma \cdot Z_0 = R + j\omega L = (2 + j5)(50 + j0) = 100 + j250$$

$$R = 100\Omega/\text{m}$$

$$L = \frac{250}{\omega} = \frac{250}{10^6} = 250\mu\text{H}/\text{m}$$

$$\frac{\gamma}{Z_0} = \frac{2 + j5}{50} = G + j\omega C = 0.04 + j0.1$$

$$G = 0.04\text{S}/\text{m}$$

$$C = \frac{0.1}{\omega} = \frac{0.1}{10^6} = 0.1\mu\text{F}/\text{m}$$

Question 74: Basics of Electromagnetics & Maxwell's Eqns (GATE EC 2016-SET-1)**NAT****1 Mark**

Concentric spherical shells of radii 2 m, 4 m, and 8 m carry uniform surface charge densities of 20 nC/m², -4 nC/m² and ρ_s , respectively. The value of ρ_s (nC/m²) required to ensure that the electric flux density $\vec{D} = \vec{0}$ at radius 10 m is _____

Detailed Solution – Question 74

$$\oint \vec{D} \cdot d\vec{s} = Q \quad (\text{charge enclosed}) \quad Q_1 + Q_2 + Q_3 = 0 \quad \text{For } D=0$$

$$\rho_{s1} \cdot 4\pi 2^2 + \rho_{s2} \cdot 4\pi \cdot 4^2 + \rho_{s3} \cdot 4\pi \cdot 8^2 = Q = 0$$

$$20 \cdot 4 - 4 \cdot 4^2 + \rho_{s3} \cdot 8^2 = 0 \quad 80 - 64 + \rho_{s3} \cdot 8^2 = 0 \quad \rho_{s3} = \frac{-16}{64} = -0.25\text{nC}/\text{m}^2$$

Question 75: Transmission Lines (GATE EC 2015-SET-3)**NAT****2 Marks**

A coaxial capacitor of inner radius 1 mm and outer radius 5 mm has a capacitance per unit length of 172 pF/m. If the ratio of outer radius to inner radius is doubled, the capacitance per unit length (in pF/m) is _____.

Detailed Solution – Question 75

$$C = \frac{2\pi \epsilon}{\ln \frac{b}{a}}$$

$$172 \times 10^{-12} = \frac{2\pi \epsilon}{\ln 5}$$

$$\epsilon = \frac{172 \times 10^{-12} \times \ln 5}{2\pi}$$

If ratio of $\frac{b}{a}$ is doubled then

$$C = \frac{2\pi \epsilon}{\ln 10} = \frac{2\pi \times \frac{172 \times 10^{-12} \times \ln 5}{2\pi}}{\ln 10}$$

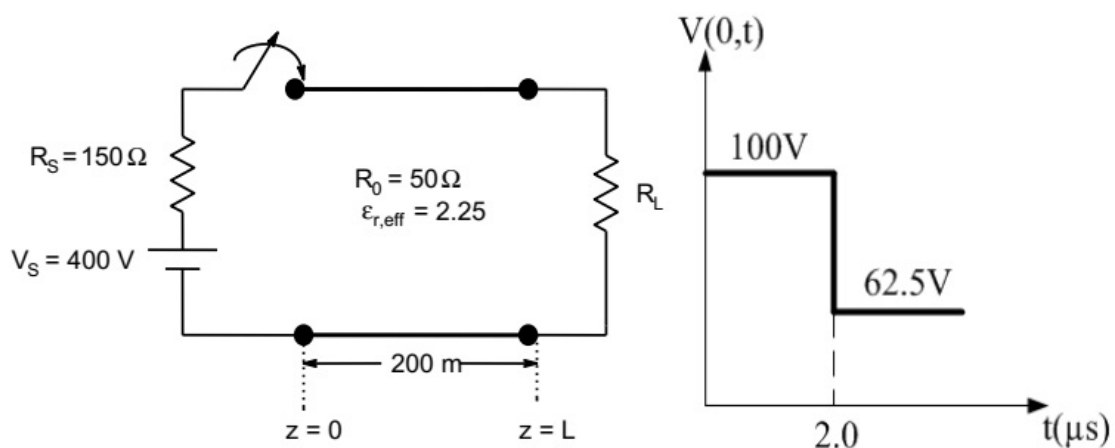
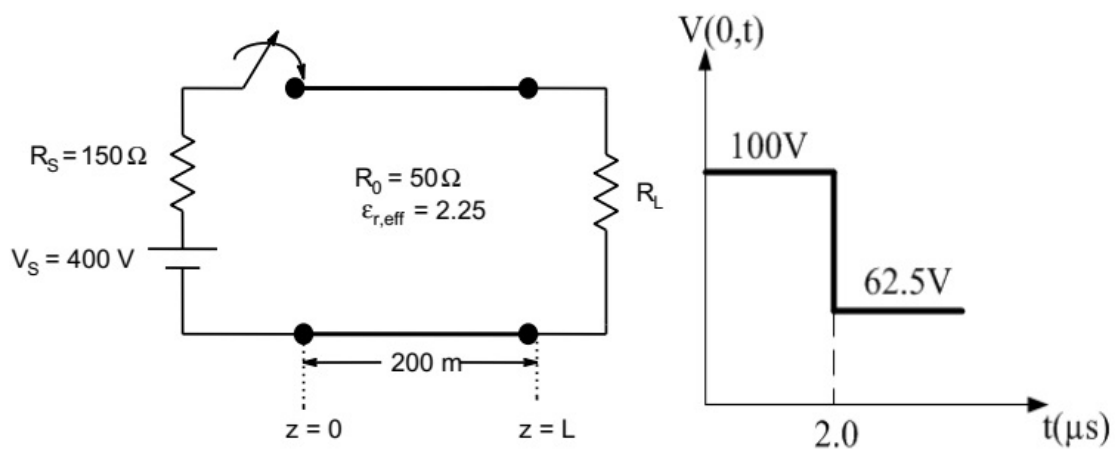
$$C = 120.22 \text{ pF}$$

Question 76: Transmission Lines (GATE EC 2015-SET-3)

NAT

2 Marks

A 200 m long transmission line having parameters shown in the figure is terminated into a load R_L . The line is connected to a 400 V source having source resistance R_S through a switch, which is closed at $t = 0$. The transient response of the circuit at the input of the line ($z = 0$) is also drawn in the figure. The value of R_L (in Ω) is _____.



Detailed Solution – Question 76

At $t = 0$

$$V_{in} = 400 \times \frac{R_0}{R_0 + R_s} = 400 \times \frac{50}{200} = 100V$$

At $t = T_0 \Gamma_L$ 100 volts reflects

At $t = 2T_0$

$$V_{in} = 100 + (1 + \Gamma_{in}) \Gamma_L 100 = 62.5$$

$$\Gamma_{in} = \frac{R_s - R_o}{R_s + R_o} = \frac{1}{2}$$

$$\text{Therefore, } \Gamma_L = -\frac{1}{4} = \frac{R_L - 50}{R_L + 50}$$

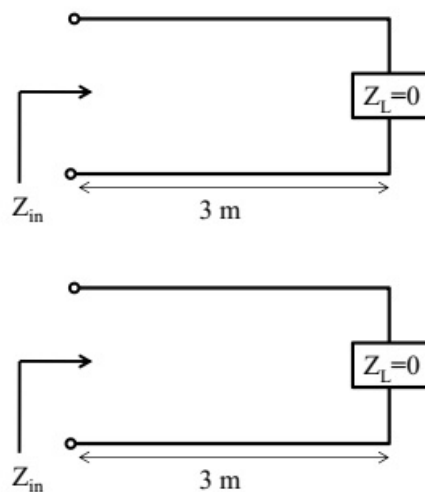
$$R_L = 30\Omega$$

Question 77: Transmission Lines (GATE EC 2015-SET-3)

MCQ

2 Marks

Consider the 3 m long lossless air-filled transmission line shown in the figure. It has a characteristic impedance of $120\pi\Omega$, is terminated by a short circuit, and is excited with a frequency of 37.5 MHz. What is the nature of the input impedance (Z_{in})?



- (A) Open
- (B) Short
- (C) Inductive
- (D) **(Correct)** Capacitive

Correct Answer: (D)

Detailed Solution – Question 77

$$\begin{aligned}
l &= 3\text{m} \\
f &= 37.5\text{MHz} \\
\Rightarrow \lambda &= \frac{3 \times 10^8}{37.5 \times 10^6} = 8\text{m} \\
Z_{\text{in}}(SC) &= jZ_0 \tan \beta l = j120\pi \tan \frac{2\pi}{8} \cdot 3 \\
&= -j120\pi\Omega
\end{aligned}$$

Question 78: Basics of Electromagnetics & Maxwell's Eqns (GATE EC 2015-SET-3)**NAT****2 Marks**

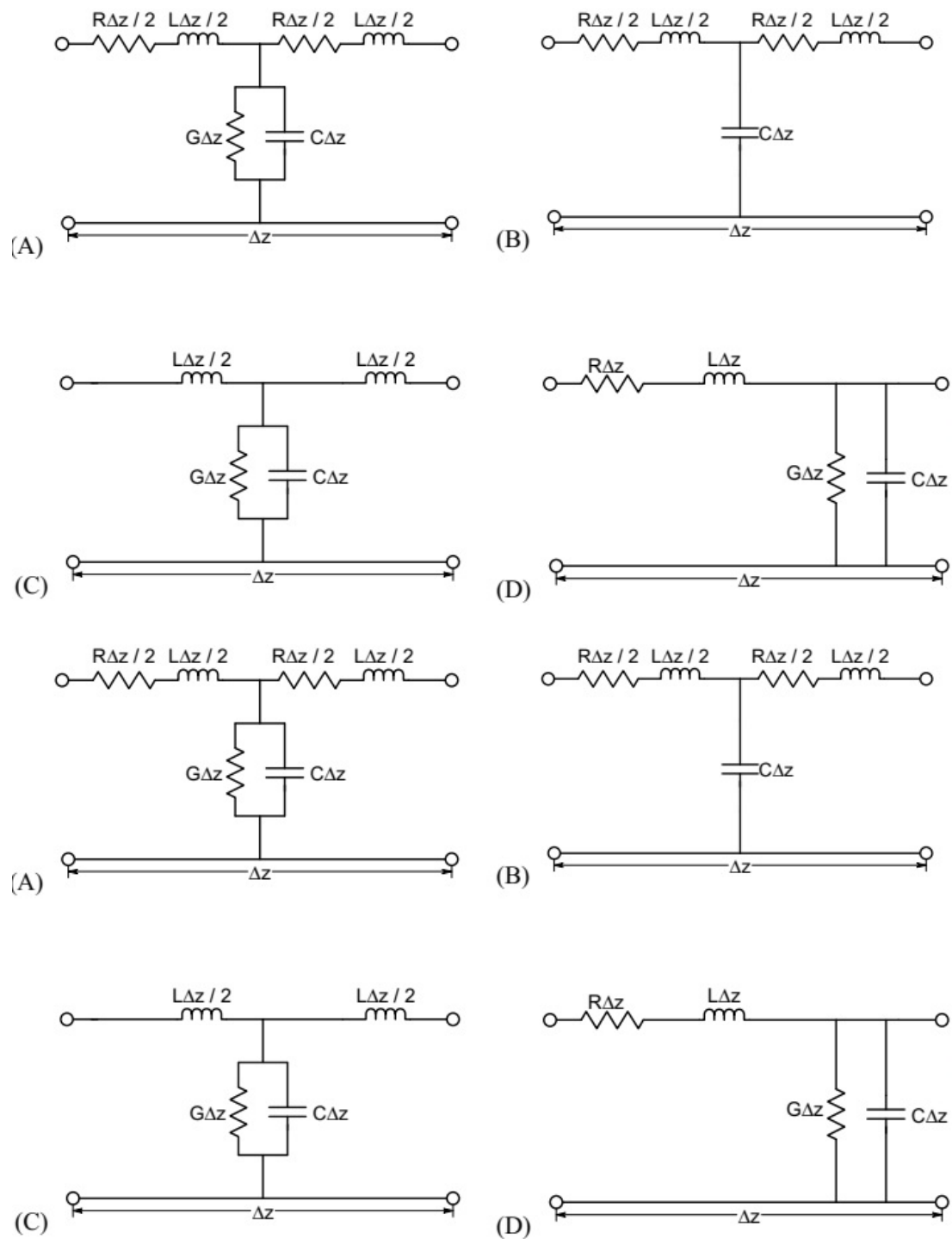
A vector field $D = 2\rho^2 a_\rho + z a_z$ exists inside a cylindrical region enclosed by the surfaces $\rho = 1$, $z = 0$ and $z = 5$. Let S be the surface bounding this cylindrical region. The surface integral of this field on S ($\oint_S D \cdot ds$) is _____.

Detailed Solution – Question 78

$$\begin{aligned}
\vec{D} &= 2\rho^2 \hat{a}_\rho + z \hat{a}_z \\
\text{surface: } \rho &= 1, z = 0 \text{ and } z = 5 \\
\oiint_S \vec{D} \cdot ds &= \int_{\text{Top surface}} \vec{D} \cdot ds \\
&\quad + \int_{\text{Bottom surface}} \vec{D} \cdot ds \\
&\quad + \int_{\text{curved surface}} \vec{D} \cdot ds \\
&= 5\pi + 0 + 20\pi = 25\pi = 78.54
\end{aligned}$$

Question 79: Transmission Lines (GATE EC 2015-SET-3)**MCQ****1 Mark**

A coaxial cable is made of two brass conductors. The spacing between the conductors is filled with Teflon ($\epsilon_r' = 2.1$, $\tan \delta = 0$). Which one of the following circuits can represent the lumped element model of a small piece of this cable having length Δz ?



- (A) A
- (B) (Correct) B
- (C) C
- (D) D

Correct Answer: (B)

Detailed Solution – Question 79

The teflon dielectric between the lines has $\tan \delta = \text{Loss tangent} = 0$ For the medium between the lines $G = 0$

Question 80: Antennas & Radiating Systems (GATE EC 2015-SET-3)

MCQ

1 Mark

The directivity of an antenna array can be increased by adding more antenna elements, as a larger number of elements

- (A) improves the radiation efficiency
- (B) **(Correct)** increases the effective area of the antenna
- (C) results in a better impedance matching
- (D) allows more power to be transmitted by the antenna

Correct Answer: (B)

Detailed Solution – Question 80

Directivity (D) is increased by adding more antenna elements in an antenna array. Effective area (A_e) and directivity (D) are related by $A_e = \frac{\lambda^2}{4\pi} D \Rightarrow D \uparrow A_e \uparrow$

Question 81: Uniform Plane Waves (GATE EC 2015-SET-2)

MCQ

2 Marks

The electric field of a plane wave propagating in a lossless non-magnetic medium is given by the following expression $E(z, t) = \hat{a}_x 5 \cos(2\pi \times 10^9 t + \beta z) + \hat{a}_y 3 \cos(2\pi \times 10^9 t + \beta z - \pi/2)$ The type of the polarization is

- (A) Right Hand Circular.
- (B) **(Correct)** Left Hand Elliptical
- (C) Right Hand Elliptical
- (D) Linear.

Correct Answer: (B)

Detailed Solution – Question 81

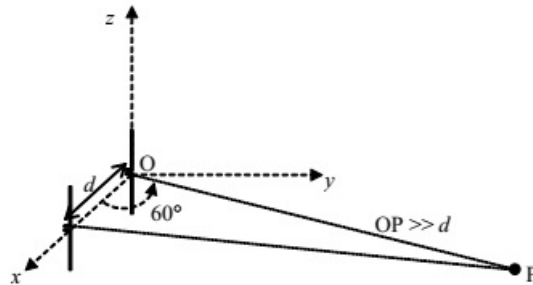
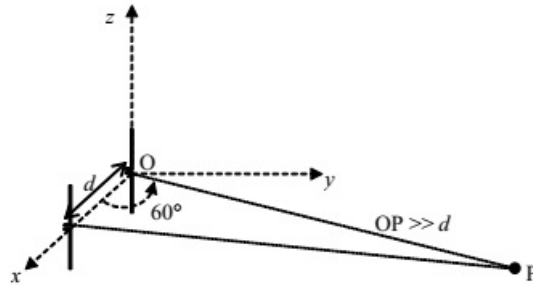
$\vec{E}(z, t) = \hat{a}_x 5 \cos(2\pi \times 10^9 t + \beta z) + \hat{a}_y 3 \cos(2\pi \times 10^9 t + \beta z - \frac{\pi}{2}) \Rightarrow$ Wave is travelling in $-\hat{a}_z$ direction. $\Rightarrow$ Wave has orthogonal components with unequal amplitudes and checking the time trace. $\therefore$ Wave is left hand elliptically polarized.

Question 82: Antennas & Radiating Systems (GATE EC 2015-SET-2)

NAT

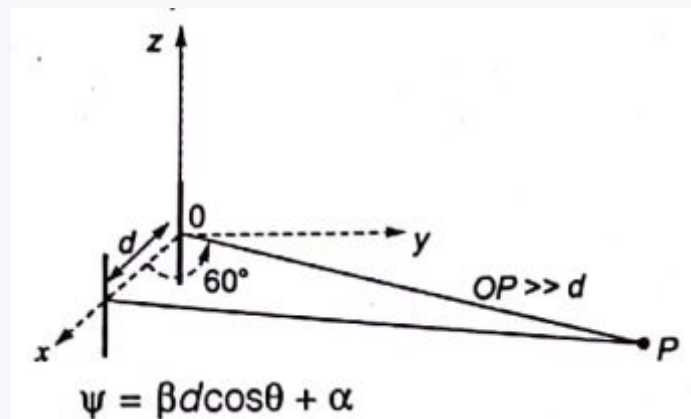
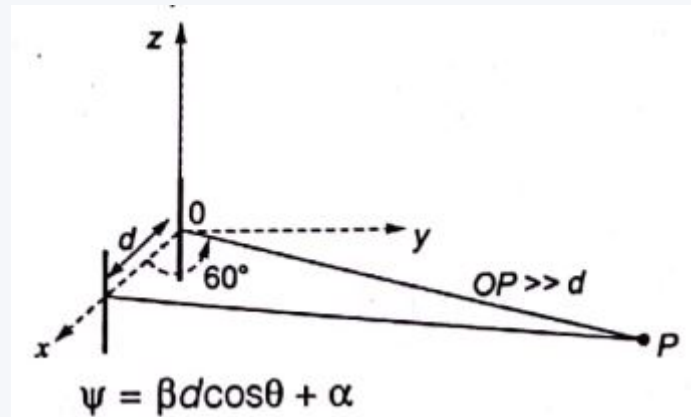
2 Marks

Two half-wave dipole antennas placed as shown in the figure are excited with sinusoidally varying currents of frequency 3 MHz and phase shift of $\pi/2$ between them (the element at the origin leads in phase). If the maximum radiated E-field at the point P in the x-y plane occurs at an azimuthal angle of 60° , the distance d (in meters) between the antennas is _____.



Detailed Solution – Question 82

Both dipole antennas have isotropic pattern in ϕ view or H plane view.



$\psi = \beta d \cos \theta + \alpha$ where,

$$\alpha = -90^\circ, \beta = \frac{2\pi}{\lambda} = \frac{2\pi}{\frac{3 \times 10^8}{3 \times 10^6}} = \frac{2\pi}{100}$$

Since maximum radiation ($\psi = 0$) is at $\theta = 60^\circ$

$$\begin{aligned} \Rightarrow 0 &= \frac{2\pi}{100} d \cos 60^\circ - 90^\circ \\ \Rightarrow 90 &= \frac{360}{100} d \frac{1}{2} \\ \Rightarrow d &= 50\text{m} \end{aligned}$$

Question 83: Waveguides & Optical Fibers (GATE EC 2015-SET-2)

NAT

2 Marks

An air-filled rectangular waveguide of internal dimensions $a \text{ cm} \times b \text{ cm}$ ($a > b$) has a cutoff frequency of 6 GHz for the dominant TE_{10} mode. For the same waveguide, if the cutoff frequency of the TM_{11} mode is 15 GHz, the cutoff frequency of the TE_{01} mode in GHz is _____.

Detailed Solution – Question 83

Given, $f_c TE_{10}$ mode is 6GHz

$$\Rightarrow \frac{c}{2a} = \frac{3 \times 10^{10}}{2 \times a} = 6 \times 10^9$$

$\Rightarrow a = 2.5\text{cm}$ Given f_c for TM_{11} mode is 15GHz

$$\begin{aligned}\Rightarrow \frac{c}{2} \sqrt{\frac{1}{a^2} + \frac{1}{b^2}} &= 15 \times 10^9 \\ \frac{3 \times 10^{10}}{2} \sqrt{\frac{1}{6.25} + \frac{1}{b^2}} &= 15 \times 10^9 \\ \Rightarrow b &= \frac{2.5}{\sqrt{5.25}} \\ \Rightarrow &2\end{aligned}$$

Now f_c for TE_{01} mode is,

$$\begin{aligned}f_c &= \frac{c}{2b} = \frac{3 \times 10^{10}}{2.5} \sqrt{5.25} \\ &= 13.75\text{GHz}\end{aligned}$$

Question 84: Uniform Plane Waves (GATE EC 2015-SET-2)

MCQ

1 Mark

The electric field of a uniform plane electromagnetic wave is $\vec{E} = (\vec{a}_x + j4\vec{a}_y) \exp[j(2\pi \times 10^7 t - 0.2z)]$
The polarization of the wave is

- (A) right handed circular
- (B) left handed circular
- (C) right handed elliptical
- (D) **(Correct)** left handed elliptical

Correct Answer: (D)

Detailed Solution – Question 84

$$\begin{aligned}\vec{E}(z, t) &= \hat{a}_x 5 \cos(2\pi \times 10^9 t + \beta z) \\ &\quad + \hat{a}_y 3 \cos\left(2\pi \times 10^9 t + \beta z - \frac{\pi}{2}\right)\end{aligned}$$

$\Rightarrow$ Wave is travelling in $-\hat{a}_z$ direction. $\Rightarrow$ Wave has orthogonal components with unequal amplitudes and checking the time trace. $\therefore$ Wave is left hand elliptically polarized.

Question 85: Basics of Electromagnetics & Maxwell's Eqns (GATE EC 2015-SET-2)

NAT

1 Mark

In a source free region in vacuum, if the electrostatic potential $\phi = 2x^2 + y^2 + cz^2$, the value of constant c must be _____.

Detailed Solution – Question 85

$\phi = 2x^2 + y^2 + cz^2$ In source-free region,

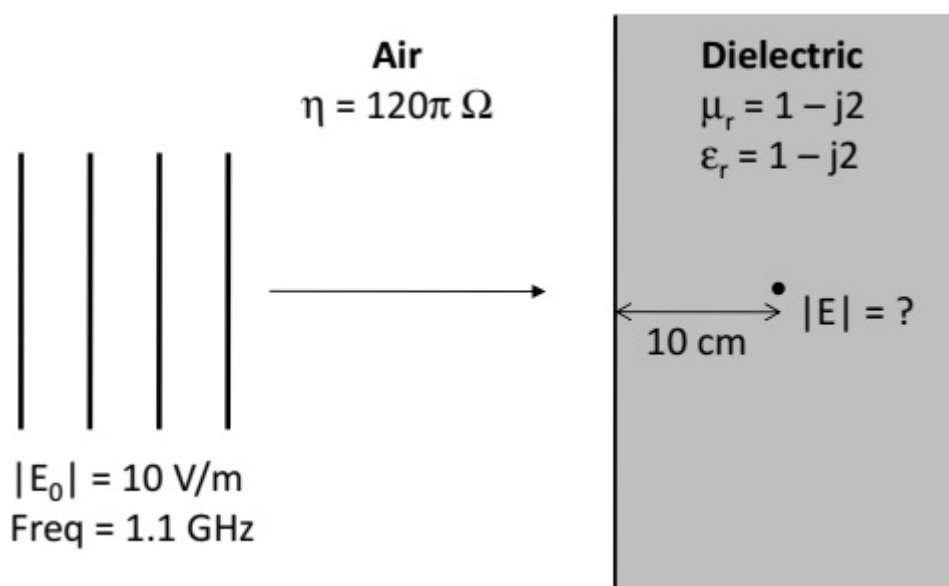
$$\begin{aligned}\nabla \cdot \vec{D} &= 0 \\ \Rightarrow \epsilon (\nabla \cdot \vec{E}) &= 0 \\ \Rightarrow \nabla \cdot \vec{E} &= 0 \\ \text{where, } \vec{E} &= -\nabla V = -\nabla \phi \\ \text{Now, } \nabla V &= \frac{\partial \phi}{\partial x} \hat{a}_x + \frac{\partial \phi}{\partial y} \hat{a}_y + \frac{\partial \phi}{\partial z} \hat{a}_z \\ \Rightarrow -\vec{E} &= 4x \hat{a}_x + 2y \hat{a}_y + 2cz \hat{a}_z \\ \text{Again } \nabla \cdot \vec{E} &= 0 \\ \Rightarrow \frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} + \frac{\partial E_z}{\partial z} &= 4 + 2 + 2c = 0 \\ c &= -3\end{aligned}$$

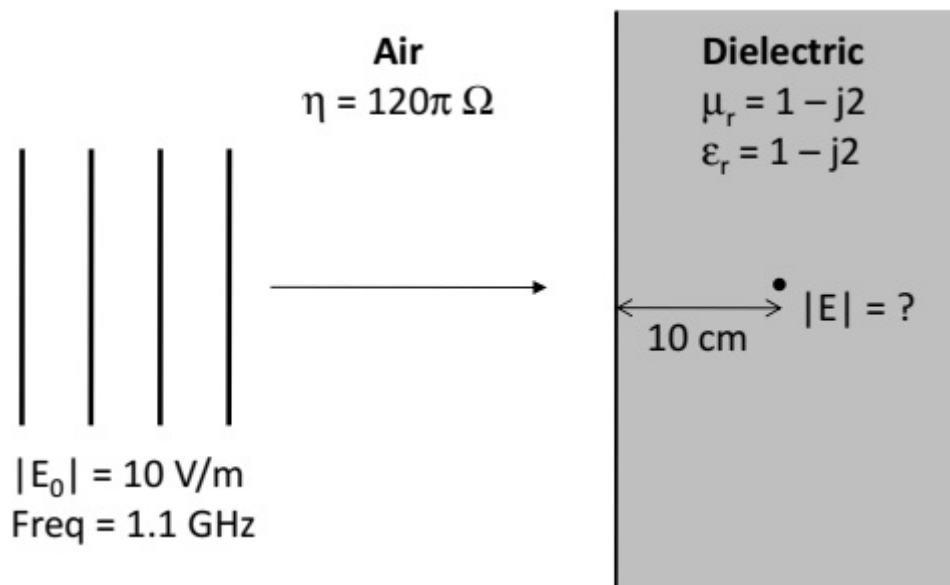
Question 86: Uniform Plane Waves (GATE EC 2015-SET-1)

NAT

2 Marks

Consider a uniform plane wave with amplitude (E_0) of 10 V/m and 1.1 GHz frequency travelling in air, and incident normally on a dielectric medium with complex relative permittivity (ϵ_r) and permeability (μ_r) as shown in the figure.





The magnitude of the transmitted electric field component (in V/m) after it has travelled a distance of 10 cm inside the dielectric region is_____.

Detailed Solution – Question 86

Material has μ_R and ϵ_R complex.

$$\eta \text{ of material} = \sqrt{\frac{j\omega\mu}{\sigma + j\omega\epsilon}} = \sqrt{\frac{\mu_o\mu_R}{\epsilon_o\epsilon_R}} = 120\pi$$

where, $\sigma = 0$ (dielectric) No reflections at the boundary.

$$\begin{aligned}\gamma &= \text{Propagation constant} \\ &= \sqrt{j\omega\mu_o\mu_R(0 + j\omega\epsilon_o\epsilon_R)} \\ &= \sqrt{j\omega\mu_o\epsilon_o(1 - j2)} \\ &= j\omega\sqrt{\mu_o\epsilon_o} + j2\omega\sqrt{\mu_o\epsilon_o}\end{aligned}$$

$$\alpha = 2\omega\sqrt{\mu_o\epsilon_o} = 4.6 \frac{1}{\text{m}}$$

$$|E| = E_o e^{-\alpha z} \text{ with } Z = 0.1\text{m}$$

$$|E| = 0.1 \text{ V/m}$$

Question 87: Uniform Plane Waves (GATE EC 2015-SET-1)

NAT

2 Marks

The electric field intensity of a plane wave traveling in free space is given by the following expression $E(x, t) = a_y 24\pi \cos(\omega t - k_0 x) \text{ (V/m)}$. In this field, consider a square area 10 cm x 10 cm on a plane $x + y = 1$. The total time-averaged power (in mW) passing through the square area is _____.

Detailed Solution – Question 87

$\vec{E}(x, t) = \hat{a}_y 24\pi \cos(\omega t - K_o x) \text{ V/m}$ Power density

$$\vec{P} = \frac{1}{2} \frac{|E|^2}{\eta} = \frac{1}{2} \frac{(24\pi)^2}{120\pi} \hat{a}_x$$

Time-averaged power

$$\begin{aligned} P_{\text{avg}} &= \vec{P} \cdot \vec{ds} \\ &= \left[\frac{1}{2} \frac{(24\pi)^2}{120\pi} \hat{a}_x \right] \cdot \left[\frac{\hat{a}_x + \hat{a}_y}{\sqrt{2}} \times 10^{-2} \right] \\ P_{\text{avg}} &= \frac{24\pi}{\sqrt{2}} = 53.29 \text{ mW} \end{aligned}$$

Question 88: Waveguides & Optical Fibers (GATE EC 2015-SET-1)

MCQ

2 Marks

The longitudinal component of the magnetic field inside an air-filled rectangular waveguide made of a perfect electric conductor is given by the following expression $H_z(x, y, z, t) = 0.1 \cos(25\pi x) \cos(30.3\pi y) \cos(12\pi \times 10^9 t - \beta z) \text{ (A/m)}$. The cross-sectional dimensions of the waveguide are given as $a = 0.08 \text{ m}$ and $b = 0.033 \text{ m}$. The mode of propagation inside the waveguide is

- (A) TM_{12}
- (B) TM_{21}
- (C) **(Correct)** TE_{21}
- (D) TE_{12}

Correct Answer: (C)

Detailed Solution – Question 88

$$\begin{aligned} H_z(x, y, z, t) &= 0.1 \cos(25\pi x) \cos(30.3\pi y) \\ &\quad \cos(12\pi \times 10^9 t - \beta z)^{-1/m} \\ a &= 0.08 \text{ m}, b = 0.033 \text{ m} \end{aligned}$$

Axial component is $H \Rightarrow$ the propagating mode is TE_{mn} , m, n can be found by

$$\begin{aligned} \frac{m\pi}{a} x &= 25\pi x \\ \Rightarrow \frac{m}{0.08} &= 25 \\ \Rightarrow m &= 2 \\ \Rightarrow \frac{n\pi}{b} y &= 30.3\pi y \\ \Rightarrow \frac{n}{0.033} &= 30.3 \\ \Rightarrow n &= 1 \end{aligned}$$

$\therefore$ The mode of propagation is TE_{21}

Question 89: Uniform Plane Waves (GATE EC 2015-SET-1)

NAT

1 Mark

The electric field component of a plane wave traveling in a lossless dielectric medium is given by $\vec{E}(z, t) = \hat{a}_y 2 \cos(10^8 t - \frac{z}{\sqrt{2}}) \text{ V/m}$. The wavelength (in m) for the wave is _____.

Detailed Solution – Question 89

$$\vec{E}(z, t) = \hat{a}_y 2 \cos\left(10^8 t - \frac{z}{\sqrt{2}}\right) \text{ V/m}$$

General form of $\vec{E}(z, t) = \hat{a}_y E_o \cos(\omega t - \beta z) \text{ V/m}$ Comparing both, we get $\beta = \frac{1}{\sqrt{2}}$ since, $\beta = \frac{2\pi}{\lambda}$

$$\Rightarrow \lambda = \frac{2\pi}{\beta} = \frac{2\pi}{\frac{1}{\sqrt{2}}} = 2\sqrt{2}\pi \text{ m}$$

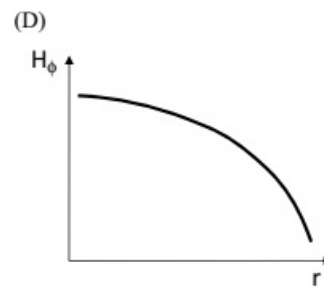
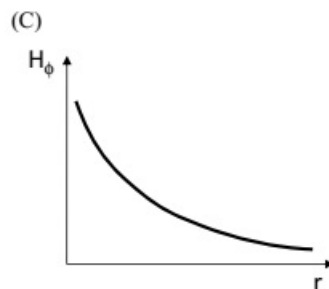
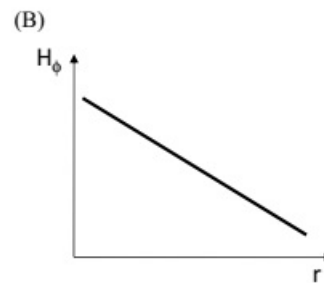
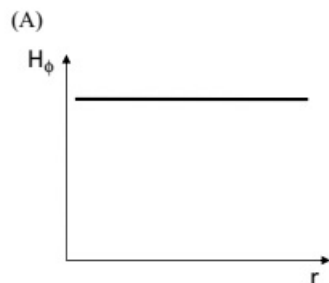
$$\Rightarrow \lambda = 8.88 \text{ m}$$

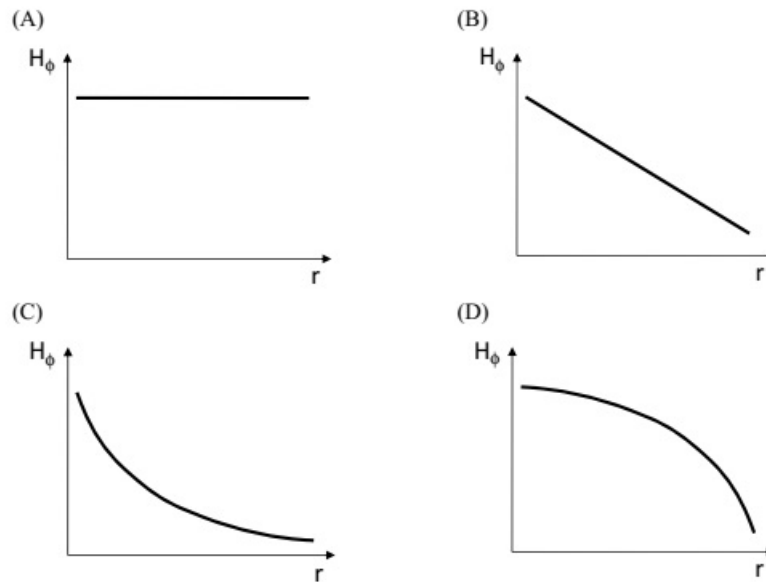
Question 90: Basics of Electromagnetics & Maxwell's Eqns (GATE EC 2015-SET-1)

MCQ

1 Mark

Consider a straight, infinitely long, current carrying conductor lying on the z-axis. Which one of the following plots (in linear scale) qualitatively represents the dependence of H_ϕ on r, where H_ϕ is the magnitude of the azimuthal component of magnetic field outside the conductor and r is the radial distance from the conductor?





- (A) A
 (B) B
 (C) **(Correct)** C
 (D) D

Correct Answer: (C)

Detailed Solution – Question 90

$$H_\phi = \frac{I}{2\pi r} \hat{a}_\phi \Rightarrow H_\phi \propto \frac{1}{r} \text{ option (c) is satisfied.}$$

Question 91: Basics of Electromagnetics & Maxwell's Eqns (GATE EC 2014-SET-4)

MCQ

2 Marks

If $\vec{E} = -9(2y^3 - 3yz^2)\hat{x} - (6xy^2 - 3xz^2)\hat{y} + (6xyz)\hat{z}$ is the electric field in a source free region, a valid expression for the electrostatic potential is

- (A) $xy^3 - yz^2$
 (B) $2xy^3 - xyz^2$
 (C) $y^3 + xyz^2$
 (D) **(Correct)** $2xy^3 - 3xyz^2$

Correct Answer: (D)

Detailed Solution – Question 91

$$E = -\frac{\partial V}{\partial x}\hat{a}_x - \frac{\partial V}{\partial y}\hat{a}_y - \frac{\partial V}{\partial z}\hat{a}_z$$

$$\frac{\partial V}{\partial x} = 2y^3 - 3yz^2 \quad V = 2xy^3 - 3xyz^2 + K_1$$

$$\frac{\partial V}{\partial y} = 6xy^2 - 3xz^2 \quad V = 2xy^3 - 3xyz^2 + K_2$$

$$\frac{\partial V}{\partial z} = -6xyz \quad V = -3xyz^2 + K_3$$

Comparing all three for uniqueness solution of potential

$$K_1 = K_2 = 0, \quad K_3 = 2xy^3$$

$$\vec{E} = -(2y^3 - 3yz^2)\hat{x} - (6xy^2 - 3xz^2)\hat{y} + (6xyz)\hat{z}$$

as we know that $E = -\nabla V$

$$E = -\left(\frac{\partial}{\partial x}\hat{x} + \frac{\partial}{\partial y}\hat{y} + \frac{\partial}{\partial z}\hat{z}\right)V \quad \dots(i)$$

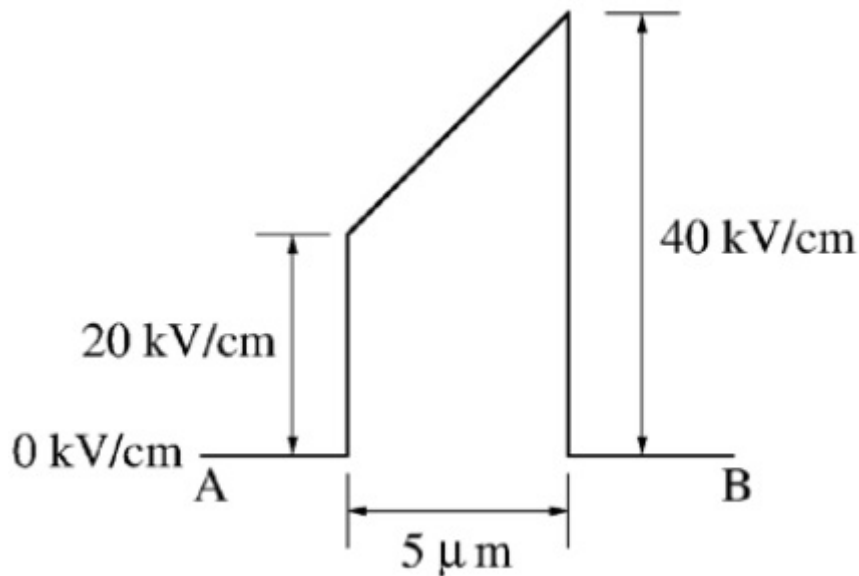
Clearly option (D) satisfies the given relation.

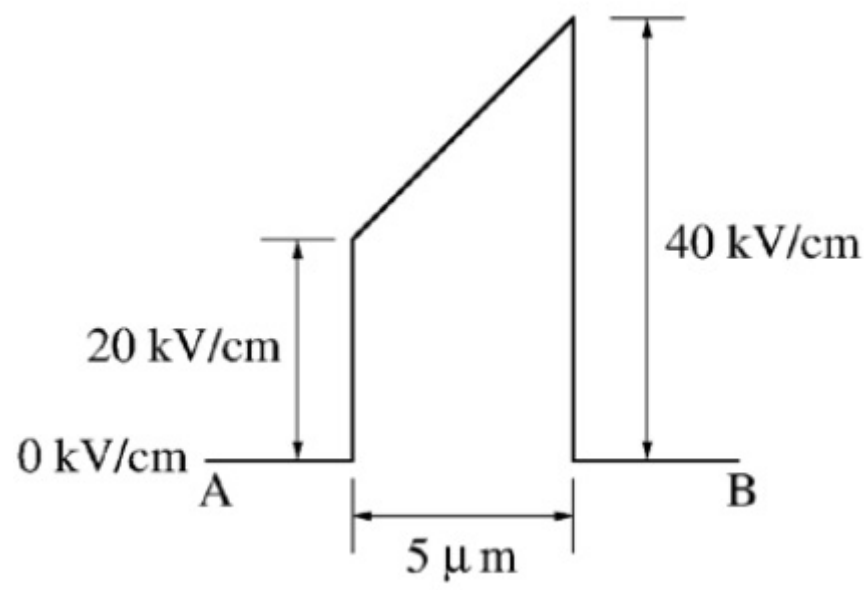
Question 92: Basics of Electromagnetics & Maxwell's Eqns (GATE EC 2014-SET-4)

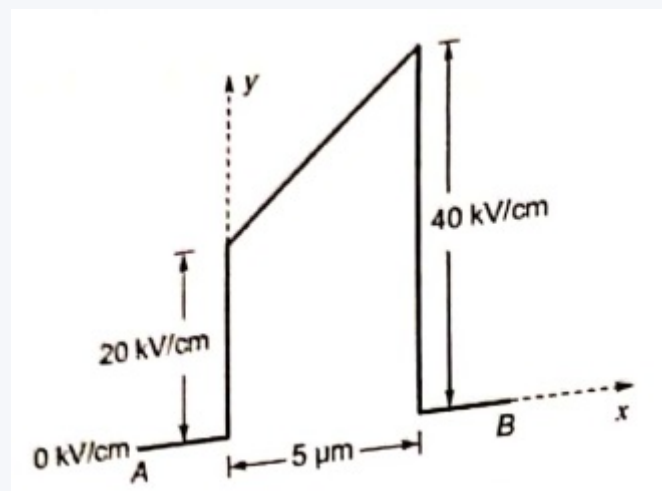
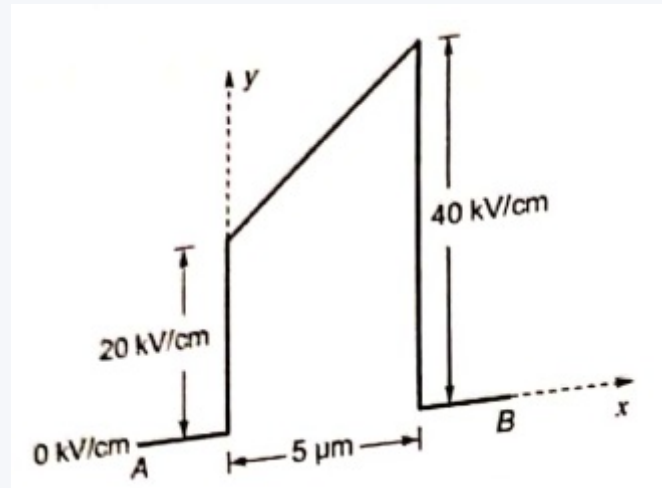
NAT

2 Marks

The electric field (assumed to be one-dimensional) between two points A and B is shown. Let ψ_A and ψ_B be the electrostatic potentials at A and B, respectively. The value of $\psi_B - \psi_A$ in Volts is







E is linear from A to B and $E = mx + C$

$$V = - \int_B^A (mx + c) dx$$

At point A : $x = 0, E = c = 2 \times 10^6 \text{ V/m} = x 10^{-6}$

At point B : $x = 5 \times 10^{-6}$

$$E = 4 \times 10^6 = m \times 5 \times 10^{-6} + 2 \times 10^6$$

$$m = 4 \times 10^{11}$$

$$- \int_{\Psi_B}^{\Psi_A} E \cdot dl = - \int_{5\mu m}^0 \left[20 \times 10^5 \times 5 \times 10^{-6} + \frac{4 \times 10^{11}}{2} [(5 \times 10^{-6}) - 0^2] \right]$$

$$\begin{aligned} \Psi_A - \Psi_B &= - \left[(20 \times 10^5 \times 5 \times 10^{-6}) + \frac{4 \times 10^{11}}{2} [(5 \times 10^{-6})^2 - 0^2] \right] \\ &= - [10 + (2 \times 10^{11} \times 25 \times 10^{-12})] \\ &= -15 \text{ V} \end{aligned}$$

Question 93: Antennas & Radiating Systems (GATE EC 2014-SET-4)

MCQ

1 Mark

Match column A with column B.

Column A

1. Point electromagnetic source
2. Dish antenna
3. Yagi-Uda antenna

Column B

- P. Highly directional
Q. End fire
R. Isotropic

Column A

1. Point electromagnetic source
2. Dish antenna
3. Yagi-Uda antenna

Column B

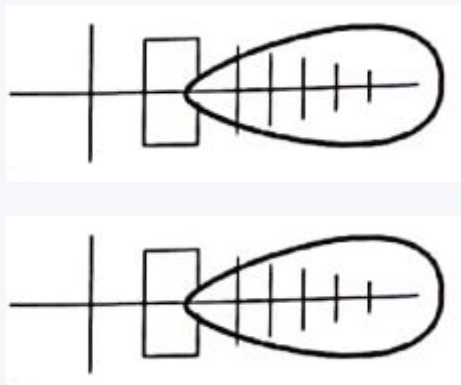
- P. Highly directional
Q. End fire
R. Isotropic

- (A) $1 \rightarrow P \ 2 \rightarrow Q \ 3 \rightarrow R$
(B) **(Correct)** $1 \rightarrow R \ 2 \rightarrow P \ 3 \rightarrow Q$
(C) $1 \rightarrow Q \ 2 \rightarrow P \ 3 \rightarrow R$
(D) $1 \rightarrow R \ 2 \rightarrow Q \ 3 \rightarrow P$

Correct Answer: (B)

Detailed Solution – Question 93

Point electromagnetic source radiates in all direction Dish antenna radiates any electromagnetic energy in any particular direction with narrow beam-width and high directivity Yagi-uda antenna is high gain antenna used for T.V. reception. Its radiation is along the axis of the antenna.



Question 94: Antennas & Radiating Systems (GATE EC 2014-SET-4)

NAT

1 Mark

For an antenna radiating in free space, the electric field at a distance of 1 km is found to be 12mV/m. Given that intrinsic impedance of the free space is $120\pi\Omega$, the magnitude of average power density due to this antenna at a distance of 2 km from the antenna (in nW/m^2) is _____.

Detailed Solution – Question 94

Given electric field, E at a distance of $1\text{km} = 12\text{mV/m}$ Also we know that $E \propto \frac{1}{r}$, where r is the distance where electric field is to be measured. So, electric field, E at a distance of $2\text{km} = \frac{12\text{mV/m}}{2} = 6\text{mV/m}$ Also, power density due to the antenna is given as,

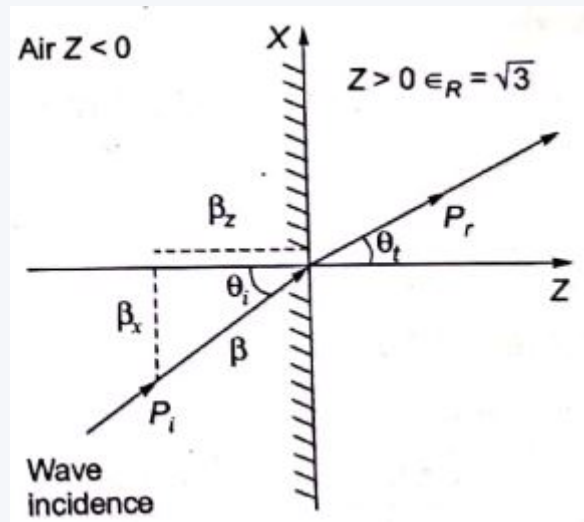
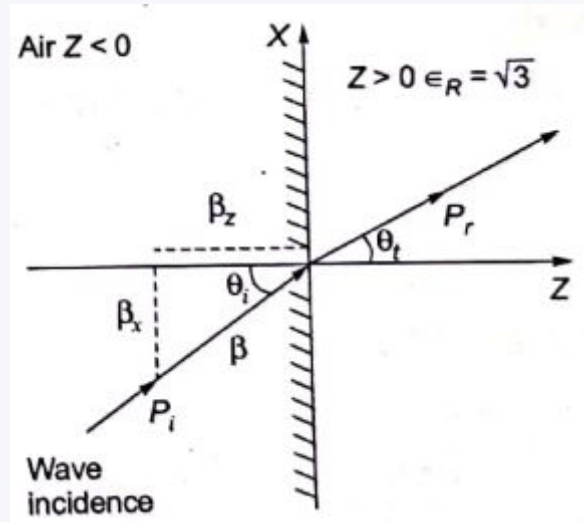
$$\begin{aligned} P_{\text{avg}} &= \frac{1}{2} \frac{E^2}{\eta} = \frac{1}{2} \times \frac{6 \times 6 \times 10^{-6}}{120\pi} \\ &= 47.7\text{nW/m}^2 \end{aligned}$$

Question 95: Uniform Plane Waves (GATE EC 2014-SET-3)

NAT

2 Marks

Assume that a plane wave in air with an electric field $\vec{E} = 10\cos(\omega t - 3x - \sqrt{3}z)\hat{a}_y\text{V/m}$ is incident on a non-magnetic dielectric slab of relative permittivity 3 which covers the region $z > 0$. The angle of transmission in the dielectric slab is _____ degrees.



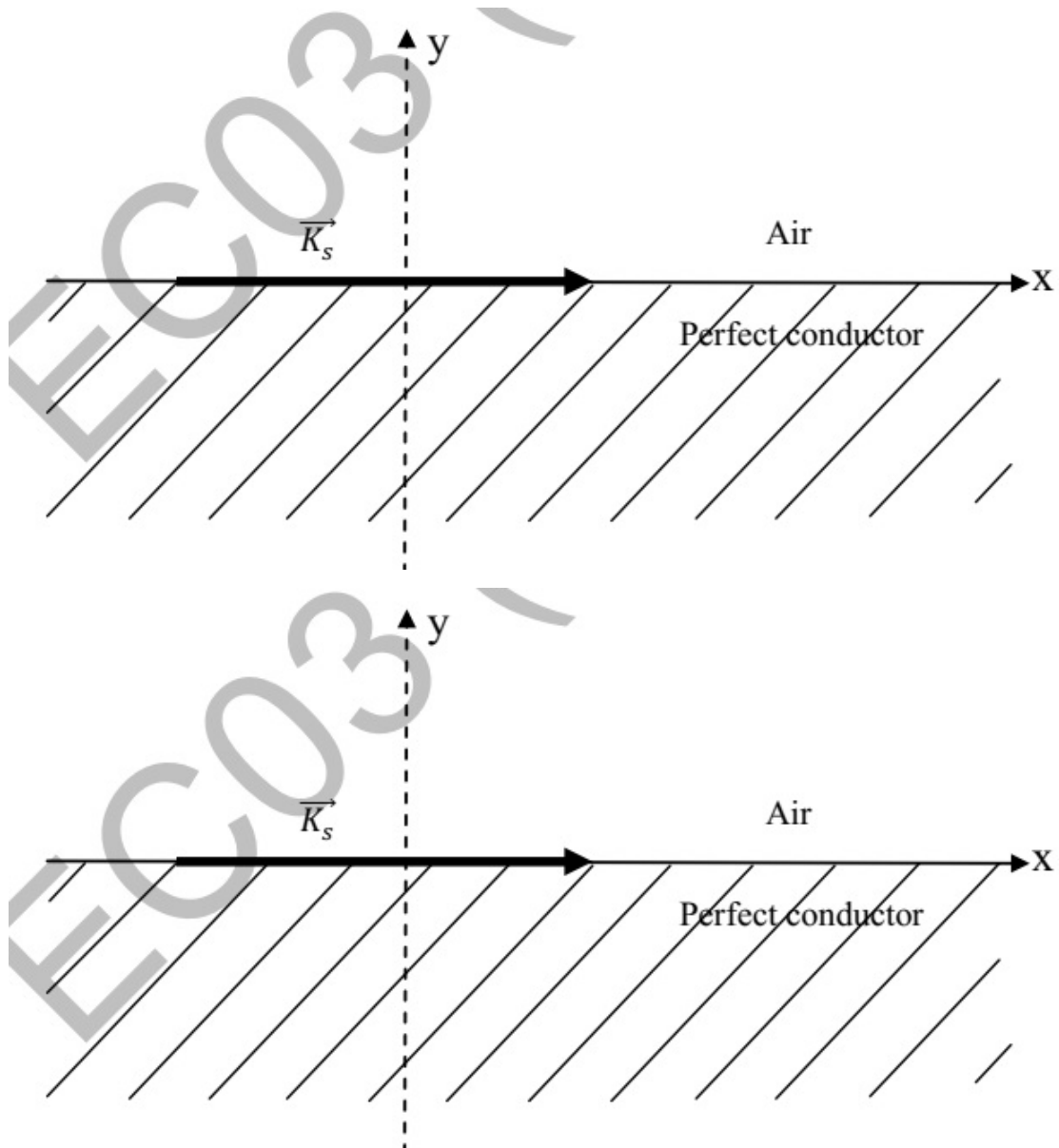
$$\begin{aligned}\beta_x &= 3, \beta_z = \sqrt{3} \text{ from } E_i \text{ of the wave} \\ \theta_i &= 60^\circ \\ \frac{\sin 60^\circ}{\sin \theta_t} &= \sqrt{3} \\ \theta_t &= 30^\circ\end{aligned}$$

Question 96: Basics of Electromagnetics & Maxwell's Eqns (GATE EC 2014-SET-3)

MCQ

2 Marks

A region shown below contains a perfect conducting half-space and air. The surface current $\vec{K}_s$ on the surface of the perfect conductor is $\vec{K}_s = \hat{x}_2$ amperes per meter. The tangential $\vec{H}$ field in the air just above the perfect conductor is



- (A) $(\hat{x} + \hat{z})2$ amperes per meter
- (B) $\hat{x}2$ amperes per meter
- (C) $-\hat{z}2$ amperes per meter
- (D) **(Correct)** $\hat{z}2$ amperes per meter

Correct Answer: (D)

Detailed Solution – Question 96

since, we know from the property of magnetic boundary conditions i.e. $\vec{H}_{t2} - \vec{H}_{t1} = \hat{K}_s \times \hat{a}_N$ In the given question the tangential field $\vec{H}$ in the air just above the perfect conductor is asked with $H=0$ inside the conductor

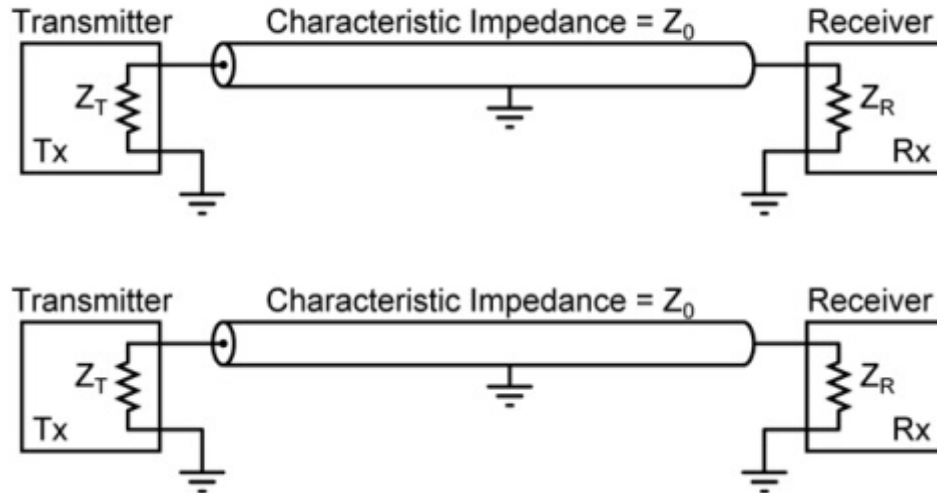
$$\begin{aligned}\vec{H}_{t1} &= 0 \\ \vec{H}_{t2} &= \hat{x}2 \times \hat{y} \\ \vec{H}_{t2} &= \hat{z}2\text{A/m}\end{aligned}$$

Question 97: Transmission Lines (GATE EC 2014-SET-3)

MCQ

1 Mark

In the following figure, the transmitter T_X sends a wideband modulated RF signal via a coaxial cable to the receiver R_X . The output impedance Z_T of T_X , the characteristic impedance Z_0 of the cable and the input impedance Z_R of R_X are all real.



- (A) The signal gets distorted if $Z_R \neq Z_0$, irrespective of the value of Z_T
- (B) The signal gets distorted if $Z_T \neq Z_0$, irrespective of the value of Z_R
- (C) **(Correct)** Signal distortion implies impedance mismatch at both ends : $Z_T \neq Z_0$ and $Z_R \neq Z_0$
- (D) Impedance mismatches do NOT result in signal distortion but reduce power transfer efficiency

Correct Answer: (C)

Detailed Solution – Question 97

The signal wave gets distorted if there is any impedance mismatch between the characteristic impedance Z_0 and the transmitting impedance Z_T and Z_0 and receiving impedance Z_R i.e. $Z_T \neq Z_0$ and $Z_R \neq Z_0$ then only distortion occurs.
As standing waveforms the signal has multi harmonics and hence distortion occurs.

Question 98: Waveguides & Optical Fibers (GATE EC 2014-SET-3)

NAT

1 Mark

Consider an air filled rectangular waveguide with a cross-section of 5cm x 3cm. For this waveguide, the cut-off frequency (in MHz) of TE_{21} mode is _____.

Detailed Solution – Question 98

Given $a = 5\text{cm} = 5 \times 10^{-2}\text{m}$

$$\begin{aligned}b &= 3\text{cm} = 3 \times 10^{-2}\text{m} \\f_c &= \frac{c}{2\pi} \sqrt{\left(\frac{nm}{a}\right)^2 + \left(\frac{rm}{b}\right)^2} \\f_{CTE_{21}} &= \frac{c}{2} \sqrt{\left(\frac{2}{5\text{cm}}\right)^2 + \left(\frac{1}{3\text{cm}}\right)^2} \\&= \frac{3 \times 10^8}{2} \sqrt{\left(\frac{2 \times 100}{5}\right)^2 + \left(\frac{100}{3}\right)^2} \\f_c &= 7810\text{MHz}\end{aligned}$$

Question 99: Waveguides & Optical Fibers (GATE EC 2014-SET-2)

NAT

2 Marks

For a rectangular waveguide of internal dimensions $a \times b$ ($a > b$), the cut-off frequency for the TE_{11} mode is the arithmetic mean of the cut-off frequencies for TE_{10} mode and TE_{20} mode. If $a = \sqrt{5}\text{ cm}$, the value of b (in cm) is_____.

Detailed Solution – Question 99

Cut-off frequency of a rectangular waveguide is

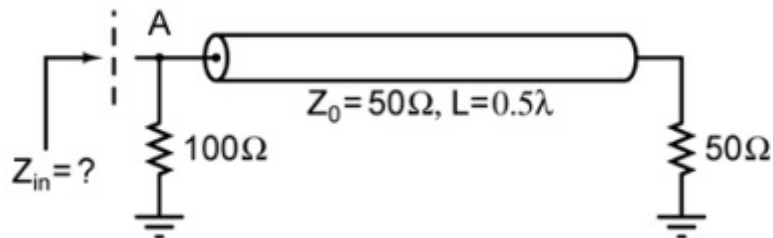
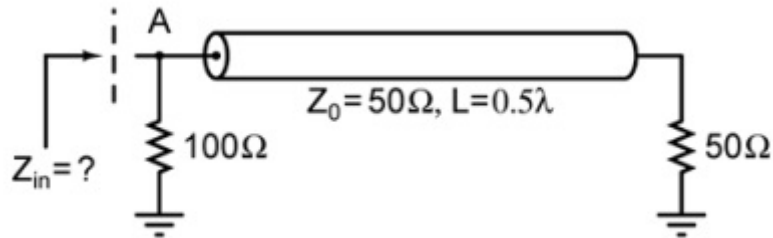
$$\begin{aligned}f_c &= \frac{c}{2\pi} \sqrt{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2} \\f_c \text{ for } TE_{10} &= \frac{c}{2\pi} \sqrt{\left(\frac{1\pi}{a}\right)^2} = \frac{c}{2a} \\f_c \text{ for } TE_{20} &= \frac{c}{2\pi} \sqrt{\left(\frac{2\pi}{a}\right)^2} = \frac{c}{a} \\f_c \text{ for } TE_{11} &= \frac{c}{2\pi} \sqrt{\left(\frac{\pi}{a}\right)^2 + \left(\frac{\pi}{b}\right)^2} \\&= \frac{c}{2} \sqrt{\left(\frac{1}{a}\right)^2 + \left(\frac{1}{b}\right)^2} \\ \text{Given, } f_{CT_{11}} &= \frac{f_{CTE_{10}} + f_{CTE_{20}}}{2} \\ \frac{c}{2} \sqrt{\left(\frac{1}{a}\right)^2 + \left(\frac{1}{b}\right)^2} &= \frac{\frac{c}{2a} + \frac{c}{a}}{2} \\ \sqrt{\left(\frac{1}{a}\right)^2 + \left(\frac{1}{b}\right)^2} &= \frac{1}{2a} + \frac{1}{a} = \frac{3}{2a} \\ \left(\frac{1}{a}\right)^2 + \left(\frac{1}{b}\right)^2 &= \frac{9}{4a^2} \\ \Rightarrow \left(\frac{1}{b}\right)^2 &= \frac{5}{4a^2} \\ b &= \frac{2a}{\sqrt{5}} = \frac{2 \times \sqrt{5}}{\sqrt{5}} \\ &= 2\text{cm}\end{aligned}$$

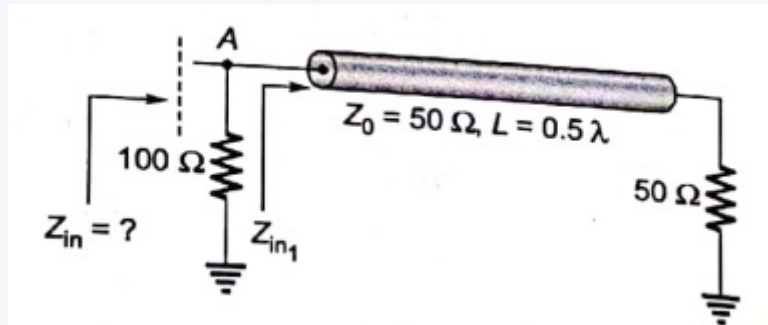
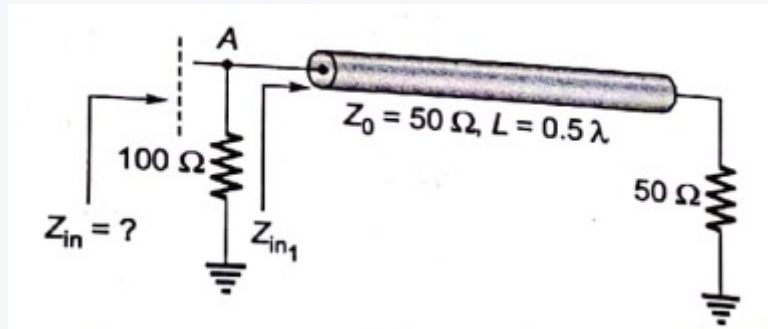
Question 100: Transmission Lines (GATE EC 2014-SET-2)

NAT

2 Marks

In the transmission line shown, the impedance Z_{in} (in ohms) between node A and the ground is _____.





$$Z_{in1} = Z_0 \left(\frac{Z_L + jZ_0 \tan \beta l}{Z_0 + jZ_L \tan \beta l} \right)$$

$$\beta l = \frac{2\pi}{\lambda} \times \frac{\lambda}{2} = \pi$$

$$Z_{in} = Z_0 \left(\frac{Z_L + j0}{Z_0 + j0} \right)$$

$$Z_{in1} = Z_L = 50\Omega$$

$$\therefore Z_{in} = (Z_{in1} \parallel 100\Omega) = \frac{100 \times 50}{150} = 33.3\Omega$$

Question 101: Uniform Plane Waves (GATE EC 2014-SET-2)

MCQ

2 Marks

If the electric field of a plane wave is $\vec{E}(z, t) = \hat{x}3\cos(\omega t - kz + 30^\circ) - \hat{y}4\sin(\omega t - kz + 45^\circ)$ (mV/m), the polarization state of the plane wave is

- (A) **(Correct)** left elliptical
- (B) left circular
- (C) right elliptical
- (D) right circular

Correct Answer: (A)

Detailed Solution – Question 101

$$\vec{E}(z, t) = 3\hat{x} \cos(\omega t - kz + 30^\circ) - 4\hat{y} \sin(\omega t - kz + 45^\circ)$$

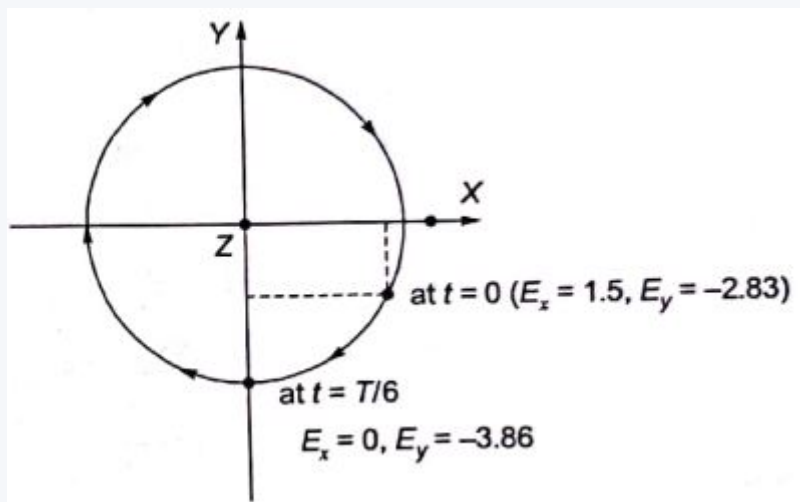
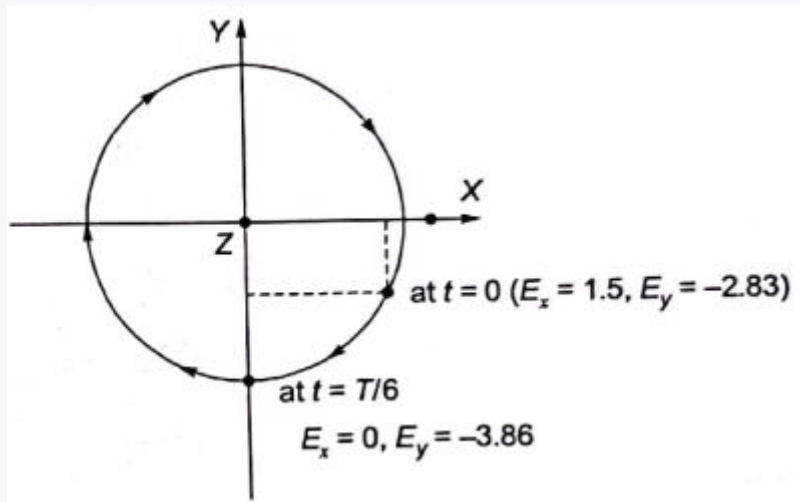
The plane wave is travelling in +z direction

$$\begin{aligned}\hat{E}_x &= 3 \cos(\omega t - kz + 30^\circ) \\ \hat{E}_y &= -4 \sin(\omega t - kz + 45^\circ)\end{aligned}$$

At $z=0$, tracing E locus for various times

$$\begin{aligned}\text{At } t = 0, E_x &= 3 \cos 30^\circ = 3 \cdot \frac{1}{2} = \frac{3}{2} = 1.5 \\ E_y &= -4 \sin 45^\circ = -2\sqrt{2} = -2.83\end{aligned}$$

$$\text{At } t = \frac{T}{6} \quad E_x = 3 \cos\left(\frac{2\pi}{T} \times \frac{T}{6} + 30^\circ\right) = 0 \quad E_y = -4 \sin(60^\circ + 45^\circ) = -3.86$$



This is left elliptical polarized plane-wave.

Question 102: Uniform Plane Waves (GATE EC 2014-SET-2)

MCQ

1 Mark

Which one of the following field patterns represents a TEM wave travelling in the positive x direction

?

- (A) $E = +8\hat{y}, H = -4\hat{z}$
(B) **(Correct)** $E = -2\hat{y}, H = -3\hat{z}$
(C) $E = +2\hat{z}, H = +2\hat{y}$
(D) $E = -3\hat{y}, H = +4\hat{z}$

Correct Answer: (B)

Detailed Solution – Question 102

Since, the TEM wave or plane wave is travelling in positive x-direction, then conditions should be

$$P = \vec{E} \times \vec{H}$$
$$\hat{a}_x = \hat{a}_y \times \hat{a}_z$$

or, $\hat{a}_x = (\hat{a}_y) \times (-\hat{a}_z)$

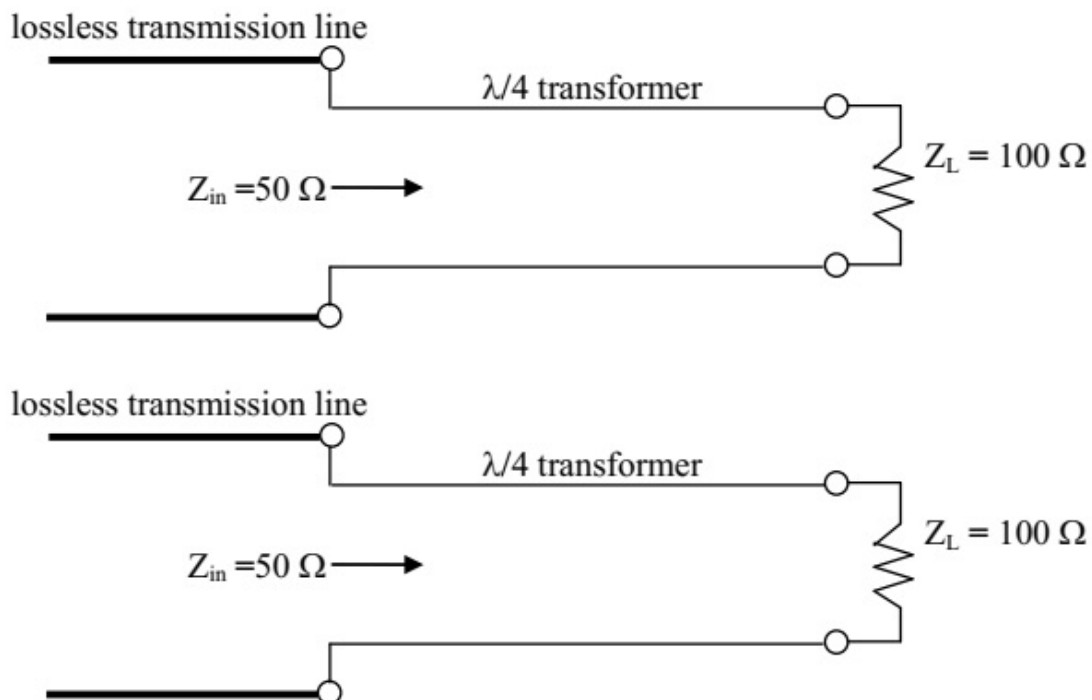
So, the option (B) satisfies the condition.

Question 103: Transmission Lines (GATE EC 2014-SET-2)

NAT

1 Mark

To maximize power transfer, a lossless transmission line is to be matched to a resistive load impedance via a $\lambda/4$ transformer as shown.



The characteristic impedance (in Ω) of the $\lambda/4$ transformer is _____

Detailed Solution – Question 103

Input impedance for quarter wave transformer is given as,

$$Z_{\text{in}} = \frac{Z_0^2}{Z_L}$$

$$\text{Given: } Z_{\text{in}} = 50\Omega, \quad Z_L = 100\Omega$$

$$\therefore Z_0^2 = 50 \times 100$$

$$Z_0 = \sqrt{50 \times 100}$$

$$Z_0 = 70.72\Omega$$

Question 104: Transmission Lines (GATE EC 2014-SET-1)

NAT

2 Marks

The input impedance of a $\frac{\lambda}{8}$ section of a lossless transmission line of characteristic impedance 50Ω is found to be real when the other end is terminated by a load $Z_L = (R + jX)\Omega$. If X is 30Ω , the value of R (in Ω) is_____.

Detailed Solution – Question 104

$$Z_{\text{in}} = Z_0 \left(\frac{Z_L + jZ_0 \tan \beta l}{Z_0 + jZ_L \tan \beta l} \right)$$

$$\text{at } l = \frac{\lambda}{8}$$

$$\Rightarrow \beta l = \frac{2\pi}{\lambda} \times \frac{\lambda}{8} = \frac{\pi}{4}$$

$$\therefore \tan \frac{\pi}{4} = 1$$

$$Z_{\text{in}} = Z_0 \left[\frac{Z_L + j50}{Z_0 + jZ_L} \right]$$

$$\text{Given, } Z_L = R + jX = R + j30 \text{ and } Z_0 = 50\Omega$$

$$30R = 80(50 + R)$$

$$\begin{aligned} Z_{\text{in}} &= 50 \left[\frac{R + j30 + j50}{50 + j(R + j30)} \right] \\ &= 50 \left[\frac{R + j80}{(50 - 30) + jR} \right] = 50 \left[\frac{R + j80}{20 + jR} \right] \\ &= 50 \left[\frac{(R + j80)(20 - jR)}{20^2 + R^2} \right] \end{aligned}$$

Since, only real part of Z_{in} exists, so imaginary part of $Z_{\text{in}} = 0$

$$\therefore Z_{\text{in}} = 50 \left[\frac{20R + 1600j - jR^2 + 50R}{20^2 + R^2} \right]$$

$$Z_{\text{imaginary}} = 0 = \frac{1600 - R^2}{20^2 + R^2}$$

$$\therefore R^2 = 1600$$

$$R = 40\Omega$$

Question 105: Transmission Lines (GATE EC 2014-SET-1)

MCQ

2 Marks

For a parallel plate transmission line, let v be the speed of propagation and Z be the characteristic impedance. Neglecting fringe effects, a reduction of the spacing between the plates by a factor of two results in

- (A) halving of v and no change in Z
- (B) **(Correct)** no changes in v and halving of Z
- (C) no change in both v and Z
- (D) halving of both v and Z

Correct Answer: (B)

Detailed Solution – Question 105

$$\begin{aligned} Z_0 &= \sqrt{\frac{L}{C}} \text{ for parallel plate line} \\ \frac{C}{l} &= \frac{\epsilon W}{d} \text{ and } \frac{L}{l} = \frac{\mu d}{W} \\ \text{where, } W &= \text{Width of the line} \\ d &= \text{Separation between the lines} \\ Z_0 &\propto d \quad (Z_0 \text{ change by half}) \\ \text{Velocity} &= \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = 3 \times 10^8 \text{ m/s} \\ &\quad (\text{No change in } V) \end{aligned}$$

Question 106: Antennas & Radiating Systems (GATE EC 2014-SET-1)

NAT

2 Marks

In spherical coordinates, let $\hat{a}_\theta, \hat{a}_\phi$ denote unit vectors along the θ, ϕ directions. $E = \frac{100}{r} \sin\theta \cos(\omega t - \beta r) \hat{a}_\theta \text{ V/m}$ and $E = \frac{0.265}{r} \sin\theta \cos(\omega t - \beta r) \hat{a}_\phi \text{ A/m}$ represent the electric and magnetic field components of the EM wave at large distances r from a dipole antenna, in free space. The average power (W) crossing the hemispherical shell located at $r = 1 \text{ km}$, $0 \leq \theta \leq \pi/2$ is _____.

Detailed Solution – Question 106

$$\begin{aligned}\vec{E} &= \text{Re} \left[\frac{100}{r} \sin \theta e^{-\beta r} e^{j\omega t} \right] \hat{a}_\theta V/m \\ &= \text{Re} \left[\frac{0.265}{r} \sin \theta e^{-\beta r} e^{j\omega t} \right] \hat{a}_\phi V/m\end{aligned}$$

Poynting vector,

$$\vec{P} = \frac{1}{2} \text{Re} [\vec{E} \times \vec{H}^*] = \frac{26.5}{2r^2} \sin^2 \theta \hat{a}_r W/m^2$$

The average power crossing the given hemispherical shell is,

$$\begin{aligned}\text{Power} &= \int_0^{\pi/2} \int_{\phi=0}^{2\pi} \vec{P} \cdot r^2 \sin \theta d\theta d\phi \hat{a}_r \\ &= \frac{26.5}{2} \times 2\pi \int_{\theta=0}^{\pi/2} \sin^3 \theta d\theta \\ &= \frac{26.5}{2} \times 2\pi \frac{2}{3} = 55.5 \text{ W}\end{aligned}$$

Question 107: Basics of Electromagnetics & Maxwell's Eqns (GATE EC 2014-SET-1)

MCQ

1 Mark

The force on a point charge $+q$ kept at a distance d from the surface of an infinite grounded metal plate in a medium of permittivity ε is

- (A) 0
- (B) $\frac{q^2}{16\pi\varepsilon d^2}$ away from the plate
- (C) **(Correct)** $\frac{q^2}{16\pi\varepsilon d^2}$ towards the plate
- (D) $\frac{q^2}{4\pi\varepsilon d^2}$ towards the plate

Correct Answer: (C)

Question 108: Miscellaneous Topics (GATE EC 2014-SET-1)

MCQ

1 Mark

A two-port network has scattering parameters given by

$$[S] = \begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix}$$

. If the port-2 of the two port is short circuited, the S_{11} parameter for the resultant one port network is

- (A) $\frac{S_{11}-S_{11}S_{22}+S_{12}S_{21}}{1+S_{22}}$

(B) **(Correct)** $\frac{S_{11}+S_{11}S_{22}-S_{12}S_{21}}{1+S_{22}}$

(C) $\frac{S_{11}+S_{11}S_{22}+S_{12}S_{21}}{1-S_{22}}$

(D) $\frac{S_{11}-S_{11}S_{22}+S_{12}S_{21}}{1-S_{22}}$

Correct Answer: (B)

Detailed Solution – Question 108

$$b_1 = s_{11} \cdot a_1 + s_{12} \cdot a_2 \quad \dots (i)$$

$$b_2 = s_{21} \cdot a_1 + s_{22}a_2 \quad \dots (ii)$$

When output is short, $a_2 = -b_2$

$$-a_2 = s_{21} \cdot a_1 + s_{22}a_2$$

$$\Rightarrow a_2 = \frac{-s_{21} \cdot a_1}{1 + s_{22}}$$

Substituting in first equation,

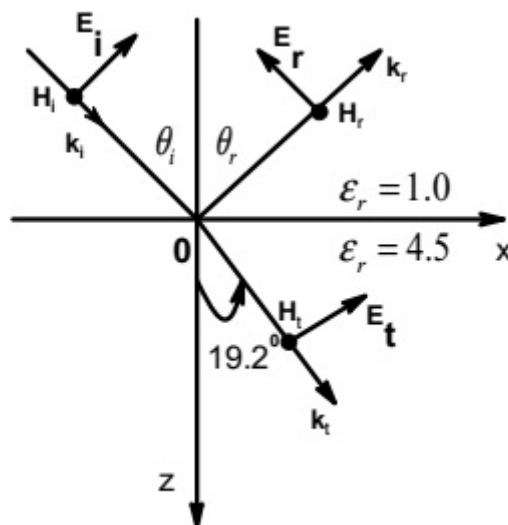
$$\begin{aligned} b_1 &= s_{11} \cdot a_1 + s_{12} \left(\frac{-s_{21} \cdot a_1}{1 + s_{22}} \right) \\ &= \frac{s_{11} \cdot a_1 + s_{11} \cdot s_{22}a_1 - s_{12} \cdot s_{21}a_1}{1 + s_{22}} \\ \frac{b_1}{a_1} &= s_{11} = \frac{s_{11} + s_{11} \cdot s_{22} - s_{12} \cdot s_{21}}{1 + s_{22}} \end{aligned}$$

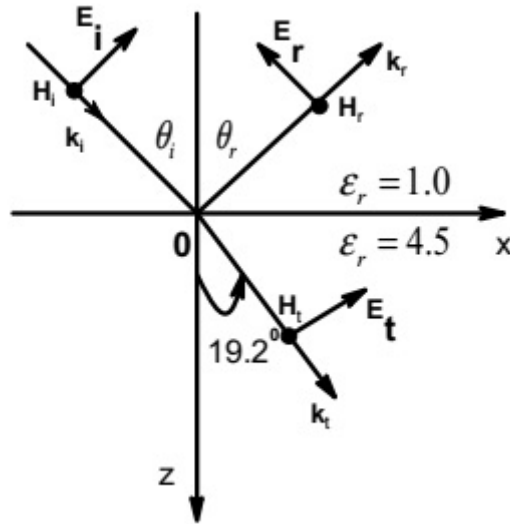
Question 109: Uniform Plane Waves (GATE EC 2013)

MCQ

1 Mark

A monochromatic plane wave of wavelength $\lambda = 600\mu m$ is propagating in the direction as shown in the figure below. $\vec{E}_i$, $\vec{E}_r$, and $\vec{E}_t$ denote incident, reflected, and transmitted electric field vectors associated with the wave





The expression for $\vec{E}_r$ is

(A) (Correct)

$$0.23 \frac{E_0}{\sqrt{2}} (\hat{a}_x + \hat{a}_z) e^{-j \frac{\pi \times 10^4 (x-z)}{3\sqrt{2}}} V/m$$

(B)

$$-\frac{E_0}{\sqrt{2}} (\hat{a}_x + \hat{a}_z) e^{j \frac{\pi \times 10^4 z}{3}} V/m$$

(C)

$$0.44 \frac{E_0}{\sqrt{2}} (\hat{a}_x + \hat{a}_z) e^{-j \frac{\pi \times 10^4 (x-z)}{3\sqrt{2}}} V/m$$

(D)

$$\frac{E_0}{\sqrt{2}} (\hat{a}_x + \hat{a}_z) e^{-j \frac{\pi \times 10^4 (x+z)}{3}} V/m$$

Correct Answer: (A)

Detailed Solution – Question 109

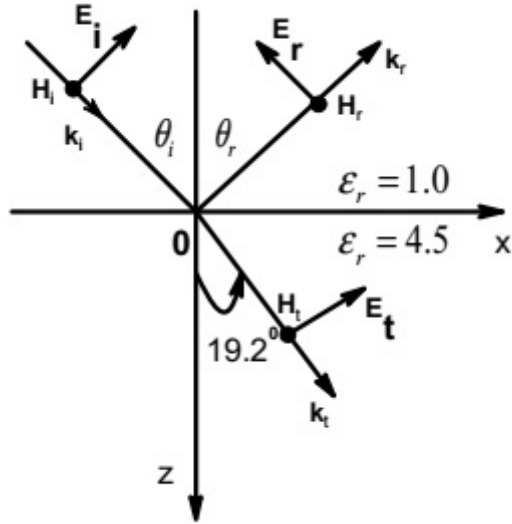
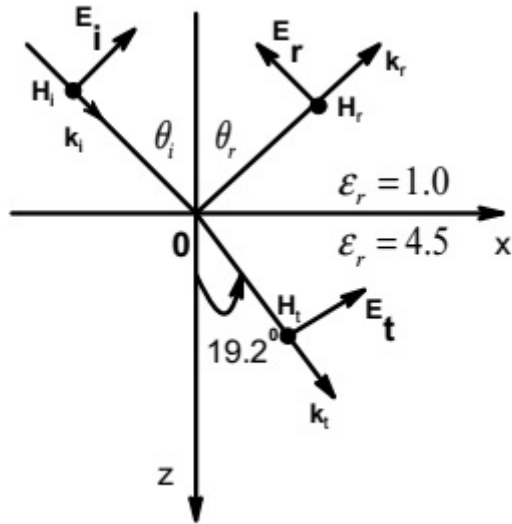
For reflected wave. E_r propagation direction changes without change in angle 45°
 E_r propagation direction = $\frac{\hat{a}_x - \hat{a}_z}{\sqrt{2}}$ Reflection coefficient $\Gamma = -0.23$ E_r field direction suitably changes satisfying $E \cdot P = 0$

Question 110: Uniform Plane Waves (GATE EC 2013)

MCQ

1 Mark

A monochromatic plane wave of wavelength $\lambda = 600 \mu m$ is propagating in the direction as shown in the figure below. $\vec{E}_i$, $\vec{E}_r$, and $\vec{E}_t$ denote incident, reflected, and transmitted electric field vectors associated with the wave



The angle of incidence θ_i and the expression for $\vec{E}_i$ are

(A) 60° and

$$\frac{E_0}{\sqrt{2}} (\hat{a}_x - \hat{a}_z) e^{-j \frac{\pi \times 10^4 (x+z)}{3\sqrt{2}}} V/m$$

(B) 45° and

$$\frac{E_0}{\sqrt{2}} (\hat{a}_x + \hat{a}_z) e^{-j \frac{\pi \times 10^4 z}{3}} V/m$$

(C) **(Correct)** 45° and

$$\frac{E_0}{\sqrt{2}} (\hat{a}_x - \hat{a}_z) e^{-j \frac{\pi \times 10^4 (x+z)}{3\sqrt{2}}} V/m$$

(D) 60° and

$$\frac{E_0}{\sqrt{2}} (\hat{a}_x - \hat{a}_z) e^{-j \frac{\pi \times 10^4 z}{3}} V/m$$

Correct Answer: (C)

Detailed Solution – Question 110

E_i propagates in ZXplane. E_i directed in ZX-plane. It is a p-plane of incidence polarized wave. Using Snell's law,

$$\frac{\sin \theta_i}{\sin 19.2^\circ} = \sqrt{\frac{4.5\epsilon_o}{\epsilon_o}} \quad \theta_i = 45^\circ$$

$$\beta_x = \beta_z = \beta \cos 45^\circ = \frac{\beta}{\sqrt{2}}$$

(for propagation direction) For field direction = $\frac{\hat{a}_x - \hat{a}_z}{\sqrt{2}}$ (45°)

Question 111: Transmission Lines (GATE EC 2013)

MCQ

1 Mark

The return loss of a device is found to be 20 dB. The voltage standing wave ratio (VSWR) and magnitude of reflection coefficient are respectively

- (A) (Correct) 1.22 and 0.1
- (B) 0.81 and 0.1
- (C) - 1.22 and 0.1
- (D) 2.44 and 0.2

Correct Answer: (A)

Detailed Solution – Question 111

$$\text{Return loss} = -20 \log |\Gamma|$$

$$-20 \log |\Gamma| = 20$$

$$\Rightarrow |\Gamma| = 10^{-1} = 0.1$$

$$\text{VSWR} = \frac{1 + |\Gamma|}{1 - |\Gamma|} = \frac{1 + 0.1}{1 - 0.1} = 1.22$$

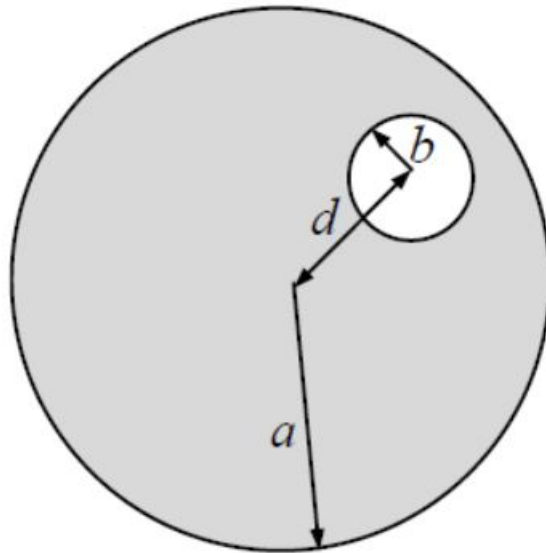
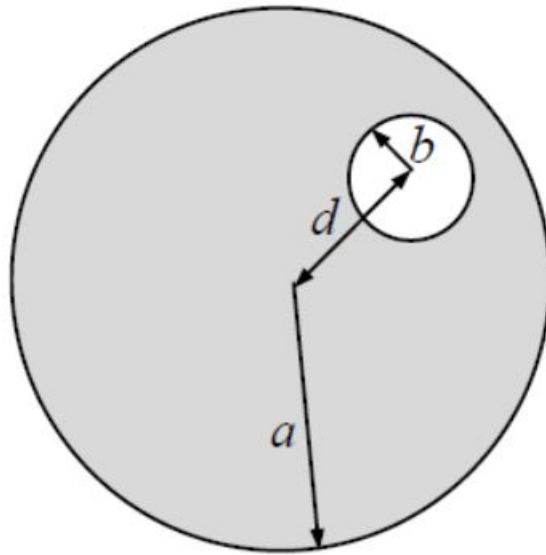
Question 112: Basics of Electromagnetics & Maxwell's Eqns (GATE EC 2012)

MCQ

1 Mark

An infinitely long uniform solid wire of radius a carries a uniform dc current of density $\vec{j}$.

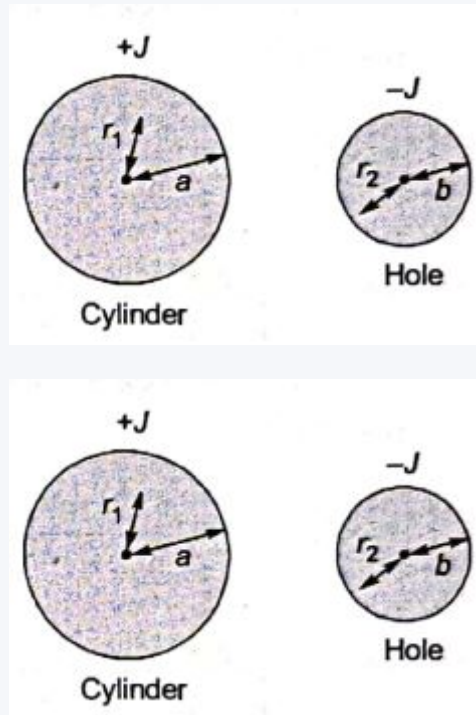
A hole of radius b ($b < a$) is now drilled along the length of the wire at a distance d from the center of the wire as shown below.



The magnetic field inside the hole is

- (A) **(Correct)** uniform and depends only on d
- (B) uniform and depends only on b
- (C) uniform and depends on both b and d
- (D) non uniform

Correct Answer: (A)



The H field due to entire cylinder without hole. $H(r_1) = \frac{Jr_1}{2}$ The Hfield due to hole only. $H(r_2) = \frac{Jr_2}{2}$ Current inside the hole is zero means $H(r) = \frac{Jr_1}{2} - \frac{Jr_2}{2} \simeq \frac{Jd}{2}$

Question 113: Basics of Electromagnetics & Maxwell's Eqns (GATE EC 2012)

MCQ

1 Mark

An infinitely long uniform solid wire of radius a carries a uniform dc current of density $\vec{j}$.

The magnetic field at a distance r from the center of the wire is proportional to

- (A) r for $r < a$ and $1/r^2$ for $r > a$
- (B) 0 for $r < a$ and $1/r$ for $r > a$
- (C) **(Correct)** r for $r < a$ and $1/r$ for $r > a$
- (D) 0 for $r < a$ and $1/r^2$ for $r > a$

Correct Answer: (C)

Detailed Solution – Question 113

We know that magnetic flux density at a distance r from the wire is,

$$|B| = \frac{\mu_0 I}{2\pi r}$$

$$\text{For } r < a \rightarrow I = J \cdot \pi r^2$$

$$\text{so } |B| = \frac{\mu_0 J r}{2}$$

$$|B| \propto r \text{ for } r < a$$

$$\text{for } r > a, \quad I = J \times \pi a^2$$

$$\text{So, } |B| = \frac{\mu_0 J \pi a^2}{2\pi r}$$

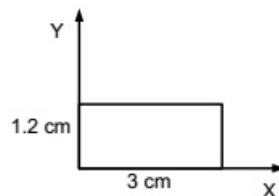
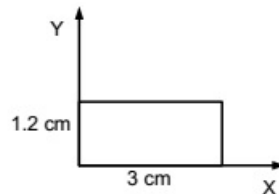
$$|B| \propto \frac{1}{r} \text{ for } r > a$$

Question 114: Waveguides & Optical Fibers (GATE EC 2012)

MCQ

1 Mark

The magnetic field along the propagation direction inside a rectangular waveguide with the cross-section shown in the figure is $H_z = 3\cos(2.094 \times 10^2 x)\cos(2.618 \times 10^2 y) \cos(6.283 \times 10^{10} t - \beta z)$



The phase velocity v_p of the wave inside the waveguide satisfies

- (A) **(Correct)** $v_p > c$
- (B) $v_p = c$
- (C) $0 < v_p < c$
- (D) $v_p = 0$

Correct Answer: (A)

Detailed Solution – Question 114

From the given data

$$\omega_c = \left(\sqrt{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2} \right) c$$

$$f_c = (\sqrt{(2.09)^2 + (2.618)^2}) \times \frac{3 \times 10^{10}}{2 \times 3.14}$$
$$= 16\text{GHz}$$

$$f = \frac{6.283 \times 10^{10}}{2 \times 3.14} = 10\text{GHz}$$

$$\text{Hence, } f < f_c, \gamma = \alpha + j0$$

$$\beta = 0, \quad v_p = \frac{\omega}{\beta} = \infty$$

Question 115: Transmission Lines (GATE EC 2012)

MCQ

1 Mark

A transmission line with a characteristic impedance of 100Ω is used to match a 50Ω section to a 200Ω section. If the matching is to be done both at 429 MHz and 1 GHz , the length of the transmission line can be approximately

- (A) 82.5 cm
- (B) 1.05 cm
- (C) **(Correct)** 1.58 cm
- (D) 1.75 cm

Correct Answer: (C)

Detailed Solution – Question 115

The 100Ω device used to match 50Ω and 200Ω devices is a quarter wave transformer. Length of matching device

$$= (2n + 1) \frac{\lambda}{4} = \text{odd multiple}$$

Where n -depends on frequency of $\lambda/4$

$$\lambda_1 \text{ of frequency } 429\text{MHz} = 0.7\text{m}$$

$$\lambda_2 \text{ of frequency } 1\text{GHz} = 0.3\text{m}$$

$$\text{Common multiple} = 2.1\text{m}$$

$$\text{Dividing by } 4 = 0.525\text{m (not available in options) Another odd multiple of } 0.525 \times 3 = 1.58\text{m}$$

Question 116: Antennas & Radiating Systems (GATE EC 2012)

MCQ

1 Mark

The radiation pattern of an antenna in spherical co-ordinates is given by $F(\theta) = \cos^4\theta; 0 \leq \theta \leq \pi/2$
The directivity of the antenna is

- (A) **(Correct)** 10 dB
 (B) 12.6 dB
 (C) 11.5 dB
 (D) 18 dB

Correct Answer: (A)

Detailed Solution – Question 116

$F(\theta) = \cos^4 \theta, 0 \leq \theta \leq \pi/2$ we know that directivity D is $D = \frac{4\pi u_{\max}}{\pi_{\text{rad}}}$... (1) $F(\theta)$ is nothing but radiation intensity $u(\theta, \phi)$ and W_{rad} is radiated power.

$$W_{\text{rad}} = \int_0^{2\pi} \int_0^{\pi/2} \cos^4 \theta \sin \theta d\theta d\phi \quad \dots (2)$$

above equation is written from the formula $W_{\text{rad}} = \iint_{\Omega} u(\theta, \phi) d\Omega$ where $d\Omega$ is solid angle and $d\Omega = \sin \theta d\theta d\phi$ So from (2)

$$W_{\text{rad}} = \int_0^{2\pi} \int_0^{\pi/2} \cos^4 \theta \sin \theta d\theta d\phi$$

$$W_{\text{rad}} = 2\pi \int_0^{\pi/2} \cos^4 \theta \sin \theta d\theta$$

$$W_{\text{rad}} = -2\pi \left[\frac{\cos^5 \theta}{5} \right]_0^{\pi/2}$$

$$W_{\text{rad}} = \frac{2\pi}{5} \quad \dots (3)$$

$$\text{So, } D = \frac{4\pi u_{\max}}{\left(\frac{2\pi}{5}\right)} = 10u_{\max} = 10$$

$$\text{In dB directivity} = 10 \log_{10} D = 10 \text{ dB}$$

Question 117: Transmission Lines (GATE EC 2012)

MCQ

1 Mark

A coaxial cable with an inner diameter of 1 mm and outer diameter of 2.4 mm is filled with a dielectric of relative permittivity 10.89. Given $\mu_0 = 4\pi \times 10^{-7} \text{ H/m}$, $\varepsilon_0 = \frac{10^{-9}}{36\pi} \text{ F/m}$, the characteristic impedance of the cable is

- (A) 330 Ω
 (B) 100 Ω
 (C) 143.3 Ω
 (D) **(Correct)** 15.3 Ω

Correct Answer: (D)

Detailed Solution – Question 117

Characteristics impedance of the co-axial cable is given by $Z_o = \frac{138}{\sqrt{\epsilon_r}} \log \frac{D}{d} = \frac{138}{\sqrt{10.89}} \log \left(\frac{2.4}{1} \right) = 15.89 \Omega$

Question 118: Uniform Plane Waves (GATE EC 2012)

MCQ

1 Mark

The electric field of a uniform plane electromagnetic wave in free space, along the positive x direction, is given by $\vec{E} = 10(\hat{a}_y + j\hat{a}_z)e^{-j25x}$. The frequency and polarization of the wave, respectively, are

- (A) **(Correct)** 1.2 GHz and left circular
- (B) 4 Hz and left circular
- (C) 1.2 GHz and right circular
- (D) 4 Hz and right circular

Correct Answer: (A)

Detailed Solution – Question 118

Out of phase by 90° and equal amplitude the wave is circularly polarized. Check the trace of E with the time which gives left circularly polarized.

$$a_y = \sin \omega t$$

$$a_z = \cos \omega t$$

Question 119: Uniform Plane Waves (GATE EC 2012)

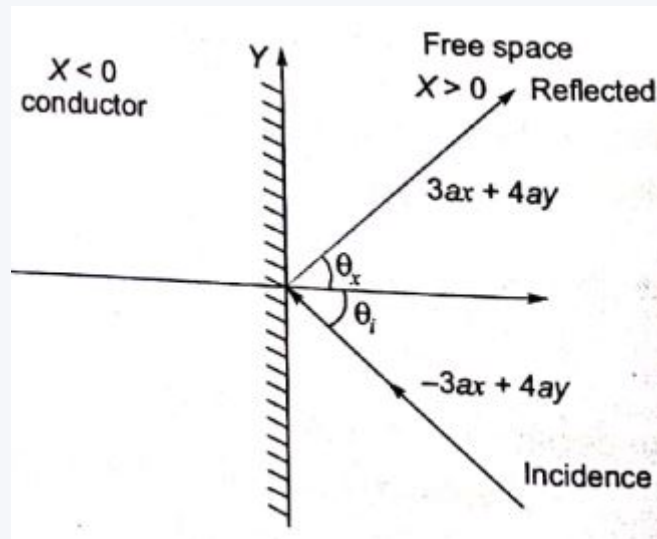
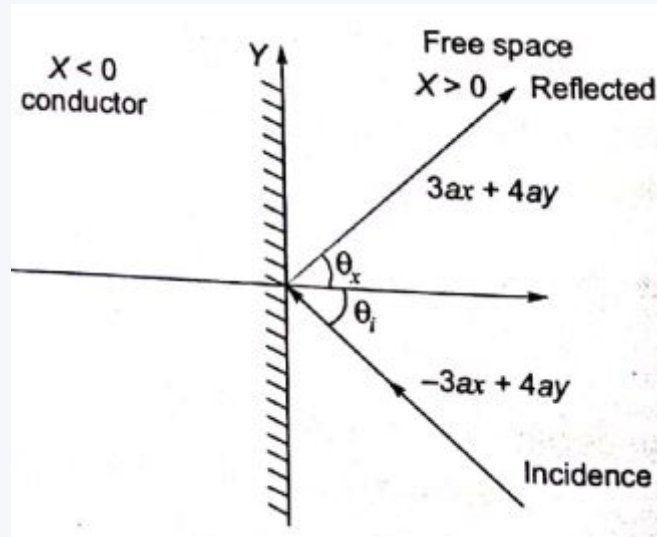
MCQ

1 Mark

A plane wave propagating in air with $\vec{E} = (8\hat{a}_x + 6\hat{a}_y + 5\hat{a}_z)e^{j(\omega t + 3x - 4y)}$ V/m is incident on a perfectly conducting slab positioned at $0 \leq x$. The E field of the reflected wave is

- (A) $(-8\hat{a}_x - 6\hat{a}_y - 5\hat{a}_z)e^{j(\omega t + 3x + 4y)}$ V/m
- (B) $(-8\hat{a}_x + 6\hat{a}_y - 5\hat{a}_z)e^{j(\omega t + 3x + 4y)}$ V/m
- (C) **(Correct)** $(-8\hat{a}_x - 6\hat{a}_y - 5\hat{a}_z)e^{j(\omega t - 3x - 4y)}$ V/m
- (D) $(-8\hat{a}_x + 6\hat{a}_y - 5\hat{a}_z)e^{j(\omega t - 3x - 4y)}$ V/m

Correct Answer: (C)



Incident wave propagation $\rightarrow -3\hat{a}_x + 4\hat{a}_y$ Reflected wave propagation $\rightarrow 3\hat{a}_x + 4\hat{a}_y$ E_i incident direction $\rightarrow 8\hat{a}_x + 6\hat{a}_y + 5\hat{a}_z$

$$\begin{aligned} E_i \text{ tangential} &\rightarrow 6\hat{a}_y + 5\hat{a}_z \\ E_r \text{ tangential} &\rightarrow -6\hat{a}_y - 5\hat{a}_z \end{aligned} \quad E_{\text{tang}} = 0$$

$$\begin{aligned} E_i \text{ normal} &= 8\hat{a}_x \\ E_r \text{ normal} &= 8\hat{a}_x \end{aligned} \quad E_{\text{normal}} = E_{\text{max}}$$

$$E_r \text{ reflected direction} = 8\hat{a}_x - 6\hat{a}_y - 5\hat{a}_z$$

Question 120: Uniform Plane Waves (GATE EC 2011)

MCQ

1 Mark

The electric and magnetic fields for a TEM wave of frequency 14 GHz in a homogeneous medium of relative permittivity ϵ_r and relative permeability $\mu_r=1$ are given by $\vec{E} = E_p e^{j(\omega t - 280\pi y)} \hat{u}_z \text{ V/m}$ $\vec{H} = 3e^{j(\omega t - 280\pi y)} \hat{u}_x \text{ V/m}$

Assuming the speed of light in free space to be $3 \times 10^8 \text{ m/s}$, the intrinsic impedance of free space

to be 120π , the relative permittivity ϵ_r of the medium and the electric field amplitude E_p are

- (A) $\epsilon_r = 3, E_p = 120\pi$
- (B) $\epsilon_r = 3, E_p = 360\pi$
- (C) $\epsilon_r = 9, E_p = 360\pi$
- (D) **(Correct)** $\epsilon_r = 9, E_p = 120\pi$

Correct Answer: (D)

Detailed Solution – Question 120

$$\begin{aligned}\text{Given: } \vec{E} &= E_p e^{i(\omega t - 280\pi y)} \hat{u}_z \text{ V/m} \\ \vec{H} &= 3e^{i(\omega t - 280\pi y)} \hat{u}_x \text{ A/m} \\ v &= 3 \times 10^8 \text{ m/s}\end{aligned}$$

From given expression we conclude that,

$$\begin{aligned}\beta &= 280\pi = \frac{2\pi}{\lambda} \\ \therefore \lambda &= \frac{1}{140} \text{ meter} \\ v &= f\lambda = 14 \times 10^9 \times \frac{1}{140} \text{ m/se} \\ v &= 1 \times 10^8 \text{ m/sec} \\ \text{but, } v &= \frac{c}{\sqrt{\epsilon_r \mu_r}} \\ \therefore 1 \times 10^8 &= \frac{3 \times 10^8}{\sqrt{1 \times \epsilon_r}} \\ \therefore \epsilon_r &= 9 \\ \frac{E_P}{H_P} &= \eta = \sqrt{\frac{\mu}{\epsilon}} = \sqrt{\frac{\mu_0 \times 1}{\epsilon_0 \times 9}} = \frac{1}{3} \sqrt{\frac{\mu_0}{\epsilon_0}} \\ \Rightarrow \frac{E_P}{3} &= \frac{1}{3} \times 120\pi \\ \therefore E_P &= 120\pi\end{aligned}$$

Question 121: Transmission Lines (GATE EC 2011)

MCQ

1 Mark

A transmission line of characteristic impedance 50Ω is terminated in a load impedance Z_L . The VSWR of the line is measured as 5 and the first of the voltage maxima in the line is observed at a distance of $\frac{\lambda}{4}$ from the load. The value of Z_L is

- (A) **(Correct)** 10Ω
- (B) 250Ω
- (C) $(19.23 + j46.15) \Omega$
- (D) $(19.23 - j 46.15) \Omega$

Correct Answer: (A)

Detailed Solution – Question 121

$VSWR = 5 = \frac{1+|\Gamma|}{1-|\Gamma|} \therefore |\Gamma| = \frac{2}{3} \Rightarrow \Gamma = \pm \frac{2}{3}$ If maximum is at $\lambda/4$ from load then minimum will be at load itself. For the phase of Γ $2\beta x_{\max} = 2n\pi + \theta$ where $x_{\max}$ is distance of $V_{\max}$ from load.

$$2 \cdot \frac{2\pi}{\lambda} \cdot \frac{\lambda}{4} = \theta$$

$$\Gamma = -\frac{2}{3}$$

$$\text{Now, } -\frac{2}{3} = \frac{Z_L - Z_0}{Z_L + Z_0} = \frac{Z_L - 50}{Z_L + 50}$$

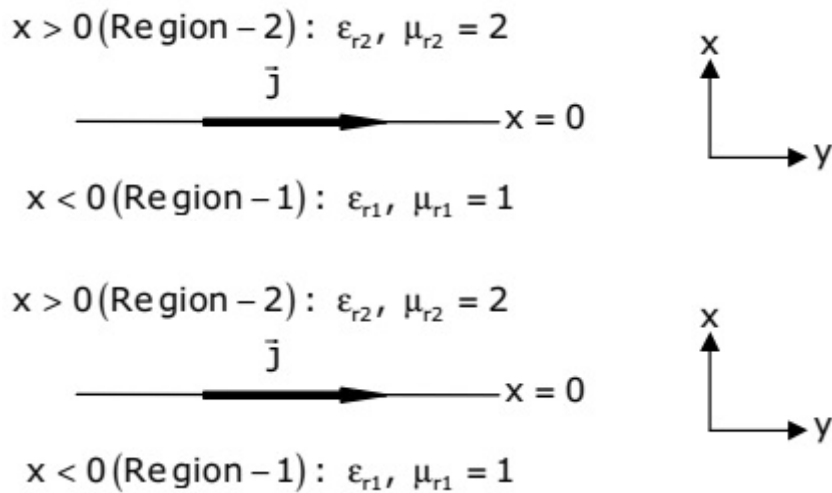
$$\Rightarrow Z_L = 10\Omega$$

Question 122: Basics of Electromagnetics & Maxwell's Eqns (GATE EC 2011)

MCQ

1 Mark

A current sheet $\vec{J} = 10\hat{u}_y$ A/m lies on the dielectric interface $x = 0$ between two dielectric media with $\epsilon_{r1}=5$, $\mu_{r1}=1$ in Region-1 ($x < 0$) and $\epsilon_{r2}=5$, $\mu_{r2}=2$ in Region-2 ($x > 0$). If the magnetic field in Region-1 at $x = 0^-$ is $\vec{H}_1 = 3\hat{u}_x + 30\hat{u}_y$ A/m. The magnetic field in Region-2 at $x = 0^+$ is



- (A) **(Correct)** $\vec{H}_2 = 1.5\hat{u}_x + 30\hat{u}_y - 10\hat{u}_z$ A/m
- (B) $\vec{H}_2 = 3\hat{u}_x + 30\hat{u}_y - 10\hat{u}_z$ A/m
- (C) $\vec{H}_2 = 1.5\hat{u}_x + 40\hat{u}_y$ A/m
- (D) $\vec{H}_2 = 3\hat{u}_x + 30\hat{u}_y + 10\hat{u}_z$ A/m

Correct Answer: (A)

Detailed Solution – Question 122

For magnetic field boundary relations are

$$\begin{aligned}\vec{B}_{n1} &= \vec{B}_{n2} \\ \text{and } \vec{H}_{t1} - \vec{H}_{t2} &= -\vec{J}_s \times \vec{a}_n \\ \Rightarrow B_{x1} &= B_{x2} \quad (\text{x is normal component}) \\ \mu_1 H_{x1} &= \mu_2 H_{x2} \\ \Rightarrow 1 \times 3 &= 2 \times H_{x2} \\ H_{x2} &= 1.5 \\ \Rightarrow \vec{H}_{11} - \vec{H}_{t2} &= -10\hat{u}_y \times \hat{u}_x = +10\hat{u}_z \\ \vec{H}_{t2} &= H_{t1} - 10\hat{u}_z \\ \vec{H}_{t2} &= 30\hat{u}_y - 10\hat{u}_z \\ \Rightarrow \vec{H}_2 &= 1.5\hat{u}_x + 30\hat{u}_y - 10\hat{u}_z \text{ A/m}\end{aligned}$$

Question 123: Waveguides & Optical Fibers (GATE EC 2011)

MCQ

1 Mark

The modes in a rectangular waveguide are denoted by $\frac{TE_{mn}}{TM_{mn}}$ where m and n are the eigen numbers along the larger and smaller dimensions of the waveguide respectively. Which one of the following statements is TRUE?

- (A) **(Correct)** The TM_{10} mode of the wave does not exist
- (B) The TE_{10} mode of the wave does not exist
- (C) The TM_{10} and TE_{10} the modes both exist and have the same cut-off frequencies
- (D) The TM_{10} and TM_{01} modes both exist and have the same cut-off frequencies

Correct Answer: (A)

Detailed Solution – Question 123

In case of rectangular wave guide TE_{rm} exists for all values of m and n except m=0 and n=0 . For TM_{mn} to exist both values of m and n must be non zero

Question 124: Transmission Lines (GATE EC 2011)

MCQ

1 Mark

A transmission line of characteristic impedance 50 Ω is terminated by a 50 Ω load. When excited by a sinusoidal voltage source at 10 GHz, the phase difference between two points spaced 2 mm apart on the line is found to be $\frac{\pi}{4}$ radians. The phase velocity of the wave along the line is

- (A) $0.8 \times 10^8 \text{ m/s}$
- (B) $1.2 \times 10^8 \text{ m/s}$

(C) **(Correct)** $1.6 \times 10^8 \text{ m/s}$

(D) $3 \times 10^8 \text{ m/s}$

Correct Answer: (C)

Detailed Solution – Question 124

$$\text{Phase difference} = \frac{2\pi}{\lambda} (\text{path difference})$$

$$\Rightarrow \frac{\pi}{4} = \frac{2\pi}{\lambda} (2 \times 10^{-3})$$

$$\therefore \lambda = 8 \times 2 \times 10^{-3}$$

$$= 16 \times 10^{-3} \text{ m}$$

$$f = 10 \text{ GHz} = 10 \times 10^9 \text{ Hz}$$

Hence the phase velocity of wave along the line is,

$$v = f\lambda = 10 \times 10^9 \times 16 \times 10^{-3} \text{ m/sec}$$

$$\therefore v = 1.6 \times 10^8 \text{ m/s}$$

Question 125: Antennas & Radiating Systems (GATE EC 2011)

MCQ

1 Mark

Consider the following statements regarding the complex Poynting vector $\vec{P}$ for the power radiated by a point source in an infinite homogeneous and lossless medium. $\text{Re}(\vec{P})$ denotes the real part of $\vec{P}$, S denotes a spherical surface whose centre is at the point source, and $\hat{n}$ denotes the unit surface normal on S . Which of the following statements is TRUE?

(A) $\text{Re}(\vec{P})$ remains constant at any radial distance from the source

(B) $\text{Re}(\vec{P})$ increases with increasing radial distance from the source

(C) **(Correct)** $\oint_S \text{Re}(\vec{P}) \cdot \hat{n} dS$ remains constant at any radial distance from the source

(D) $\oint_S \text{Re}(\vec{P}) \cdot \hat{n} dS$ decreases with increasing radial distance from the source

Correct Answer: (C)

Detailed Solution – Question 125

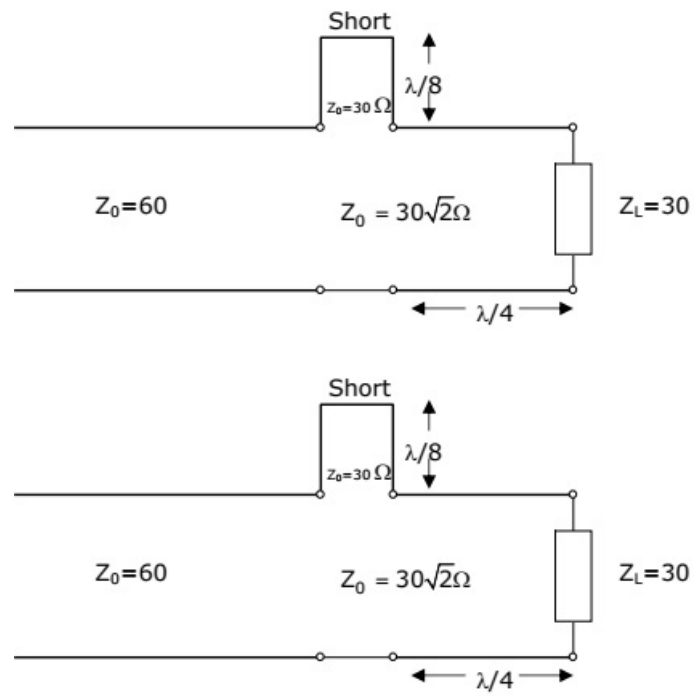
Power density of any point source decreases with distance i.e. the density decreases and area of cross-over increases with the product being constant.

Question 126: Transmission Lines (GATE EC 2010)

MCQ

1 Mark

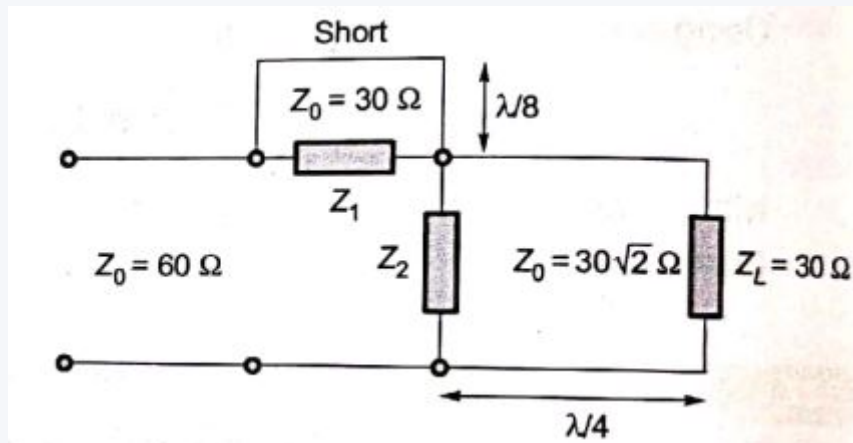
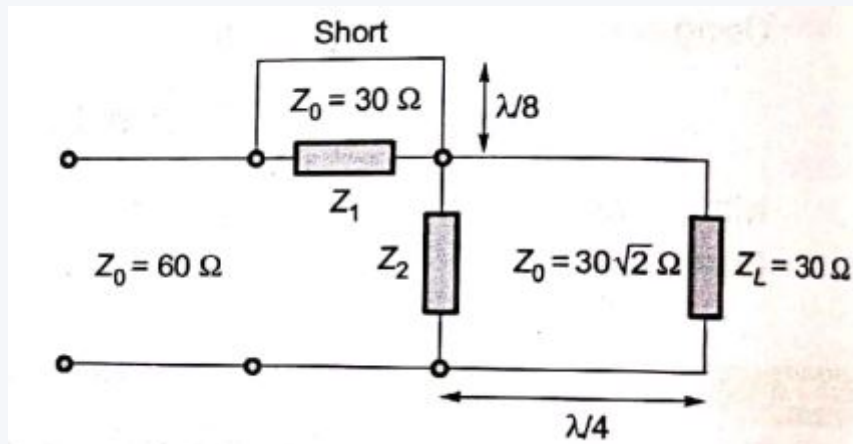
In the circuit shown, all the transmission line sections are lossless. The Voltage Standing Wave Ratio (VSWR) on the 60Ω line is



- (A) 1.00
- (B) **(Correct)** 1.64
- (C) 2.50
- (D) 3.00

Correct Answer: (B)

$$Z_{in} = Z_0 \left(\frac{Z_L + jZ_0 \tan \beta l}{Z_0 + jZ_L \tan \beta l} \right)$$



Input impedance,

$$\begin{aligned} Z_1 &= 30 \left[\frac{0 + j30 \tan \left(\frac{2\pi}{\lambda} \right) \left(\frac{\lambda}{8} \right)}{30 + 0} \right] \\ \Rightarrow Z_1 &= j30 \\ Z_2 &= 30\sqrt{2} \left[\frac{30 + j30\sqrt{2} \tan \left(\frac{2\pi}{\lambda} \right) \left(\frac{\lambda}{4} \right)}{30\sqrt{2} + j30 \tan \left(\frac{2\pi}{\lambda} \right) \left(\frac{\lambda}{4} \right)} \right] \\ &= 30\sqrt{2} \left[\frac{\frac{30}{\tan \pi/2} + j30\sqrt{2}}{\frac{30\sqrt{2}}{\tan \pi/2} + j30} \right] \\ &= 60\Omega \\ \text{or, } Z_2 &= \frac{(30\sqrt{2})^2}{30} = 60\Omega \end{aligned}$$

Load impedance, $Z_L = Z_1 + Z_2 = j30 + 60$ Magnitude of reflection coefficient,

$$\begin{aligned} |\Gamma| &= \left| \frac{Z_L - Z_0}{Z_L + Z_0} \right| = \left| \frac{60 + j30 - 60}{60 + j30 + 60} \right| \\ &= \left| \frac{j30}{120 + j30} \right| = \left| \frac{j1}{4 + j} \right| \\ &= \frac{1}{\sqrt{16 + 1}} = \frac{1}{\sqrt{17}} \end{aligned}$$

VSWR on 60 Ω line,

$$\text{VSWR} = \frac{1 + |\Gamma|}{1 - |\Gamma|} = \frac{1 + \frac{1}{\sqrt{17}}}{1 - \frac{1}{\sqrt{17}}} = 1.24$$

Question 127: Uniform Plane Waves (GATE EC 2010)

MCQ

1 Mark

A plane wave having the electric field components $\vec{E}_i = 24 \cos(3 \times 10^8 t - \beta y) \hat{a}_z \text{ V/m}$ and traveling in free space is incident normally on a lossless medium with $\mu = \mu_0$ and $\epsilon = 9\epsilon_0$ which occupies the region $y \geq 0$. The reflected magnetic field component is given by

- (A) **(Correct)** $\frac{1}{10\pi} \cos(3 \times 10^8 t + y) \hat{a}_x \text{ A/m}$
- (B) $\frac{1}{20\pi} \cos(3 \times 10^8 t + y) \hat{a}_x \text{ A/m}$
- (C) $-\frac{1}{20\pi} \cos(3 \times 10^8 t + y) \hat{a}_x \text{ A/m}$
- (D) $-\frac{1}{10\pi} \cos(3 \times 10^8 t + y) \hat{a}_x \text{ A/m}$

Correct Answer: (A)

Detailed Solution – Question 127

$\vec{E}_i = 24 \cos(3 \times 10^8 t - \beta y) \hat{a}_z \text{ V/m}$ The wave is travelling in +y direction.

$$\begin{aligned} H_i &= \frac{1}{\eta} (\hat{a}_y \times \vec{E}_i) \\ &= \frac{24 \cos}{120\pi} (3 \times 10^8 t - \beta y) \hat{a}_x \\ &= \frac{1}{5\pi} \cos(3 \times 10^8 t - \beta y) \hat{a}_x \\ \frac{H_r}{H_i} &= \frac{\eta_1 - \eta_2}{\eta_1 + \eta_2} = \frac{\sqrt{\frac{\mu_0}{\epsilon_1}} - \sqrt{\frac{\mu_0}{\epsilon_2}}}{\sqrt{\frac{\mu_0}{\epsilon_1}} + \sqrt{\frac{\mu_0}{\epsilon_2}}} \\ &= \frac{\sqrt{\epsilon_2} - \sqrt{\epsilon_1}}{\sqrt{\epsilon_2} + \sqrt{\epsilon_1}} = \frac{\sqrt{9} - \sqrt{1}}{\sqrt{9} + \sqrt{1}} = \frac{3 - 1}{3 + 1} = \frac{2}{4} = \frac{1}{2} \end{aligned}$$

Therefore, reflected magnetic field component,

$$\vec{H}_r = \frac{1}{10\pi} \cos(3 \times 10^8 t + \beta y) \hat{a}_x$$

Due to change in propagation direction to -y

Question 128: Uniform Plane Waves (GATE EC 2010)

MCQ

1 Mark

The electric field component of a time harmonic plane EM wave traveling in a nonmagnetic lossless dielectric medium has an amplitude of 1 V/m. If the relative permittivity of the medium is 4, the magnitude of the time-average power density vector (in W/m²) is

- (A) $\frac{1}{30\pi}$
- (B) $\frac{1}{60\pi}$
- (C) **(Correct)** $\frac{1}{120\pi}$

(D) $\frac{1}{240\pi}$

Correct Answer: (C)

Detailed Solution – Question 128

Magnitude of the time-average power density vector

$$P_{av} = \frac{E^2}{2\eta}$$

where, $\eta = \sqrt{\mu/\epsilon}$

$$\eta = \sqrt{\frac{\mu_0}{4\epsilon_0}} = \frac{\eta_0}{2} = \frac{120\pi}{2}$$
$$P_{av} = \frac{1^2}{2\left(\frac{120\pi}{2}\right)} = \frac{1}{120\pi} \text{ W/m}^2$$

Question 129: Transmission Lines (GATE EC 2010)

MCQ

1 Mark

A transmission line has a characteristic impedance of 50Ω and a resistance of $0.1\Omega/\text{m}$. If the line is distortion less, the attenuation constant(in Np/m) is

- (A) 500
(B) 5
(C) 0.014
(D) **(Correct)** 0.002

Correct Answer: (D)

Detailed Solution – Question 129

For distortionless transmission line,

$$LG = RC$$
$$\Rightarrow \frac{L}{C} = \frac{R}{G}$$

Characteristic impedance, $Z_0 = \sqrt{\frac{L}{C}} = \sqrt{\frac{R}{G}}$ Attenuation constant,

$$\alpha = \sqrt{RG} = \sqrt{R} \cdot \frac{\sqrt{R}}{Z_0} = \frac{R}{Z_0} = \frac{0.1}{50}$$
$$= 0.002 \text{ Np/m}$$

Question 130: Miscellaneous Topics (GATE EC 2010)

MCQ

1 Mark

If the scattering matrix $[S]$ of a two port network is

$$|S| = \begin{bmatrix} 0.2\angle 0^\circ & 0.9\angle 90^\circ \\ 0.9\angle 90^\circ & 0.1\angle 90^\circ \end{bmatrix}$$

then the network is

- (A) lossless and reciprocal
- (B) lossless but not reciprocal
- (C) **(Correct)** not lossless but reciprocal
- (D) neither lossless nor reciprocal

Correct Answer: (C)

Detailed Solution – Question 130

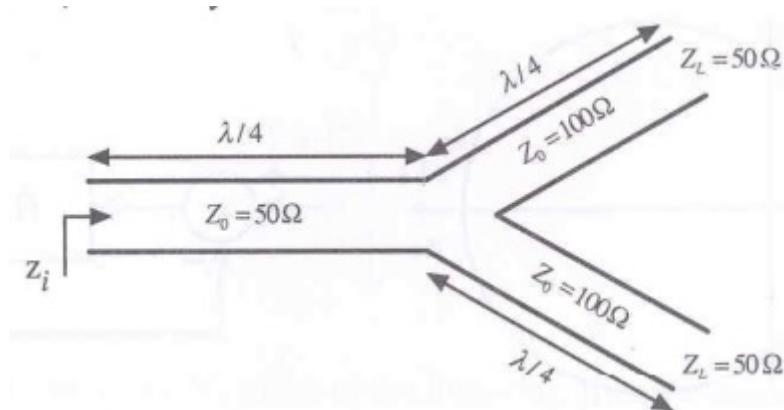
For reciprocal networks, $S_{12} = S_{21}$ For symmetrical networks, $S_{11} = S_{22}$ For antisymmetrical networks, $S_{11} = -S_{22}$ For lossless reciprocal networks, $|S_{11}| = |S_{22}|$ and $|S_{11}|^2 + |S_{12}|^2 = 1$

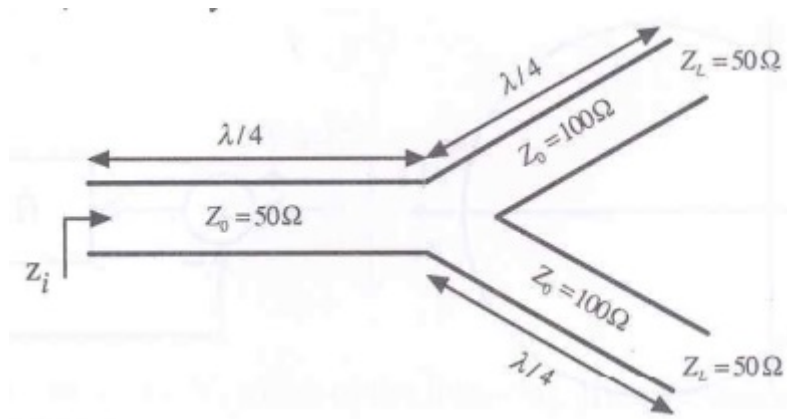
Question 131: Transmission Lines (GATE EC 2009)

MCQ

1 Mark

A transmission line terminates in two branches, each of length $\frac{\lambda}{4}$, as shown. The branches are terminated by 50Ω loads. The lines are lossless and have the characteristic impedances shown. Determine the impedance Z_i as seen by the source





- (A) 200Ω
- (B) 100Ω
- (C) 50Ω
- (D) **(Correct)** 25Ω

Correct Answer: (D)

Detailed Solution – Question 131

$$\begin{aligned}
 z_1 &= \frac{Z_0^2}{Z_L} = \frac{(100)^2}{50} = 200\Omega \\
 Z_2 &= \frac{Z_0^2}{Z_1} = \frac{(100)^2}{200} = 50\Omega \\
 Z_L &= Z_1 \parallel Z_2 = 100\Omega \\
 Z_i &= \frac{Z_0^2}{Z_L} = \frac{(50)^2}{100} = 25\Omega
 \end{aligned}$$

Question 132: Basics of Electromagnetics & Maxwell's Eqns (GATE EC 2009)

MCQ

1 Mark

A magnetic field in air is measured to be $\vec{B} = B_0(\frac{x}{x^2+y^2}\hat{y} - \frac{y}{x^2+y^2}\hat{x})$ What current distribution leads to this field ? [Hint : The algebra is trivial in cylindrical coordinates.]

- (A) $\vec{J} = \frac{B_0\hat{z}}{\mu_0}(\frac{1}{x^2+y^2}), r \neq 0$
- (B) $\vec{J} = \frac{B_0\hat{z}}{\mu_0}(\frac{2}{x^2+y^2}), r \neq 0$
- (C) **(Correct)** $\vec{J} = 0, r \neq 0$
- (D) $\vec{J} = \frac{B_0\hat{z}}{\mu_0}(\frac{1}{x^2+y^2}), r \neq 0$

Correct Answer: (C)

Detailed Solution – Question 132

Applying $\nabla \cdot H = J$

$$\nabla \times B = \begin{vmatrix} \hat{a}_x & \hat{a}_y & \hat{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{-y}{x^2+y^2} & \frac{x}{x^2+y^2} & 0 \end{vmatrix} = 0$$
$$J = 0$$

Question 133: Basics of Electromagnetics & Maxwell's Eqns (GATE EC 2009)

MCQ

1 Mark

If a vector field $\vec{V}$ is related to another vector field $\vec{A}$ through $\vec{V} = \nabla * \vec{A}$, which of the following is true? (Note : C and S_C refer to any closed contour and any surface whose boundary is C.)

(A)

$$\oint_C \vec{V} \cdot d\vec{l} = \int_{S_C} \vec{A} \cdot d\vec{S}$$

(B) (Correct)

$$\oint_C \vec{A} \cdot d\vec{l} = \int_{S_C} \vec{V} \cdot d\vec{S}$$

(C)

$$\oint_C \nabla \times \vec{V} \cdot d\vec{l} = \int_{S_C} \nabla \times \vec{A} \cdot d\vec{S}$$

(D)

$$\oint_C \nabla \times \vec{A} \cdot d\vec{l} = \int_{S_C} \vec{V} \cdot d\vec{S}$$

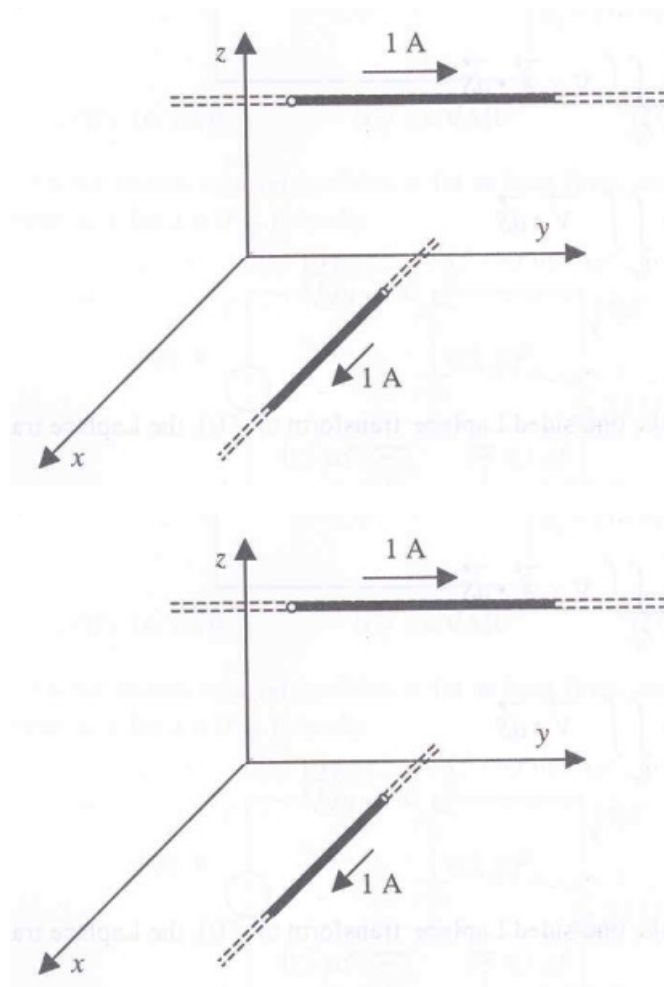
Correct Answer: (B)

Question 134: Basics of Electromagnetics & Maxwell's Eqns (GATE EC 2009)

MCQ

1 Mark

Two infinitely long wires carrying current are as shown in the figure below. One wire is in the y-z plane and parallel to the y-axis. The other wire is in the x-y plane and parallel to the x-axis. Which components of the resulting magnetic field are non-zero at the origin ?



- (A) x,y, z components
- (B) x,y components
- (C) y,z components
- (D) **(Correct)** x,z components

Correct Answer: (D)

Detailed Solution – Question 134

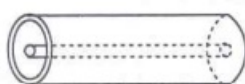
$H_{direction}$ due to any current wire $= I \text{ flow direction} \times (\text{cross}) \text{ radial vector from current to point of consideration}$.

Question 135: Waveguides & Optical Fibers (GATE EC 2009)

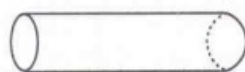
MCQ

1 Mark

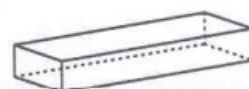
Which of the following statements is true regarding the fundamental mode of the metallic waveguides shown ?



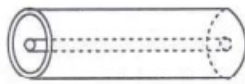
P: Coaxial



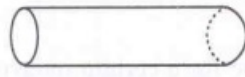
Q: Cylindrical



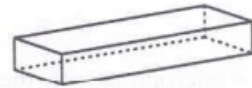
R: Rectangular



P: Coaxial



Q: Cylindrical



R: Rectangular

- (A) **(Correct)** Only P has no cutoff-frequency
- (B) Only Q has no cutoff-frequency
- (C) Only R has no cutoff-frequency
- (D) All three have cutoff-frequencies

Correct Answer: (A)

Detailed Solution – Question 135

P is coaxial line and support TEM wave $\therefore$ P has no cutoff frequency

Q and R are wave-guides and cutoff frequency each depends upon their dimensions.

Question 136: Antennas & Radiating Systems (GATE EC 2008)

MCQ

1 Mark

At 20 GHz, the gain of a parabolic dish antenna of 1 meter and 70

- (A) 15 dB
- (B) 25 dB
- (C) 35 dB
- (D) **(Correct)** 45 dB

Correct Answer: (D)

Detailed Solution – Question 136

$$\text{Gain} = \frac{4\pi}{\lambda^2} A_e = \frac{4\pi}{\lambda^2} \pi R^2$$

Along with illumination or efficiency

$$\begin{aligned} \text{Gain} &= \eta \pi^2 \frac{(2R)^2}{\lambda^2} = 7 \left(\frac{D}{\lambda} \right)^2 \\ &= 0.7 \pi^2 \left(\frac{1 \times 20 \times 10^9}{3 \times 10^8} \right)^2 \\ &= 30705.4359 \\ G(\text{dB}) &= 10 \log_{10} G \\ &= 10 \log_{10}(30705.4359) = 45\text{dB} \end{aligned}$$

Question 137: miscellaneous Topics (GATE EC 2008)

MCQ

1 Mark

In the design of a single mode step index optical fibre close to upper cut-off, the single-mode operation is not preserved if

- (A) radius as well as operating wavelength are halved
- (B) radius as well as operating wavelength are doubled
- (C) **(Correct)** radius is halved and operating wavelength is doubled
- (D) radius is doubled and operating wavelength is halved

Correct Answer: (C)

Question 138: Uniform Plane Waves (GATE EC 2008)

MCQ

1 Mark

A uniform plane wave in the free space is normally incident on an infinitely thick dielectric slab (dielectric constant $\epsilon_r=9$). The magnitude of the reflection coefficient is

- (A) 0
- (B) 0.3
- (C) **(Correct)** 0.5
- (D) 0.8

Correct Answer: (C)

Detailed Solution – Question 138

Dielectric constant of free space, $\epsilon_{r_1} = 1$ Dielectric constant of dielectric slab, $\epsilon_{r_2} = 9$ Magnitude of reflection coefficient

$$|\Gamma| = \left| \frac{\sqrt{\epsilon_{r_1}} - \sqrt{\epsilon_{r_2}}}{\sqrt{\epsilon_{r_1}} + \sqrt{\epsilon_{r_2}}} \right| = \left| \frac{1 - 3}{1 + 3} \right| = \frac{2}{4} = 0.5$$

Question 139: Transmission Lines (GATE EC 2008)

MCQ

1 Mark

One end of a loss-less transmission line having the characteristic impedance of 75Ω and length of 1 cm is short-circuited. At 3 GHz, the input impedance at the other end of transmission line is

- (A) 0
- (B) Resistive
- (C) Capacitive
- (D) **(Correct)** Inductive

Correct Answer: (D)

Detailed Solution – Question 139

It is a short circuit line with $l = 1\text{cm}$. As $\lambda = \frac{3 \times 10^8}{3 \times 10^9} = 0.3 = 30\text{cm}$ $l < \frac{\lambda}{4}$ The line has inductive nature. As $Z_{SC} = jZ_o \tan \beta l$

Question 140: Waveguides & Optical Fibers (GATE EC 2008)

MCQ

1 Mark

A rectangular waveguide of internal dimensions ($a=4\text{cm}$ and $b=3\text{cm}$) is to be operated in TE_{11} mode. The minimum operating frequency is

- (A) **(Correct)** 6.25 GHz
- (B) 6.0 GHz
- (C) 5.0 GHz
- (D) 3.75 GHz

Correct Answer: (A)

Detailed Solution – Question 140

Minimum operating frequency,

$$f_0 = \frac{c}{2\pi} \sqrt{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2}$$

For TE_{11} mode

$$f_0 = \frac{3 \times 10^8}{2} \sqrt{\left(\frac{1}{0.04}\right)^2 + \left(\frac{1}{0.03}\right)^2}$$
$$= \frac{3 \times 10^8}{2} \times \frac{0.05}{0.04 \times 0.03} = 6.25\text{GHz}$$

Question 141: Basics of Electromagnetics & Maxwell's Eqns (GATE EC 2008)

MCQ

1 Mark

For static electric and magnetic fields in an inhomogeneous source-free medium, which of the following represents the correct form of Maxwell's equations ?

- (A) $\nabla \cdot E = 0, \nabla \times B = 0$
- (B) $\nabla \cdot E = 0, \nabla \cdot B = 0$
- (C) $\nabla \times E = 0, \nabla \times B = 0$
- (D) **(Correct)** $\nabla \times E = 0, \nabla \cdot B = 0$

Correct Answer: (D)

Detailed Solution – Question 141

Intensity - curl - line integral Density divergence - surface integral This is order of understanding Maxwell's equations. For any material

$$\begin{aligned}\nabla \times E &= 0 & \nabla \cdot B &= 0 \\ \nabla \times H &= J & \nabla \cdot D &= \rho_v\end{aligned}$$

For source free like charge or current free conditions $\nabla \times E = 0$ $\nabla \cdot B = 0$ $\nabla \times H = 0$ $\nabla \cdot D = 0$
If $D = \epsilon E$ and $B = \mu H$ and ϵ or μ are constant with distance or direction then

$$\begin{aligned}\nabla \cdot (\epsilon E) &= 0 & \nabla \times \frac{B}{\mu} &= 0 \\ \nabla \cdot E &= 0 & \nabla \times B &= 0\end{aligned}$$

These are called as homogeneous/isotropic medium. For non-homogeneous material. However they are not valid.

Question 142: Antennas & Radiating Systems (GATE EC 2008)

MCQ

1 Mark

For a Hertz dipole antenna, the half power beam width (HPBW) in the E-plane is

- (A) 360°
- (B) 180°
- (C) **(Correct)** 90°
- (D) 45°

Correct Answer: (C)

Detailed Solution – Question 142

E-plane mean θ -plane or vertical plane $E \propto \sin \theta$ in Hertzian dipole

$$\begin{aligned}E_{\max} &\text{ when } \theta = 90^\circ \\ E_{\min} &\text{ when } \theta = 0\end{aligned}$$

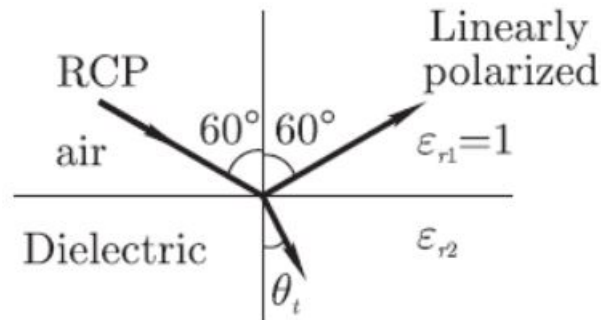
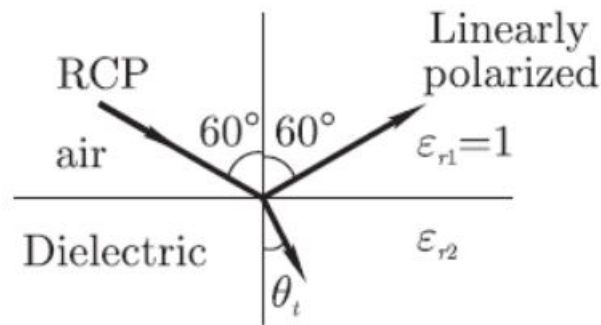
E has half power of maximum when $\theta = 45^\circ$ and 180° HPBW in θ side $= 135^\circ - 45^\circ = 90^\circ$

Question 143: Uniform Plane Waves (GATE EC 2007)

MCQ

1 Mark

A right circularly polarized (RCP) plane wave is incident at an angle 60° to the normal, on an air-dielectric interface. If the reflected wave is linearly polarized, the relative dielectric constant ϵ_{r2} is.



- (A) $\sqrt{2}$
- (B) $\sqrt{3}$
- (C) 2
- (D) **(Correct)** 3

Correct Answer: (D)

Detailed Solution – Question 143

The reflected wave is linearly polarized when circularly polarized wave is incident under Brewster's angle of incidence.

$$\tan \theta_B = \sqrt{\frac{\epsilon_{r2}}{\epsilon_{r1}}} = \tan 60^\circ$$

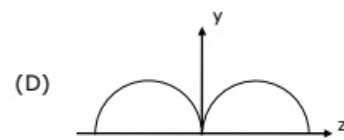
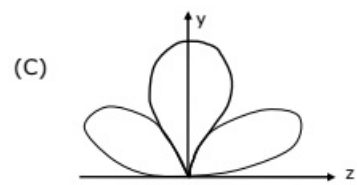
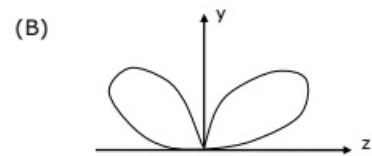
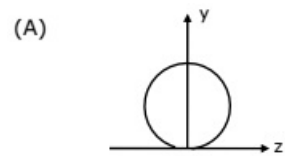
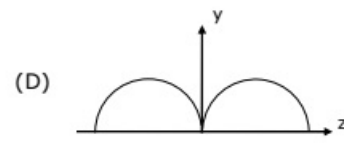
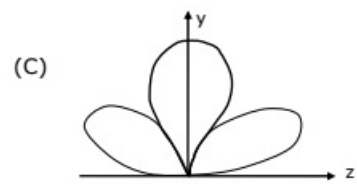
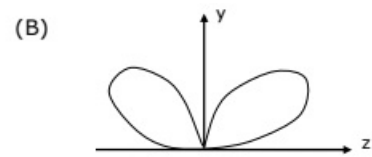
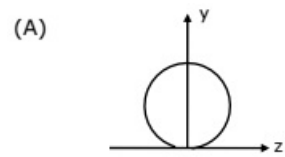
$$\epsilon_{r2} = 3$$

Question 144: Antennas & Radiating Systems (GATE EC 2007)

MCQ

1 Mark

A $\frac{\lambda}{2}$ dipole is kept horizontally at a height of $\frac{\lambda_0}{2}$ above a perfectly conducting infinite ground plane. The radiation pattern in the plane of the dipole ($\vec{E}$ plane) looks approximately as



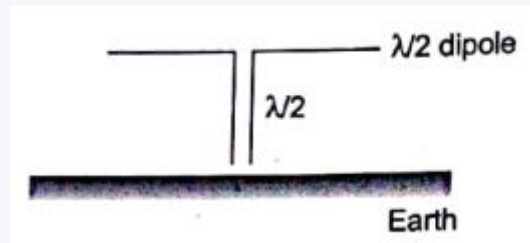
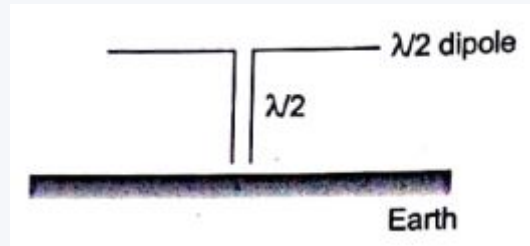
(A) A

(B) (Correct) B

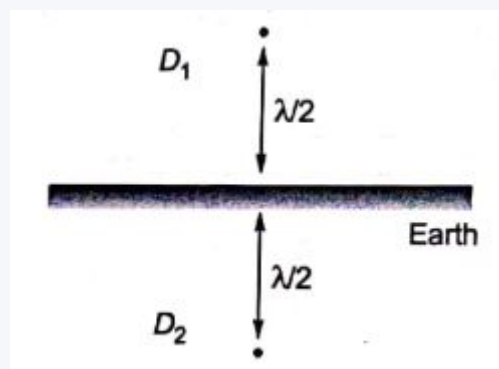
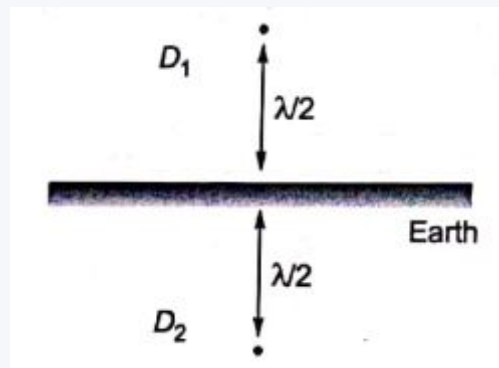
(C) C

(D) D

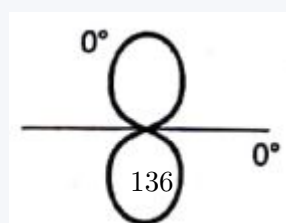
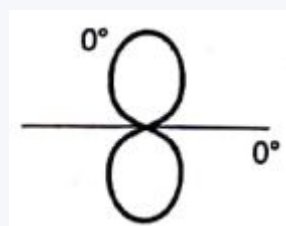
Correct Answer: (B)



A horizontally placed dipole has an image in the earth and interference with the image formed gives the final pattern of radiation



Dipole pattern

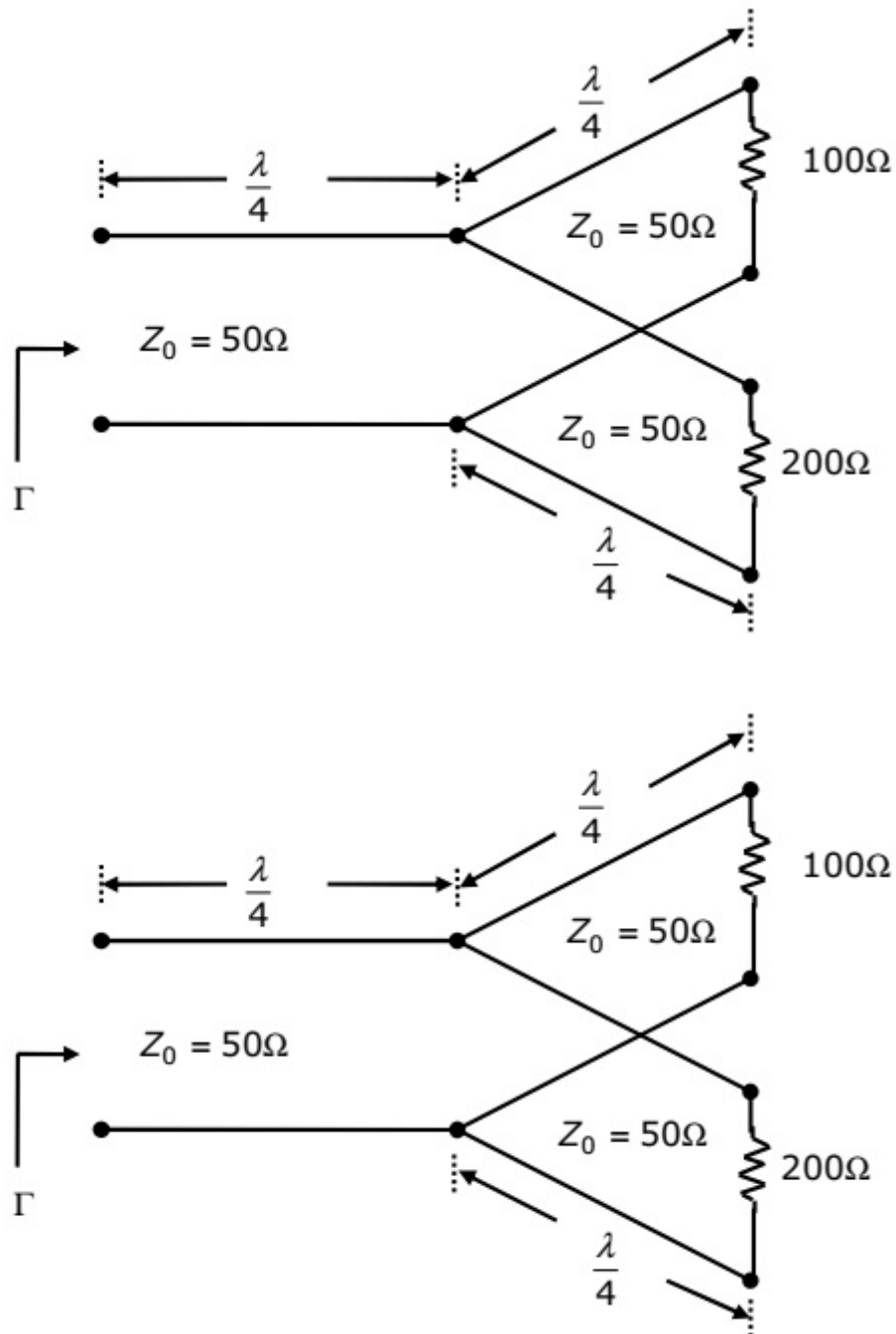


Question 145: Transmission Lines (GATE EC 2007)

MCQ

1 Mark

The parallel branches of a 2-wire transmission line are terminated in 100Ω and 200Ω resistors as shown in the figure. The characteristic impedance of the line is $Z_0 = 50\Omega$ and each section has a length of $\frac{\lambda}{4}$. The voltage reflection coefficient Γ at the input is



(A) $-j\frac{7}{5}$

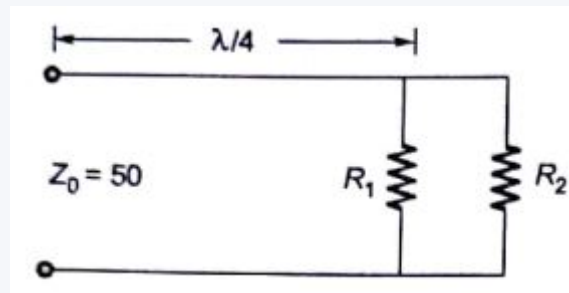
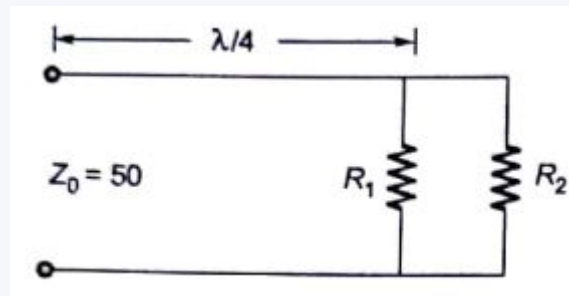
(B) $\frac{-5}{7}$

(C) $j\frac{5}{7}$

(D) (Correct) $\frac{5}{7}$

Correct Answer: (D)

Detailed Solution – Question 145



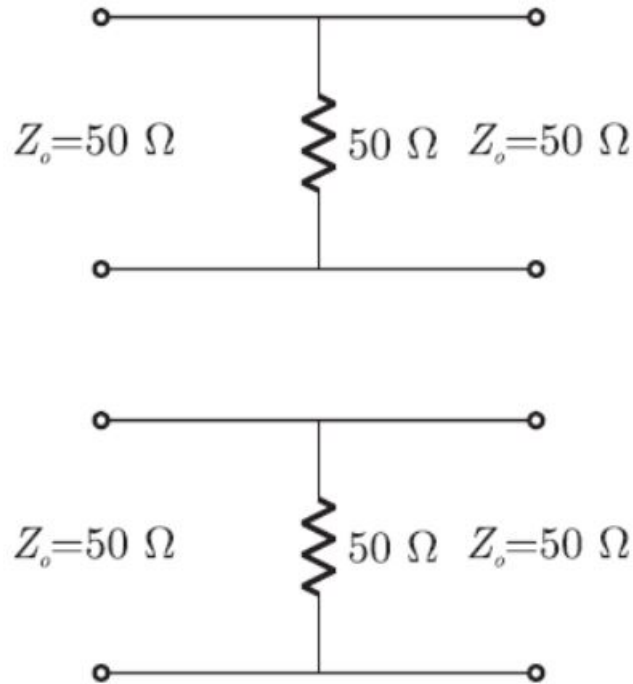
$$\begin{aligned} Z_{\text{in}} &= \frac{Z_0^2}{Z_L}; \text{ if } l = \frac{\lambda}{4} \\ R_1 \text{ due to } 100\Omega &= \frac{50^2}{100} = 25 \\ R_2 \text{ due to } 200\Omega &= \frac{50^2}{200} = \frac{25}{2} \\ \Rightarrow R_1 \parallel R_2 &= 25 \parallel \frac{25}{2} = \frac{25}{3} \\ Z_S &= \frac{(50)^2}{25/3} = 300\Omega \\ \therefore \Gamma &= \frac{Z_S - Z_0}{Z_S + Z_0} = \frac{300 - 50}{300 + 50} = \frac{5}{7} \end{aligned}$$

Question 146: miscellaneous Topics (GATE EC 2007)

MCQ

1 Mark

A load of 50Ω is connected in shunt in a 2-wire transmission line of $Z_0 = 50\Omega$ as shown in the figure. The 2-port scattering parameter matrix (s-matrix) of the shunt element is



(A)

$$\begin{bmatrix} -\frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{bmatrix}$$

(B)

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

(C) **(Correct)**

$$\begin{bmatrix} -\frac{1}{3} & \frac{2}{3} \\ \frac{2}{3} & -\frac{1}{3} \end{bmatrix}$$

(D)

$$\begin{bmatrix} \frac{1}{4} & -\frac{3}{4} \\ -\frac{3}{4} & \frac{1}{4} \end{bmatrix}$$

Correct Answer: (C)

Detailed Solution – Question 146

The line is terminated with 50Ω at the center and so, matched on both the sides.

Question 147: Waveguides & Optical Fibers (GATE EC 2007)

MCQ

1 Mark

The $\vec{E}$ field in a rectangular waveguide of inner dimension $a \times b$ is given by

$$\vec{E} = \frac{\omega\mu}{h^2} \left(\frac{\pi}{a}\right) H_0 \sin\left(\frac{2\pi x}{a}\right)^2 \sin(\omega t - \beta z) \hat{y}$$

where H_0 is a constant, and a and b are the dimensions along the x -axis and the y -axis respectively. The mode of propagation in the waveguide is

- (A) **(Correct)** TE_{20}
- (B) TM_{11}
- (C) TM_{20}
- (D) TE_{10}

Correct Answer: (A)

Detailed Solution – Question 147

The wave is $E(x,z,t)$ with no propagation in y -direction. This is possible only in TE waves when $n = 0$ ($\cos \frac{n\pi}{b}y = 1$)

Question 148: Uniform Plane Waves (GATE EC 2007)

MCQ

1 Mark

The $\vec{H}$ field (in A/m) of a plane wave propagating in free space is given by

$$\vec{H} = \hat{x} \frac{5\sqrt{3}}{\eta_0} \cos(\omega t - \beta z) + \hat{y} \frac{5}{\eta_0} \sin(\omega t - \beta z + \frac{\pi}{2})$$

. The time average power flow density in Watts is

- (A) $\frac{\eta_0}{100}$
- (B) $\frac{100}{\eta_0}$
- (C) $50\eta_0^2$
- (D) **(Correct)** $\frac{50}{\eta_0}$

Correct Answer: (D)

Detailed Solution – Question 148

For free space,

$$\begin{aligned} P &= \frac{E^2}{2\eta_0}, E = \eta_0 H \\ P &= \frac{\eta_0^2 H^2}{2\eta_0} = \frac{\eta_0 H^2}{2} \\ H &= \frac{1}{\eta_0} \sqrt{(5\sqrt{3})^2 + (5)^2} = \frac{10}{\eta_0} \\ P &= \frac{\eta_0}{2} \times \frac{100}{\eta_0^2} = \frac{50}{\eta_0} \text{ watts} \end{aligned}$$

Question 149: Waveguides & Optical Fibers (GATE EC 2007)

MCQ

1 Mark

An air-filled rectangular waveguide has inner dimensions of 3cm x 2cm. The wave impedance of the TE₂₀ mode of propagation in the waveguide at a frequency of 30 GHz is (free space impedance $\eta_0 = 377 \Omega$)

- (A) 308Ω
- (B) 355Ω
- (C) **(Correct)** 400Ω
- (D) 461Ω

Correct Answer: (C)

Detailed Solution – Question 149

For TE₂₀, $\lambda_C = a = 3 \times 10^{-2} \text{m}$

$$f_C = \frac{C}{\lambda_C} = \frac{3 \times 10^8}{3 \times 10^{-2}} = 10^{10} \text{Hz}$$

$$\eta_{TE} = \frac{\eta}{\sqrt{1 - \left(\frac{f_c}{f}\right)^2}} = \frac{377}{\sqrt{1 - \left(\frac{10^{10}}{3 \times 10^{10}}\right)^2}}$$

$$\eta_{TE} = 400\Omega$$

Question 150: Basics of Electromagnetics & Maxwell's Eqns (GATE EC 2007)

MCQ

1 Mark

If C is closed curve enclosing a surface S, then the magnetic field intensity $\vec{H}$, the current density $\vec{J}$ and the electric flux density $\vec{D}$ are related by

- (A)

$$\int \int_S \vec{H} \cdot d\vec{s} = \oint_c \left(\vec{J} + \frac{\partial \vec{D}}{\partial t} \right) \cdot d\vec{l}$$

- (B)

$$\int_S \vec{H} \cdot d\vec{l} = \oint \oint_S \left(\vec{J} + \frac{\partial \vec{D}}{\partial t} \right) \cdot d\vec{s}$$

- (C)

$$\oint \oint_S \vec{H} \cdot d\vec{s} = \int_C \left(\vec{J} + \frac{\partial \vec{D}}{\partial t} \right) \cdot d\vec{l}$$

- (D) **(Correct)**

$$\oint_C \vec{H} \cdot d\vec{l} \oint_c = \int \int_S \left(\vec{J} + \frac{\partial \vec{D}}{\partial t} \right) \cdot d\vec{s}$$

Correct Answer: (D)

Question 151: Uniform Plane Waves (GATE EC 2007)

MCQ

1 Mark

A plane wave of wavelength λ is traveling in a direction making an angle 30° with positive x-axis and 90° with positive y-axis. The $\vec{E}$ field of the plane wave can be represented as (E_0 is constant)

(A) (Correct)

$$\vec{E} = \hat{y}E_0 e^{j(\omega t - \frac{\sqrt{3}\pi}{\lambda}x - \frac{\pi}{\lambda}z)}$$

(B)

$$\vec{E} = \hat{y}E_0 e^{j(\omega t - \frac{\pi}{\lambda}x - \frac{\sqrt{3}\pi}{\lambda}z)}$$

(C)

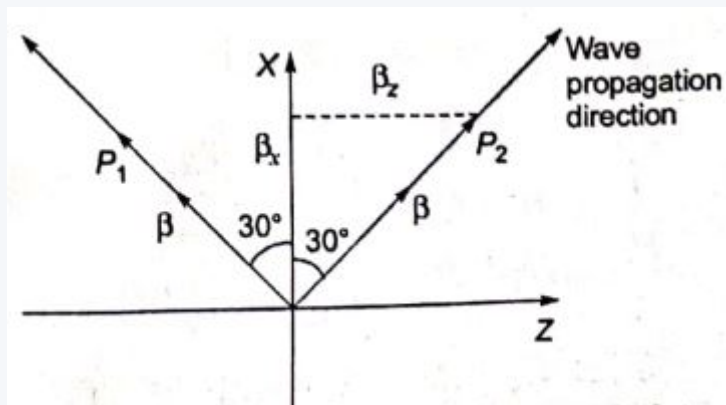
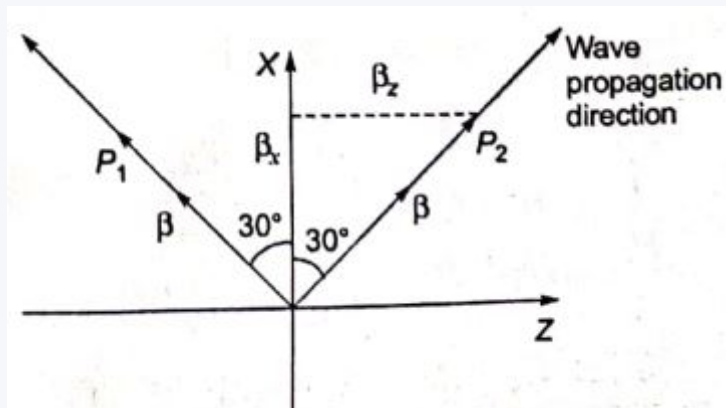
$$\vec{E} = \hat{y}E_0 e^{j(\omega t + \frac{\sqrt{3}\pi}{\lambda}x + \frac{\pi}{\lambda}z)}$$

(D)

$$\vec{E} = \hat{y}E_0 e^{j(\omega t - \frac{\pi}{\lambda}x + \frac{\sqrt{3}\pi}{\lambda}z)}$$

Correct Answer: (A)

Detailed Solution – Question 151



Wave is travelling in the ZX-plane (90° to Y-axis) and 30° to X-axis.

$$\beta_x = \beta \cos 30^\circ = \frac{2\pi}{\lambda} \cdot \frac{\sqrt{3}}{2} = \frac{\sqrt{3}\pi}{\lambda}$$

$$\beta_z = \beta \sin 30^\circ = \frac{\pi}{\lambda}$$

The two solutions possible are shown as P_1 and P_2 paths in the diagrams. Option (A) is P_2

Question 152: Transmission Lines (GATE EC 2006)

MCQ

1 Mark

A 30-Volts battery with zero source resistance is connected to a coaxial line of characteristic impedance of 50 Ohms at $t=0$ second terminated in an unknown resistive load. The line length is that it takes $400\ \mu\text{s}$ for an electromagnetic wave to travel from source end to load end and vice-versa. At $t=400\ \mu\text{s}$, the voltage at the load end is found to be 40 Volts.

The steady-state current through the load resistance is

- (A) 1.2 Amps
- (B) **(Correct)** 0.3 Amps
- (C) 0.6 Amps
- (D) 0.4 Amps

Correct Answer: (B)

Detailed Solution – Question 152

In steady state, $V_L = 30\text{V}$ $I_L = \frac{30}{100} = 0.3\text{A}$

Question 153: Transmission Lines (GATE EC 2006)

MCQ

1 Mark

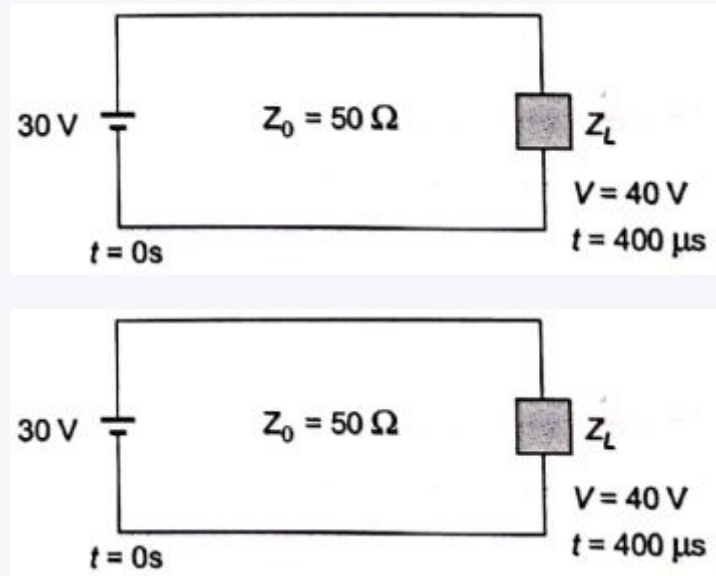
A 30-Volts battery with zero source resistance is connected to a coaxial line of characteristic impedance of 50 Ohms at $t=0$ second terminated in an unknown resistive load. The line length is that it takes $400\ \mu\text{s}$ for an electromagnetic wave to travel from source end to load end and vice-versa. At $t=400\ \mu\text{s}$, the voltage at the load end is found to be 40 Volts.

The load resistance is

- (A) 25 Ohms
- (B) 50 Ohms
- (C) 75 Ohms
- (D) **(Correct)** 100 Ohms

Correct Answer: (D)

Detailed Solution – Question 153



10V reflects to add to 30V in phase and gives 40V

$$\begin{aligned}\Gamma &= \frac{V_r}{V_i} = \frac{10}{30} = \frac{1}{3} \\ \Gamma &= \frac{Z_L - Z_0}{Z_L + Z_0} \\ \Rightarrow \frac{1}{3} &= \frac{Z_L - 50}{Z_L + 50} \\ \Rightarrow Z_L &= 100\Omega\end{aligned}$$

Question 154: Antennas & Radiating Systems (GATE EC 2006)

MCQ

1 Mark

A mast antenna consisting of a 50 meter long vertical conductor operates over a perfectly conducting ground plane. It is base-fed at a frequency of 600 kHz. The radiation resistance of the antenna is Ohms is

- (A) (Correct) $\frac{2\pi^2}{5}$
- (B) $\frac{\pi^2}{5}$
- (C) $\frac{4\pi^2}{5}$
- (D) $20\pi^2$

Correct Answer: (A)

Detailed Solution – Question 154

A mast antenna is a tower antenna or monopole vertically erected on the ground for low frequency ground wave communication. $l = 50\text{m}$, $\lambda = 500\text{ m}$, $l = \frac{\lambda}{10}$ electrically short monopole $R_r = 20\pi^2 \left(\frac{dl}{\lambda}\right)^2 \times 2$ This 2 in the multiplication is due to image in conducting earth. $R_r = \frac{2\pi^2}{5} \simeq 4\Omega$

Question 155: Waveguides & Optical Fibers (GATE EC 2006)

MCQ

1 Mark

A rectangular wave guide having TE_{10} mode as dominant mode is having a cut off frequency 18-GHz for the mode TE_{30} . The inner broad-wall dimension of the rectangular wave guide is

- (A) $\frac{5}{3} \text{ cm}$
- (B) 5 cm
- (C) **(Correct)** $\frac{5}{2} \text{ cm}$
- (D) 10 cm

Correct Answer: (C)

Detailed Solution – Question 155

$$\begin{aligned} \text{TE}_{30} \quad \lambda_c &= \frac{2a}{m} = \frac{2a}{3} \\ \lambda_c &= \frac{c}{f} = \frac{3 \times 10^8}{18 \times 10^9} = \frac{1}{60} \\ \frac{1}{60} &= \frac{2a}{3} \\ \Rightarrow a &= \frac{1}{40} = 2.5 \text{ cm} = \frac{5}{2} \text{ cm} \end{aligned}$$

Question 156: Uniform Plane Waves (GATE EC 2006)

MCQ

1 Mark

When a plane wave traveling in free-space is incident normally on a medium having $\epsilon_r = 4.0$, the fraction of power transmitted into the medium is given by

- (A) **(Correct)** $\frac{8}{9}$
- (B) $\frac{1}{2}$
- (C) $\frac{1}{3}$
- (D) $\frac{5}{6}$

Correct Answer: (A)

Detailed Solution – Question 156

$$\begin{aligned}
 P_t &= (1 - \Gamma^2) P_i \\
 \Gamma &= \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} = \frac{\sqrt{\frac{\mu_0}{\epsilon_0 \epsilon_r}} - \sqrt{\frac{\mu_0}{\epsilon_0}}}{\sqrt{\frac{\mu_0}{\epsilon_0 \epsilon_r}} + \sqrt{\frac{\mu_0}{\epsilon_0}}} \\
 \Gamma &= \frac{\frac{1}{2} - 1}{\frac{1}{2} + 1} = \frac{-1}{3} \\
 \tau &= 1 - \Gamma^2 = 1 - \frac{1}{9} = \frac{8}{9} \\
 \therefore P_t &= \tau \cdot P_i \\
 \frac{P_t}{P_i} &= \frac{8}{9}
 \end{aligned}$$

Question 157: Basics of Electromagnetics & Maxwell's Eqns (GATE EC 2006)

MCQ

1 Mark

A medium is divide into regions I and II about $x = 0$ plane, as shown in the figure below.

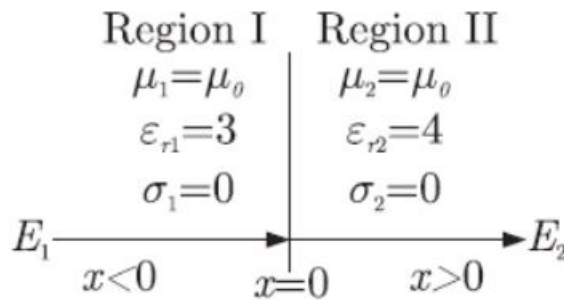


Fig Q.67

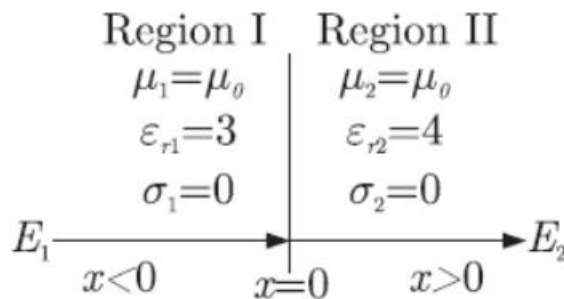


Fig Q.67

An electromagnetic wave with electric field $E_1 = 4\hat{a}_x + 3\hat{a}_y + 5\hat{a}_z$ is incident normally on the interface from region I. The electric field E_2 in region II at the interface is

- (A) $E_2 = E_1$
- (B) $4\hat{a}_x + 0.75\hat{a}_y - 1.25\hat{a}_z$

(C) **(Correct)** $3\hat{a}_x + 3\hat{a}_y + 5\hat{a}_z$

(D) $-3\hat{a}_x + 3\hat{a}_y + 5\hat{a}_z$

Correct Answer: (C)

Detailed Solution – Question 157

$$\begin{aligned}\vec{E}_{t_1} &= 3\hat{a}_y + 5\hat{a}_z = \vec{E}_{t_2} \\ \vec{D}_{n_1} &= \vec{D}_{n_2} \\ \epsilon_2 \vec{E}_{n_2} &= \epsilon_1 \vec{E}_{n_1} \\ \vec{E}_{n_2} &= \frac{3 \times 4\hat{a}_x}{4} = 3\hat{a}_x \\ \therefore \vec{E}_2 &= \vec{E}_{t_2} + \vec{E}_{n_2} \\ &= 3\hat{a}_y + 5\hat{a}_z + 3\hat{a}_x\end{aligned}$$

Question 158: Uniform Plane Waves (GATE EC 2006)

MCQ

1 Mark

A medium of relative permittivity $\epsilon_{r2} = 2$ forms an interface with free-space. A point source of electromagnetic energy is located in the medium at a depth of 1 meter from the interface. Due to the total internal reflection, the transmitted beam has a circular cross-section over the interface. The area of the beam crosssection at the interface is given by

(A) $2\pi m^2$

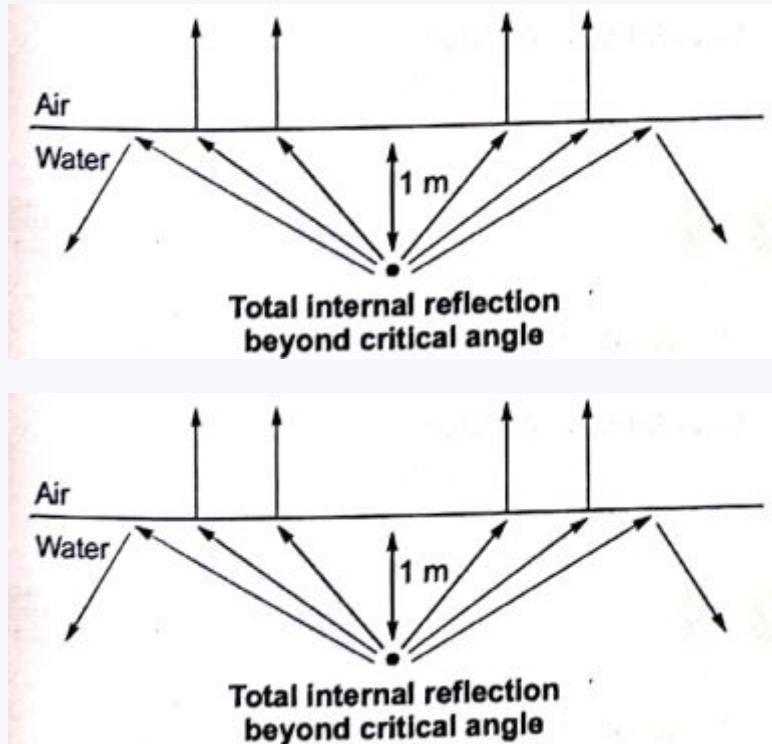
(B) $\pi^2 m^2$

(C) $\frac{\pi}{2} m^2$

(D) **(Correct)** πm^2

Correct Answer: (D)

Detailed Solution – Question 158



Circular cross-section beam for rays incident below critical angle. Calculating critical angle $\sin \theta_c = \sqrt{\frac{\epsilon_o}{2\epsilon_o}}$ $\theta_c = 45^\circ$ 1m – depth; 1m – radius Area = $\pi R^2 = \pi$

Question 159: Antennas & Radiating Systems (GATE EC 2006)

MCQ

1 Mark

A transmission line is feeding 1 watt of power to a horn antenna having a gain of 10 dB. The antenna is matched to the transmission line. The total power radiated by the horn antenna into the free space is

- (A) 10 Watts
- (B) **(Correct)** 1 Watts
- (C) 0.1 Watts
- (D) 0.01 Watts

Correct Answer: (B)

Detailed Solution – Question 159

Gain of antenna is directive gain i.e. different from that of amplifier. It radiates same power. So 1 W.

Question 160: Uniform Plane Waves (GATE EC 2006)

MCQ

1 Mark

The electric field of an electromagnetic wave propagation in the positive direction is given by $E = \hat{a}_x \sin(\omega t - \beta z) + \hat{a}_y \sin(\omega t - \beta z + \pi/2)$. The wave is

- (A) Linearly polarized in the z -direction
- (B) Elliptically polarized
- (C) **(Correct)** Left-hand circularly polarized
- (D) Right-hand circularly polarized

Correct Answer: (C)

Detailed Solution – Question 160

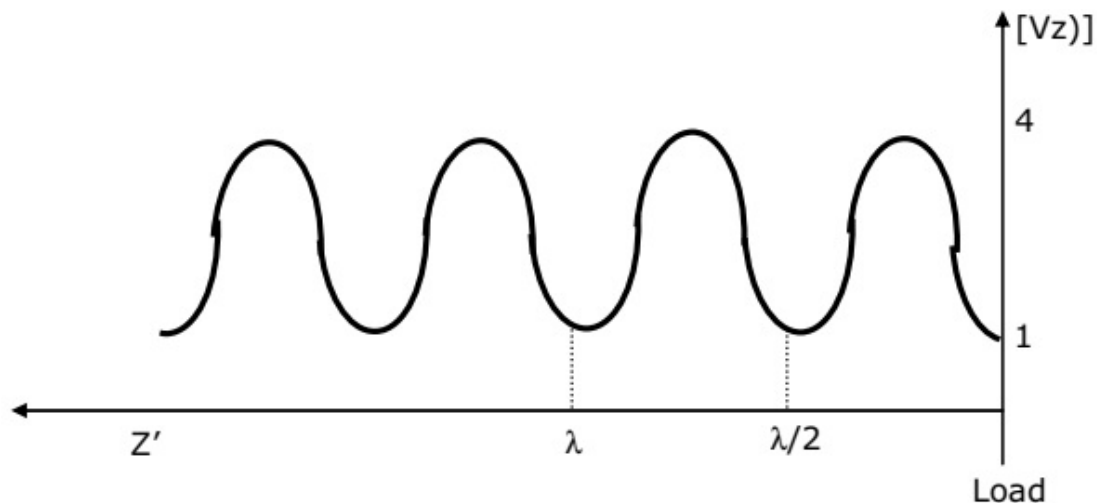
Equal amplitudes and out of phase by 90° , wave is circularly polarized. Tracing the time advancing E on the $z = 0$ plane at $t = 0$, $t = T/4$, $t = T/2$ gives left/right notations

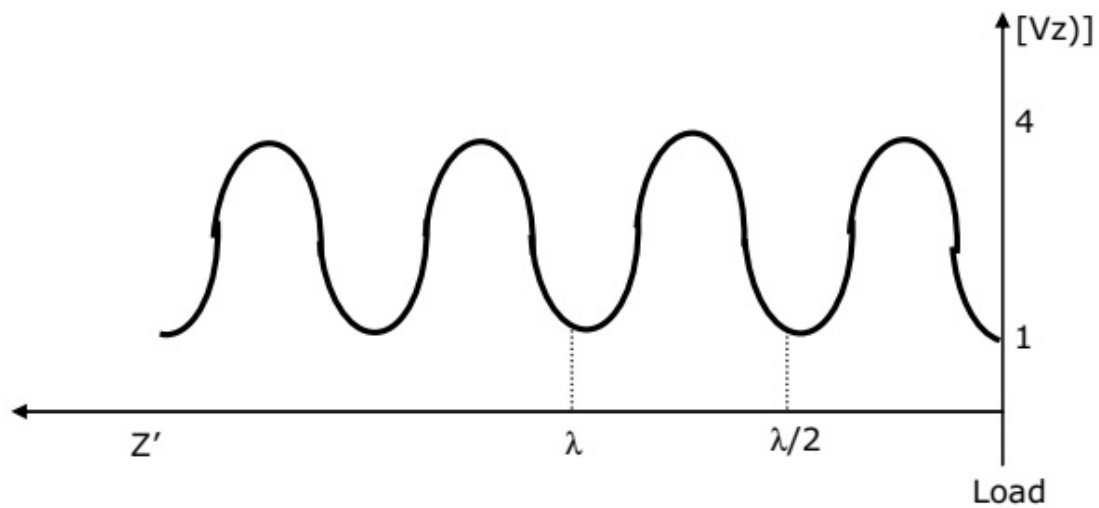
Question 161: Transmission Lines (GATE EC 2005)

MCQ

1 Mark

Voltage standing wave pattern in a lossless transmission line with characteristic impedance 50Ω and a resistive load is shown in figure.





The reflection coefficient is given by

- (A) **(Correct)** -0.6
- (B) -1
- (C) 0.6
- (D) 0

Correct Answer: (A)

Detailed Solution – Question 161

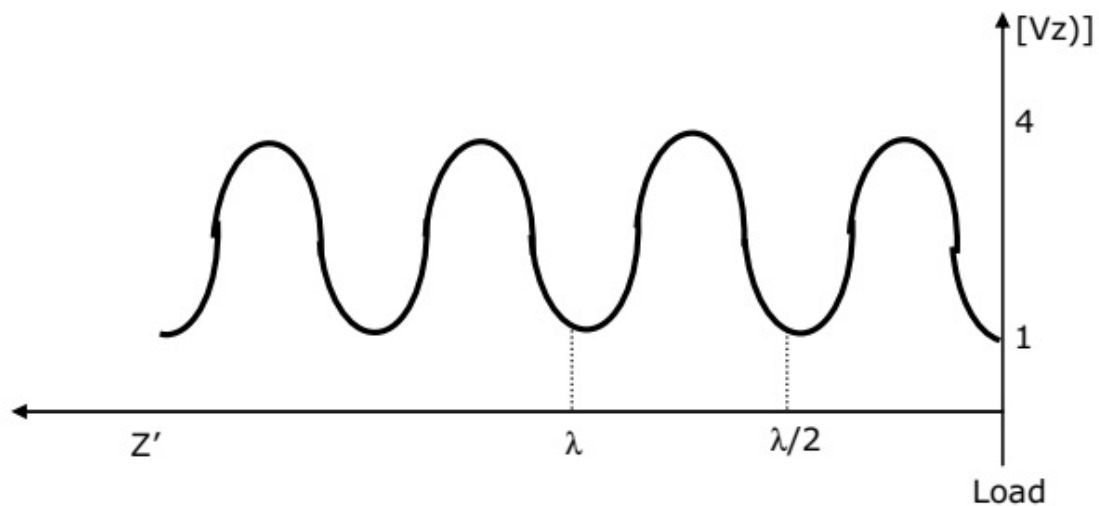
$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0} = \frac{12.5 - 50}{12.5 + 50} = -0.6$$

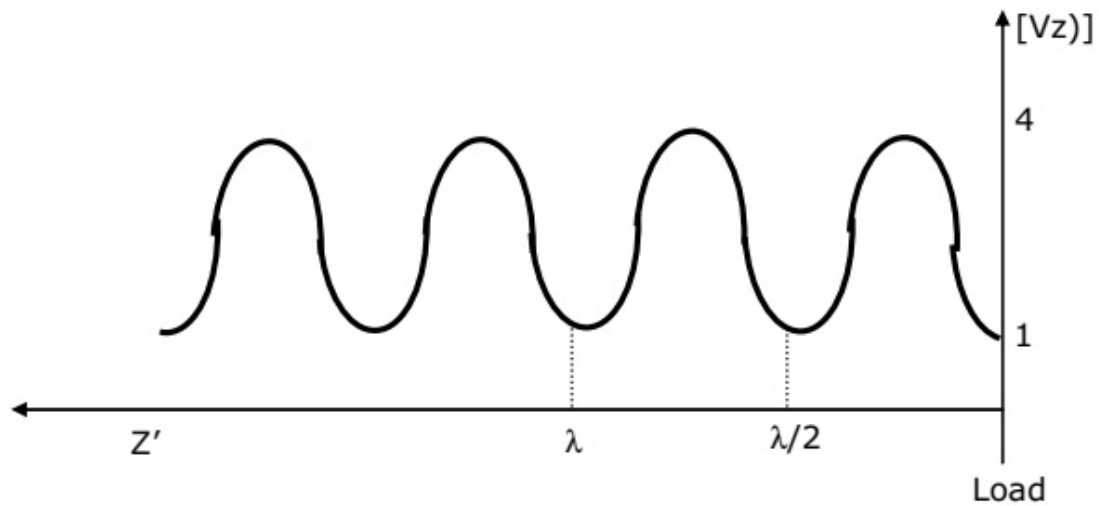
Question 162: Transmission Lines (GATE EC 2005)

MCQ

1 Mark

Voltage standing wave pattern in a lossless transmission line with characteristic impedance 50Ω and a resistive load is shown in figure.





The value of the load resistance is

- (A) 50Ω
- (B) 200Ω
- (C) **(Correct)** 12.5Ω
- (D) 0

Correct Answer: (C)

Detailed Solution – Question 162

$$S = \frac{V_{\max}}{V_{\min}} = \frac{4}{1} = 4$$

$$S = \frac{1 + \Gamma}{1 - \Gamma}$$

$$Z_{\max} = Z_0 \cdot S$$

$$Z_{\min} = Z_0 / S$$

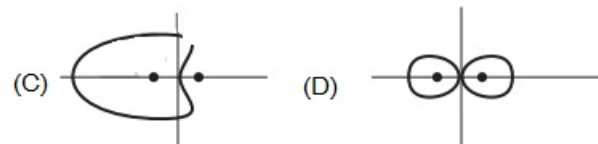
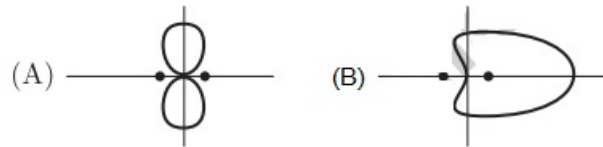
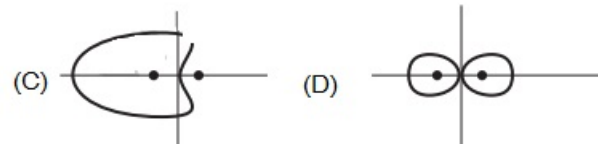
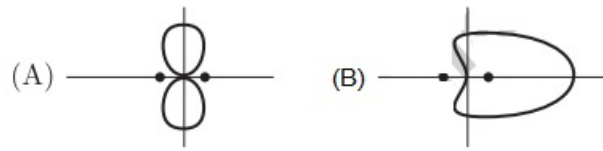
The diagram shows that minima is at load $\therefore Z_L = Z_{\min} = \frac{50}{4} = 12.5 \Omega$

Question 163: Antennas & Radiating Systems (GATE EC 2005)

MCQ

1 Mark

Two identical and parallel dipole antennas are kept apart by a distance of $\frac{\lambda}{4}$ in the H-plane. They are fed with equal currents but the right most antenna has a phase shift of $+90^\circ$. The radiation pattern is given as.



(A) A

(B) B

(C) **(Correct)** C

(D) D

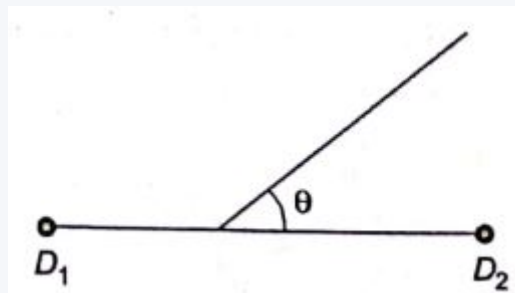
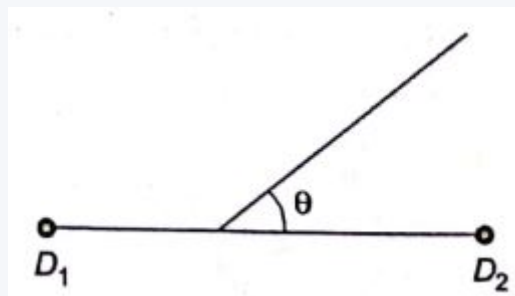
Correct Answer: (C)

Detailed Solution – Question 163

H-plane means ϕ pattern. Both dipoles are horizontally isotropic in the H-plane.

$$\begin{aligned}
 E &= 2E_0 \cos(\psi/2) \\
 \psi &= \alpha + \beta d \cos \theta \\
 &= \frac{\pi}{2} + \frac{2\pi}{\lambda} \cdot \frac{\lambda}{4} \cos \theta \\
 \psi &= \frac{\pi}{2} + \frac{\pi}{2} \cos \theta \\
 \psi = 0 \quad \cos \theta &= -1 & \theta_{\max} &= 180^\circ \\
 \psi = \pi \quad \cos \theta &= 1 & \theta_{NP} &= 0^\circ \\
 \psi = \frac{\pi}{2} \quad \cos \theta &= 0 & \theta_{\text{HPP}} &= 90^\circ / 270^\circ
 \end{aligned}$$

Option (C) when θ is considered as shown in array calculations



Question 164: Transmission Lines (GATE EC 2005)

MCQ

1 Mark

Characteristic impedance of a transmission line is 50Ω . Input impedance of the open-circuited line $Z_{oc} = 100 + j150 \Omega$. when the transmission line a short circuited, then value of the input impedance will be.

- (A) 50Ω
- (B) $100 + j150 \Omega$
- (C) $7.69 + j11.54 \Omega$
- (D) **(Correct)** $7.69 - j11.54 \Omega$

Correct Answer: (D)

Detailed Solution – Question 164

$$z_0^2 = z_{oc} \cdot z_{sc}$$

$$z_{sc} = \frac{50 \times 50}{100 + j150} = \frac{50}{2 + 3j} = \frac{50(2 - 3j)}{13}$$

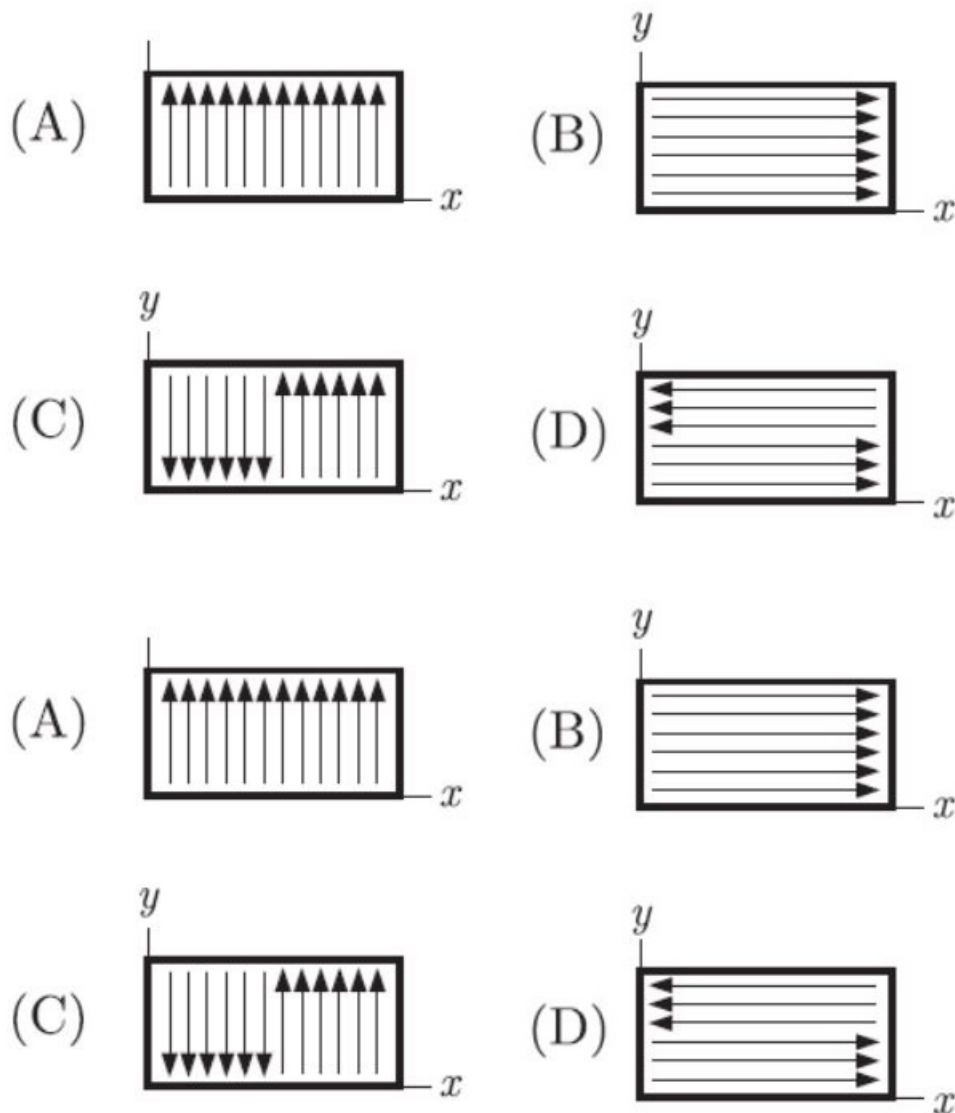
$$z_{sc} = 7.69 - j11.54$$

Question 165: Waveguides & Optical Fibers (GATE EC 2005)

MCQ

1 Mark

Which one of the following does represent the electric field lines for the TE_{02} mode in the cross-section of a hollow rectangular metallic waveguide ?



(A) A

(B) B

(C) C

(D) (Correct) D

Correct Answer: (D)

Detailed Solution – Question 165

Two maximas in the field along y-direction identifies TE_{02}

Question 166: Uniform Plane Waves (GATE EC 2005)

MCQ

1 Mark

Refractive index of glass is 1.5. Find the wavelength of a beam of light with frequency of 10^{14} Hz in glass. Assume velocity of light is 3×10^8 m/s in vacuum

- (A) $3 \mu\text{m}$
- (B) 3 mm
- (C) (Correct) $2 \mu\text{m}$
- (D) $1 \mu\text{m}$

Correct Answer: (C)

Detailed Solution – Question 166

Refractive index $= \sqrt{\frac{\epsilon_2}{\epsilon_1}}$ For glass with respect to air $= \sqrt{\epsilon_R}$

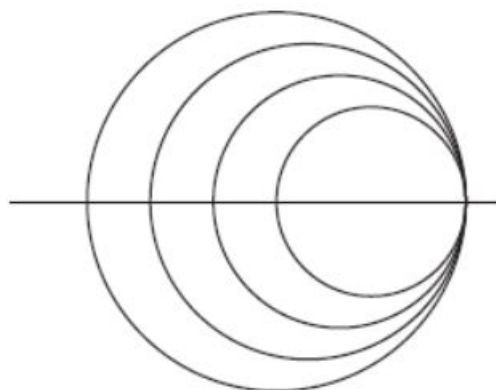
$$V_p = \frac{3 \times 10^8}{\sqrt{\epsilon_R} \times f} = \frac{3 \times 10^8}{1.5 \times 10^{14}} = 2 \mu\text{m}$$

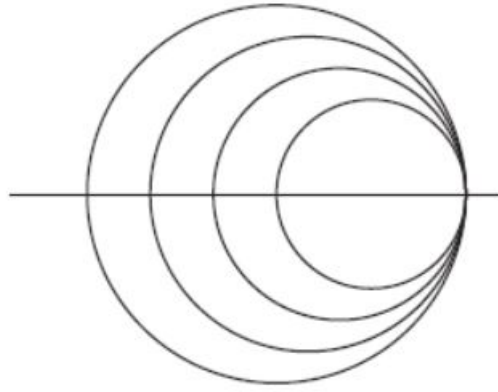
Question 167: Transmission Lines (GATE EC 2005)

MCQ

1 Mark

Many circles are drawn in a Smith Chart used for transmission line calculations. The circles shown in the figure represent





- (A) Unit circles
- (B) Constant resistance circles
- (C) **(Correct)** Constant reactance circles
- (D) Constant reflection coefficient circles.

Correct Answer: (C)

Question 168: Uniform Plane Waves (GATE EC 2005)

MCQ

1 Mark

The magnetic field intensity vector of a plane wave is given by $\vec{H}(x, y, z, t) = 10\sin(50000t + 0.004x + 30)\hat{a}_y$ where $\hat{a}_y$, denotes the unit vector in y direction. The wave is propagating with a phase velocity.

- (A) $5 \times 10^4 \text{ m/s}$
- (B) $-3 \times 10^8 \text{ m/s}$
- (C) **(Correct)** $-1.25 \times 10^7 \text{ m/s}$
- (D) $3 \times 10^8 \text{ m/s}$

Correct Answer: (C)

Detailed Solution – Question 168

$$\omega = 50,000; \beta = 0.004$$

$$v_p = \frac{\omega}{\beta} = \frac{5 \times 10^4}{4 \times 10^{-3}} = 1.25 \times 10^7 \text{ m/s}$$

Question 169: Transmission Lines (GATE EC 2004)

MCQ

1 Mark

A lossless transmission line is terminated in a load which reflects a part of the incident power. The measured VSWR is 2. The percentage of the power that is reflected back is

- (A) 57.73
 (B) 33.33
 (C) 0.11
 (D) **(Correct)** 11.11

Correct Answer: (D)

Detailed Solution – Question 169

$\text{SWR} = \frac{1+\Gamma}{1-\Gamma} \Rightarrow 2 = \frac{1+\Gamma}{1-\Gamma}$ Voltage reflection coefficient $\Rightarrow \Gamma = \frac{1}{3}$ Power reflection coefficient

$$\therefore \frac{P_{\text{ref}}}{P_{\text{inc}}} = \Gamma^2 = \frac{1}{9}$$

$\Rightarrow 11.11\%$ of P_i is reflected

Question 170: Transmission Lines (GATE EC 2004)

MCQ

1 Mark

A plane electromagnetic wave propagating in free space is incident normally on a large slab of loss-less, non-magnetic, dielectric material with $\epsilon > \epsilon_0$. Maxima and minima are observed when the electric field is measured in front of the slab. The maximum electric field is found to be 5 times the minimum field. The intrinsic impedance of the medium should be

- (A) $120\pi\Omega$
 (B) $60\pi\Omega$
 (C) $600\pi\Omega$
 (D) **(Correct)** $24\pi\Omega$

Correct Answer: (D)

Detailed Solution – Question 170

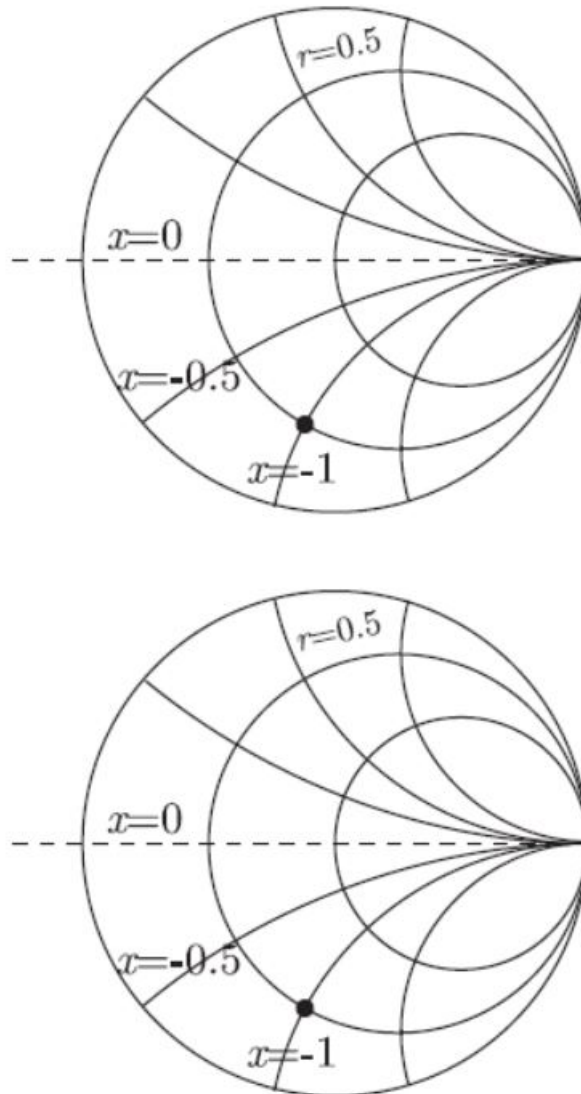
$$\begin{aligned} \text{SWR} &= \frac{E_{\text{max}}}{E_{\text{min}}} \\ \Rightarrow s &= \frac{5E_{\text{min}}}{E_{\text{min}}} = 5 \\ 5 &= \frac{1+|\Gamma|}{1-|\Gamma|} \quad |\Gamma| = \frac{2}{3} \\ \eta_2 &= \frac{120\pi}{\sqrt{\epsilon_r}} \text{ and } \eta_2 = 120\pi \\ \text{As } \eta_2 &< \eta_1 \\ \Rightarrow -\frac{2}{3} &= \frac{\eta_2 - 120\pi}{\eta_2 + 120\pi} \\ \Rightarrow \eta_2 &= 24\pi \end{aligned}$$

Question 171: Transmission Lines (GATE EC 2004)

MCQ

1 Mark

Consider an impedance $Z = R + jX$ marked with point P in an impedance Smith chart as shown in Fig. The movement from point P along a constant resistance circle in the clockwise direction by an angle 45° is equivalent to



- (A) **(Correct)** adding an inductance in series with Z
- (B) adding a capacitance in series with Z
- (C) adding an inductance in shunt across Z
- (D) adding a capacitance in shunt across Z

Correct Answer: (A)

Detailed Solution – Question 171

Movement on constant resistance circle by an $\angle 45^\circ$ in clockwise direction.

R is same and reactance increases i.e. addition of an inductance in series with Z.

Question 172: Uniform Plane Waves (GATE EC 2004)

MCQ

1 Mark

If $\vec{E} = (\hat{a}_x + j\hat{a}_y)e^{jkz-k\omega t}$ and $\vec{H} = (k/\omega\mu)(\hat{a}_y + j\hat{a}_x)e^{jkz-j\omega t}$, the time-averaged Poynting vector is

- (A) **(Correct)** null vector
- (B) $(k/\omega\mu)\hat{a}_z$
- (C) $(2k/\omega\mu)\hat{a}_z$
- (D) $(k/2\omega\mu)\hat{a}_z$

Correct Answer: (A)

Detailed Solution – Question 172

$$\begin{aligned}\vec{P}_{\text{avg}} &= \frac{1}{2} \text{Re} \left[\vec{E} \times \vec{H}^* \right] \\ &= \frac{1}{2} \text{Re} \left[(\hat{a}_x + j\hat{a}_y) e^{jkz-j\omega t} \right. \\ &\quad \left. \times \frac{k}{\omega\mu} (\hat{a}_x + j\hat{a}_y) e^{jkz-j\omega t} \right] \\ \vec{P}_{\text{avg}} &= \frac{k}{2\mu\omega} [0] = 0 \\ \begin{bmatrix} 1 & j \\ -j & 1 \end{bmatrix} &= 1 + j^2 = 1 - 1 = 0\end{aligned}$$

Question 173: Miscellaneous Topics (GATE EC 2004)

MCQ

1 Mark

In a microwave test bench, why is the microwave signal amplitude modulated at 1 kHz

- (A) To increase the sensitivity of measurement
- (B) To transmit the signal to a far-off place
- (C) To study amplitude modulations
- (D) **(Correct)** Because crystal detector fails at microwave frequencies

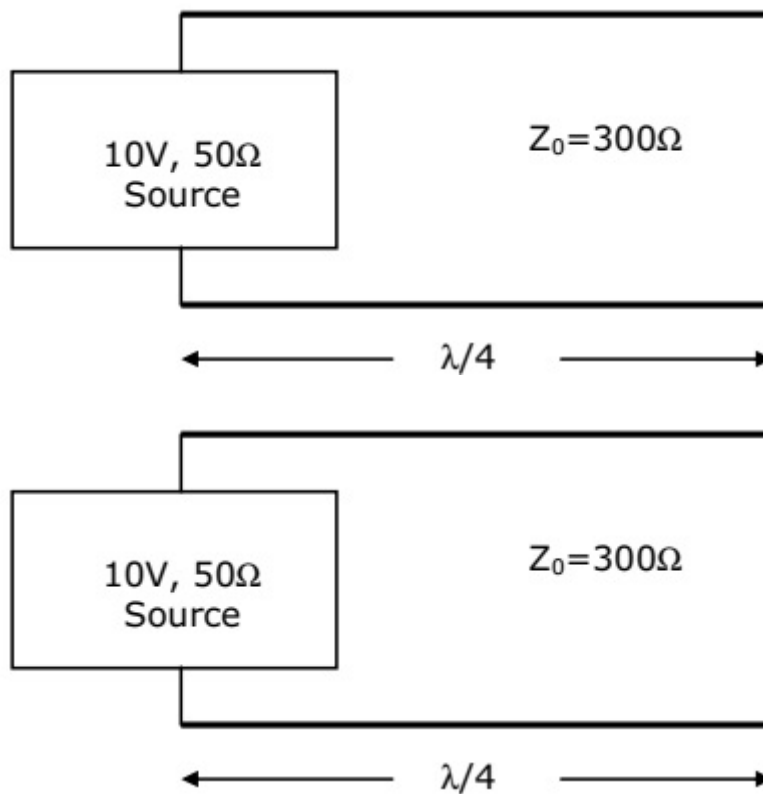
Correct Answer: (D)

Question 174: Transmission Lines (GATE EC 2004)

MCQ

1 Mark

Consider a 300Ω , quarter-wave long (at 1 GHz) transmission line as shown in Fig. It is connected to a 10V, 50Ω source at one end and is left open circuited at the other end. The magnitude of the voltage at the open circuit end of the line is



- (A) 10 V
- (B) 5 V
- (C) **(Correct)** 60 V
- (D) 60/7 V

Correct Answer: (C)

Detailed Solution – Question 174

The $\lambda/4$ line has $Z'_{in} = 0$ as $Z_L = \infty$. Current I_{in} into the line $= \frac{V}{50} = 0.2A$. Using current equation on the line $I(x) = I_{in} = I_L \cos \beta x + \frac{jV_L}{Z_0} \sin \beta x$ with $x = \lambda/4$, $I_L = 0$ as $Z_L = \infty$. Open circuit end $V_L = 60V$.

Question 175: Uniform Plane Waves (GATE EC 2004)

MCQ

1 Mark

A parallel plate air-filled capacitor has plate area of $10^{-4}m^2$ and plate separation of 10^{-3} m. It is connected to a 0.5V, 3.6 GHz source. The magnitude of the displacement current is ($\epsilon = \frac{1}{36\pi}10^{-9}F/m$)

- (A) **(Correct)** 10 mA
- (B) 100 mA
- (C) 10A
- (D) 1.59 mA

Correct Answer: (A)

Detailed Solution – Question 175

Displacement current,

$$\begin{aligned}I_d &= AJ_d \\&= A \frac{\partial D}{\partial t} = A \epsilon \frac{\partial E}{\partial t} \\|I_d| &= |A \epsilon \omega E| = A \omega \epsilon \frac{V}{d}\end{aligned}$$

After putting values, we get, $I_d = 10\text{mA}$

Question 176: Antennas & Radiating Systems (GATE EC 2004)

MCQ

1 Mark

Consider a lossless antenna with a directive gain of +6dB. If 1 mW of power is fed to it the total power radiated by the antenna will be

- (A) **(Correct)** 4 mW
- (B) 1 mW
- (C) 7 mW
- (D) 1/4 mW

Correct Answer: (A)

Detailed Solution – Question 176

When an antenna is lossless then the antenna efficiency is 100%. Power fed to antenna is radiated in the form of radiation. Therefore, if power fed to antenna is 1 mW then total radiated power is also 1 mW.

Question 177: Waveguides & Optical Fibers (GATE EC 2004)

MCQ

1 Mark

The phase velocity of an electromagnetic wave propagating in a hollow metallic rectangular waveguide in the TE_{10} mode is

- (A) equal to its group velocity
- (B) less than the velocity of light in free space
- (C) equal to the velocity of light in free space
- (D) **(Correct)** greater than the velocity of light in free space

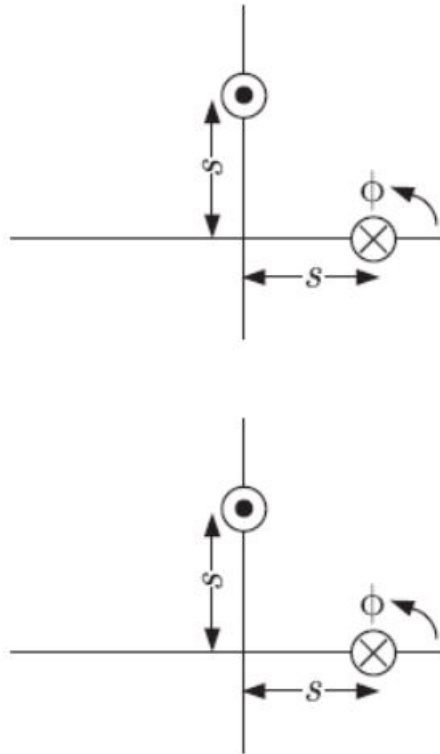
Correct Answer: (D)

Question 178: Antennas & Radiating Systems (GATE EC 2003)

MCQ

1 Mark

Two identical antennas are placed in the $\theta = \pi/2$ plane as shown in Fig. The elements have equal amplitude excitation with 180° polarity difference, operating at wavelength λ . The correct value of the magnitude of the far-zone resultant electric field strength normalized with that of a single element, both computed for $\phi = 0$, is

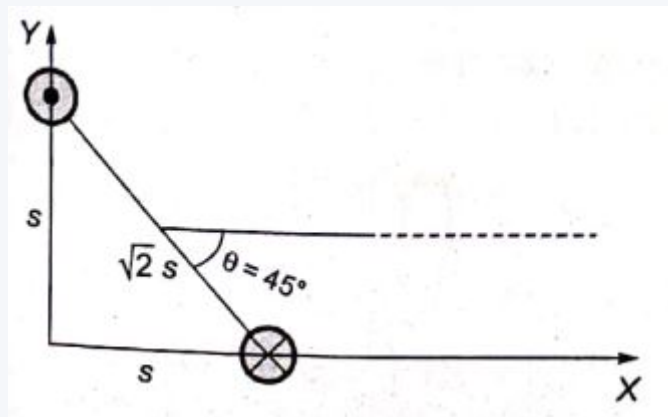
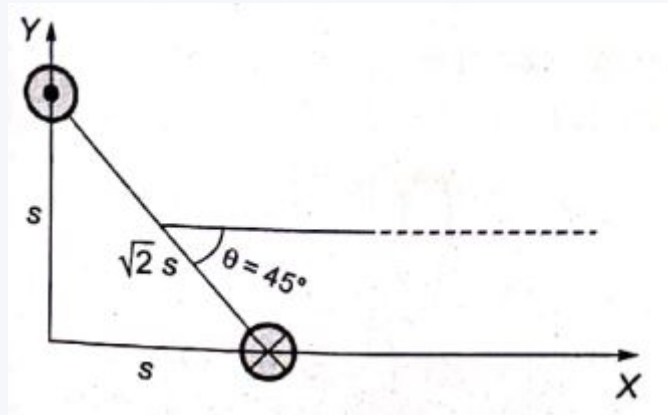


- (A) $2 \cos(\frac{2\pi s}{\lambda})$
- (B) $2 \sin(\frac{2\pi s}{\lambda})$
- (C) $2 \cos(\frac{\pi s}{\lambda})$
- (D) **(Correct)** $2 \sin(\frac{\pi s}{\lambda})$

Correct Answer: (D)

Detailed Solution – Question 178

$\phi = 90^\circ$ means XY plane means $Z = 0$ plane Far zone calculations of field at $\phi = 0$ means point on X-axis. Angle made by the array axis to the point of consideration, if taken as θ .



$$\begin{aligned}
 \text{Normalized array factor} &= 2 \left| \cos \frac{\psi}{2} \right| \\
 \psi &= \beta d \cos \theta + \alpha \\
 d &= \sqrt{2}s, \theta = 45^\circ, \alpha = 180^\circ \\
 \Rightarrow 2 \left| \cos \frac{\psi}{2} \right| &= 2 \cos \left[\frac{\beta d \cos \theta + \alpha}{2} \right] \\
 &= 2 \cos \left[\frac{2\pi}{\lambda} \cdot \sqrt{2}s \cos 45^\circ + \frac{180^\circ}{2} \right] \\
 &= 2 \cos \left[\frac{\pi s}{\lambda} + 90^\circ \right] \\
 &= 2 \sin \left(\frac{\pi s}{\lambda} \right)
 \end{aligned}$$

Question 179: Waveguides & Optical Fibers (GATE EC 2003)

MCQ

1 Mark

A rectangular metal wave guide filled with a dielectric material of relative permittivity $\epsilon_r = 4$ has the inside dimensions $3.0 \text{ cm} \times 1.2 \text{ cm}$. The cut-off frequency for the dominant mode is

- (A) (Correct) 2.5 GHz

- (B) 5.0 GHz
- (C) 10.0 GHz
- (D) 12.5 GHz

Correct Answer: (A)

Detailed Solution – Question 179

$$v = \frac{c}{\sqrt{\epsilon_r}} = \frac{3 \times 10^8}{2} = 1.5 \times 10^8$$

$$f_c = \frac{v}{\lambda_c} = \frac{1.5 \times 10^8}{6 \times 10^{-2}} = 2.5 \text{GHz}$$

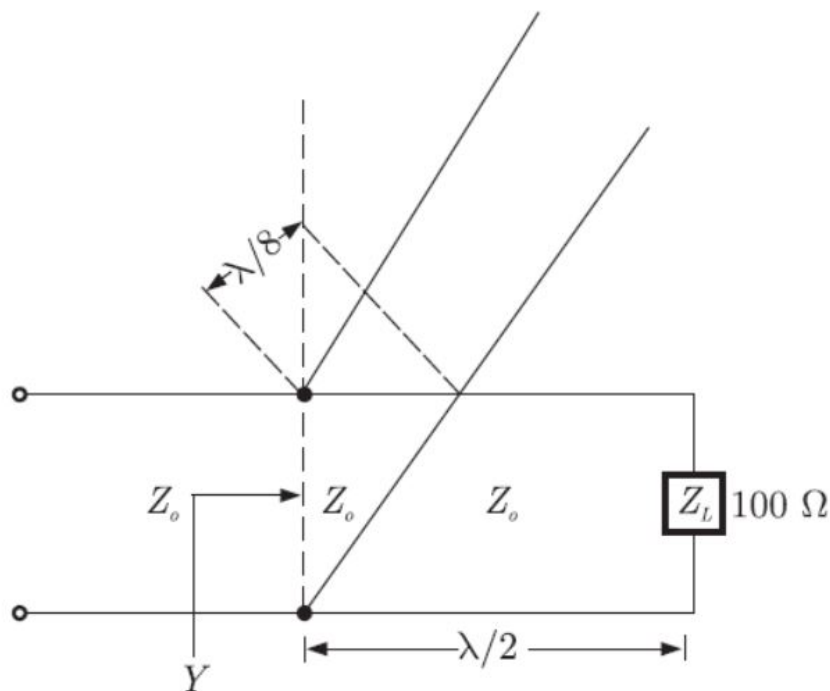
$$\text{or, } f_c = \frac{v}{2a} = \frac{1.5 \times 10^8}{6 \times 10^{-2}} = 2.5 \text{GHz}$$

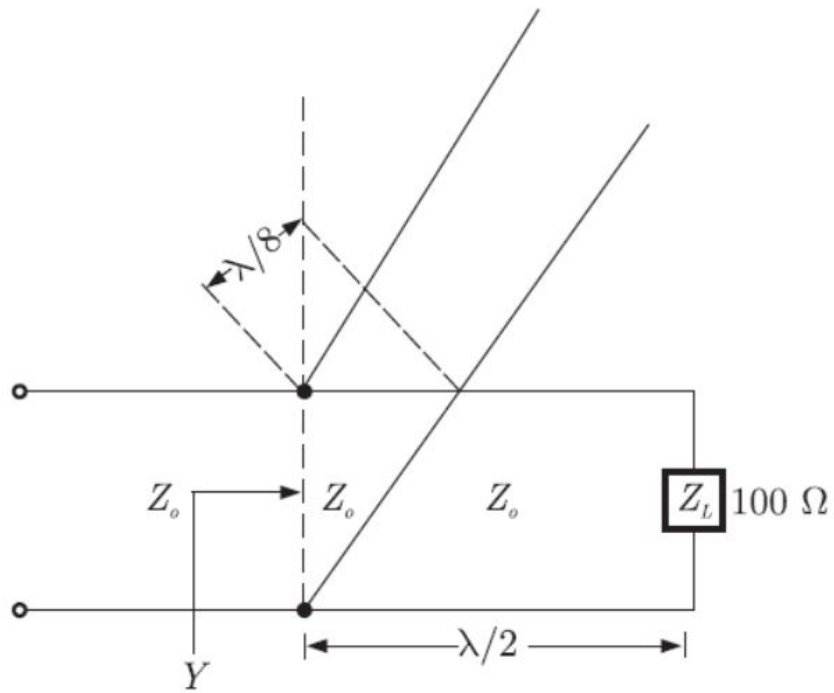
Question 180: Transmission Lines (GATE EC 2003)

MCQ

1 Mark

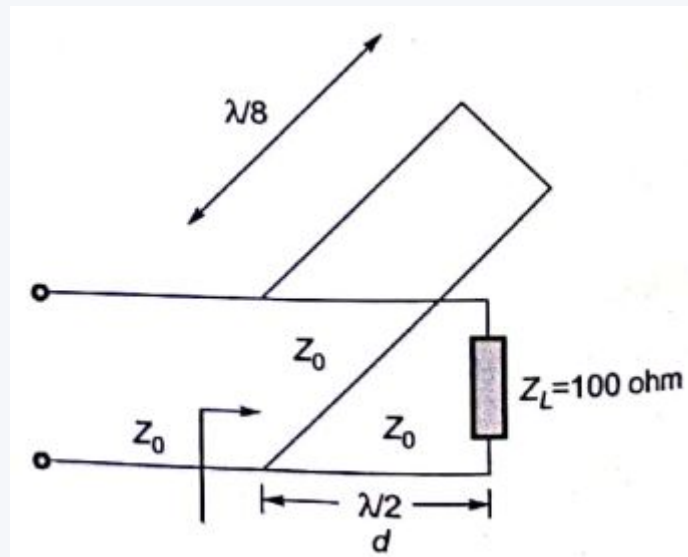
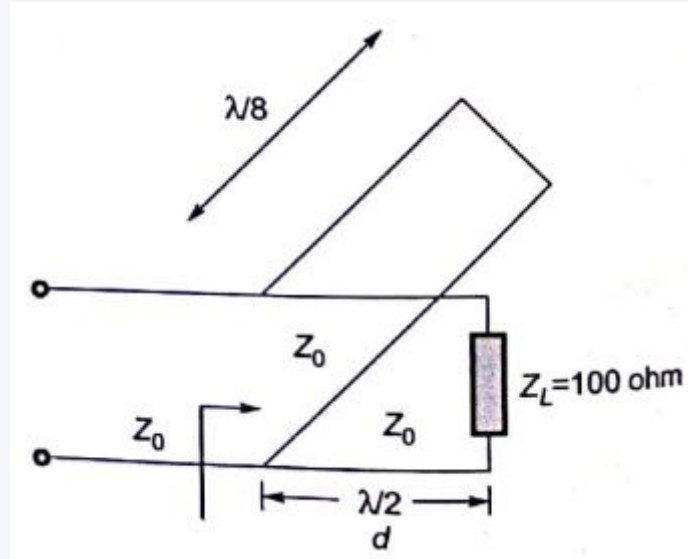
A short - circuited stub is shunt connected to a transmission line as shown in fig. If $Z_0 = 50 \text{ ohm}$, the admittance Y seen at the junction of the stub and the transmission line is





- (A) **(Correct)** $(0.01 - j0.02) \text{ mho}$
- (B) $(0.02 - j0.01) \text{ mho}$
- (C) $(0.04 - j0.02) \text{ mho}$
- (D) $(0.02 + j0) \text{ mho}$

Correct Answer: (A)



$$\beta = \frac{2\pi}{\lambda}$$

$$\text{For } Y_d \beta d = \frac{2\pi}{\lambda} \times \frac{\lambda}{2} = \pi$$

$$Z_d = \frac{Z_0 [Z_L + jZ_0 \tan \beta d]}{Z_0 + jZ_L \tan \beta d}$$

$$Z_d = \frac{50[100 + j50 \tan \pi]}{50 + j100 \tan \pi} = 100$$

$$Y_d = \frac{1}{Z_d} = \frac{1}{100} = 0.01$$

$$\text{For } Y_s \quad \beta d = \frac{2\pi}{\lambda} \times \frac{\lambda}{8} = \frac{\pi}{4}, Z_L \rightarrow 0$$

$$Z_s = \frac{Z_0 [Z_L + jZ_0 \tan \frac{\pi}{4}]}{[Z_0 + jZ_L \tan \frac{\pi}{4}]} = jZ_0$$

$$Y_s = \frac{1}{Z_s} = \frac{1}{jZ_0} = -0.02j$$

$$\therefore Y = Y_d + Y_s = 0.01 - 0.02j$$

Question 181: Uniform Plane Waves (GATE EC 2003)

MCQ

1 Mark

If the electric field intensity associated with a uniform plane electromagnetic wave traveling in a perfect dielectric medium is given by $E(z, t) = 10 \cos(2\pi 10^7 t - 0.1\pi z)$ V/m, then the velocity of the traveling wave is

- (A) 3.00×10^8 m/sec
- (B) **(Correct)** 2.00×10^8 m/sec
- (C) 6.28×10^7 m/sec
- (D) 2.00×10^7 m/sec

Correct Answer: (B)

Detailed Solution – Question 181

$$\begin{aligned}\omega &= 2\pi \times 10^7 \\ \beta &= 0.1\pi\end{aligned}$$

Comparing with $A \cos(\omega t - \beta z)$

$$v = \frac{\omega}{\beta} = \frac{2\pi \times 10^7}{0.1\pi} = 2 \times 10^8 \text{ m}$$

Question 182: Uniform Plane Waves (GATE EC 2003)

MCQ

1 Mark

A uniform plane wave traveling in air is incident on the plane boundary between air and another dielectric medium with $\epsilon_r = 4$. The reflection coefficient for the normal incidence, is

- (A) zero
- (B) $0.5 \angle 180^\circ$
- (C) $0.333 \angle 0^\circ$
- (D) **(Correct)** $0.333 \angle 180^\circ$

Correct Answer: (D)

Detailed Solution – Question 182

$$\begin{aligned}\Gamma &= \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} \\ &= \frac{\sqrt{\frac{\mu_0}{\epsilon_0 \epsilon_r}} - \sqrt{\frac{\mu_0}{\epsilon_0}}}{\sqrt{\frac{\mu_0}{\epsilon_0 \epsilon_r}} + \sqrt{\frac{\mu_0}{\epsilon_0}}} = \frac{\frac{1}{\sqrt{\epsilon_r}} - 1}{\frac{1}{\sqrt{\epsilon_r}} + 1}, \epsilon_r = 4 \\ &= \frac{-1/2}{3/2} = \frac{-1}{3} \Rightarrow 0.333 \angle 180^\circ\end{aligned}$$

Question 183: Basics of Electromagnetics & Maxwell's Eqns (GATE EC 2003)

MCQ

1 Mark

If the electric field intensity is given by $E = (xu_x + yu_y + zu_z)$ volt/m, the potential difference between X(2,0,0) and Y(1,2,3) is

- (A) +1 volt
- (B) -1 volt
- (C) **(Correct)** +5 volt
- (D) +6 volt

Correct Answer: (C)

Detailed Solution – Question 183

$$\begin{aligned} V &= - \int \vec{E} \cdot \vec{d} \\ &= - \left[\int_1^2 x dx \vec{u}_x + \int_2^0 y dy \vec{u}_y + \int_3^0 z dz \vec{u}_z \right] \\ &= - \left[\frac{x^2}{2} \Big|_1^2 + \frac{y^2}{2} \Big|_2^0 + \frac{z^2}{2} \Big|_3^0 \right] \\ &= - \frac{1}{2} [2^2 - 1^2 + 0^2 - 2^2 + 0^2 - 3^2] \\ &= - \frac{1}{2} (-10) = 5V \end{aligned}$$

Question 184: Basics of Electromagnetics & Maxwell's Eqns (GATE EC 2003)

MCQ

1 Mark

Medium 1 has the electrical permittivity $\epsilon_1 = 1.5\epsilon_0$ farad/m and occupies the region to the left of $x = 0$ plane. Medium 2 has the electrical permittivity $\epsilon_2 = 2.5\epsilon_0$ farad/m and occupies the region to the right of $x = 0$ plane. If E_1 in medium 1 is $E_1 = (2u_x - 3u_y + 1u_z)$ volt/m, then E_2 in medium 2 is

- (A) $(2.0u_x - 7.5u_y + 2.5u_z)$ volt/m
- (B) $(2.0u_x - 2.0u_y + 0.6u_z)$ volt/m
- (C) **(Correct)** $(1.2u_x - 3.0u_y + 1.0u_z)$ volt/m
- (D) $(1.2u_x - 2.0u_y + 0.6u_z)$ volt/m

Correct Answer: (C)

Detailed Solution – Question 184

$$\begin{aligned}
 \vec{E}_1 &= 2u_x - 3u_y + 1u_z \\
 \vec{E}_{1t} &= -3u_y + u_z = E_{2t} & (x = 0 \text{ plane}) \\
 \vec{E}_{1n} &= 2u_x \\
 \vec{D}_{1n} &= \vec{D}_{2n} = \epsilon \vec{E} \\
 \Rightarrow \epsilon_1 \vec{E}_{1n} &= \epsilon_2 \vec{E}_{2n} \\
 &= 1.5\epsilon_0 \cdot 2u_x = 2.5\epsilon_0 \cdot \vec{E}_{2n} \\
 \vec{E}_{2n} &= \frac{3}{2.5}u_x = 1.2u_x \\
 \vec{E}_2 &= \vec{E}_{2t} + \vec{E}_{2n} \\
 \vec{E}_2 &= -3u_y + u_z + 1.2u_x \\
 &= 1.2u_x - 3.0u_y + 1.0u_z
 \end{aligned}$$

Question 185: Uniform Plane Waves (GATE EC 2003)

MCQ

1 Mark

The depth of penetration of electromagnetic wave in a medium having conductivity σ at a frequency of 1 MHz is 25 cm. The depth of penetration at a frequency of 4 MHz will be

- (A) 6.25 dm
- (B) **(Correct)** 12.50 cm
- (C) 50.00 cm
- (D) 100.00 cm

Correct Answer: (B)

Detailed Solution – Question 185

$$\begin{aligned}
 \text{For conductors, } \delta &= \sqrt{\frac{2}{\omega\mu\sigma}} \\
 \delta &\propto \frac{1}{\sqrt{f}} \\
 \frac{\delta_2}{\delta_1} &= \sqrt{\frac{f_1}{f_2}} \\
 \delta_2 &= \sqrt{\frac{1\text{MHz}}{4\text{MHz}}} \times 25\text{cm} = 12.5\text{cm}
 \end{aligned}$$

Question 186: Basics of Electromagnetics & Maxwell's Eqns (GATE EC 2003)

MCQ

1 Mark

The unit of $\nabla \times H$ is

- (A) Ampere
- (B) Ampere/meter
- (C) **(Correct)** *Ampere/meter²*
- (D) Ampere-meter

Correct Answer: (C)

Detailed Solution – Question 186

$$\nabla \times \vec{H} = \frac{\partial \vec{D}}{\partial t} + \vec{J}$$

$$\frac{1}{m} \times \frac{\text{Ampere}}{m} = \frac{A}{m^2}$$

Question 187: Antennas & Radiating Systems (GATE EC 2002)

MCQ

1 Mark

A person with a receiver is 5 Km away from the transmitter. What is the distance that this person must move further to detect a 3-dB decrease in signal strength?

- (A) 942m
- (B) **(Correct)** 2070m
- (C) 4978m
- (D) 5320m

Correct Answer: (B)

Detailed Solution – Question 187

$$\text{Signal strength} = \frac{P}{4\pi R^2}$$

$$P_1 = \frac{P}{4\pi(5000)^2} \text{ and } P_2 = \frac{P}{4\pi(x)^2}$$

Given, $\frac{P_2}{P_1} = \frac{1}{2}$ (3 dB decrease)

$$\Rightarrow \left(\frac{5000}{x} \right)^2 = \frac{1}{2}$$

$$\Rightarrow x = 5000\sqrt{2}$$

$\therefore$ Required distance = $5000\sqrt{2} - 5000 = 2071\text{m}$

Question 188: Basics of Electromagnetics & Maxwell's Eqns (GATE EC 2002)

MCQ

1 Mark

The electric field on the surface of a perfect conductor is 2 V/m. The conductor is immersed in water with $\epsilon = 80\epsilon_0$. The surface charge density on the conductor is ($\epsilon = 10^{-9}/(36\pi)F/m$)

- (A) 0 C/m^2
- (B) 2 C/m^2
- (C) $1.8 \times 10^{-11} \text{ C/m}^2$
- (D) **(Correct)** $1.14 \times 10^{-9} \text{ C/m}^2$

Correct Answer: (D)

Detailed Solution – Question 188

The surface charge density is equally to normal flux density

$$D_{\text{normal}} = \rho_s$$

$$D = \rho_s = \epsilon E = 80 \cdot \epsilon \cdot 2$$

Question 189: Uniform Plane Waves (GATE EC 2002)

MCQ

1 Mark

Distilled water at 25°C is characterized by $\sigma = 1.7 \times 10^{-4} \text{ mho/m}$ and $\epsilon = 78\epsilon_0$ at a frequency of 3 GHz. Its loss tangent $\tan\delta$ is ($\epsilon = 10^{-9}/(36\pi) \text{ F/m}$)

- (A) **(Correct)** 1.3×10^{-5}
- (B) 1.3×10^{-3}
- (C) $1.7 \times 10^{-4}/78$
- (D) $1.7 \times 10^{-4}/(78\epsilon_0)$

Correct Answer: (A)

Detailed Solution – Question 189

$$\begin{aligned} \text{Loss tangent} &= \frac{\sigma}{\omega \epsilon} \\ &= \frac{1.7 \times 10^{-4}}{2\pi \times 3 \times 10^9 \times 78\epsilon_0} \\ &= \frac{1.7 \times 10^{-4} \times 9 \times 10^9}{3 \times 10^9 \times 39} \\ &= 0.13 \times 10^{-4} = 1.3 \times 10^{-5} \end{aligned}$$

Question 190: Uniform Plane Waves (GATE EC 2002)

MCQ

1 Mark

A plane wave is characterized by $\vec{E} = (0.5\hat{x} + \hat{y}e^{\frac{j\pi}{2}})e^{j\omega t - jkz}$. This wave is

- (A) linearly polarized
- (B) circularly polarized

- (C) **(Correct)** elliptically polarized
- (D) unpolarized

Correct Answer: (C)

Detailed Solution – Question 190

E_x and E_y out of phase by 90° , unequal amplitudes. This is elliptical polarization.

Question 191: Waveguides & Optical Fibers (GATE EC 2002)

MCQ

1 Mark

The phase velocity for the TE₁₀-mode in an air-filled rectangular waveguide is (c is the velocity of plane waves in free space)

- (A) less than c
- (B) equal to c
- (C) **(Correct)** greater than c
- (D) none of the above

Correct Answer: (C)

Detailed Solution – Question 191

$$V_p = \frac{c}{\cos \theta} = \frac{c}{\sqrt{1 - \left(\frac{f_c}{f}\right)^2}}$$

with $f > f_c$, $V_p > c$ always.

Question 192: Transmission Lines (GATE EC 2002)

MCQ

1 Mark

In an impedance Smith chart, a clockwise movement along a constant resistance circle gives rise to

- (A) a decrease in the value of reactance
- (B) **(Correct)** an increase in the value of reactance
- (C) no change in the reactance value
- (D) no change in the impedance value

Correct Answer: (B)

Detailed Solution – Question 192

Moving along clockwise direction is an inductive increase in the load impedance

Question 193: Transmission Lines (GATE EC 2002)

MCQ

1 Mark

The VSWR can have any value between

- (A) 0 and 1
- (B) -1 and +1
- (C) 0 and ∞
- (D) **(Correct)** 1 and ∞

Correct Answer: (D)

Question 194: Antennas & Radiating Systems (GATE EC 2002)

MCQ

1 Mark

The line-of-sight communication requires the transmit and receive antenna to face each other. If the transmit antenna is vertically polarized, for best reception the receive antenna should be

- (A) horizontally polarized
- (B) **(Correct)** vertically polarized
- (C) at 45° with respect to horizontal polarization
- (D) at 45° with respect to vertical polarization

Correct Answer: (B)

Detailed Solution – Question 194

Both transmitting and receiving antennas should be oriented correctly for maximum induction in the receiver. Transmitting antenna is vertical produced E field is vertical inducing Efield is vertical so receiving antenna has to be vertical.

Question 195: Antennas & Radiating Systems (GATE EC 2001)

MCQ

1 Mark

In a uniform linear array, four isotropic radiating elements are spaced $\frac{\lambda}{4}$ apart. The progressive phase shift between the elements required for forming the main beam at 60° off the end-fire is:

- (A) $-\pi$ radians
- (B) $-\frac{\pi}{2}$ radians
- (C) **(Correct)** $-\frac{\pi}{4}$ radians
- (D) $-\frac{\pi}{8}$ radians

Correct Answer: (C)

Detailed Solution – Question 195

$\theta_{\max} = 60^\circ$ from end fire means 60° from array axis.

Condition for maximum, $\psi \rightarrow 0$

$$\begin{aligned}E_T &= E_O \frac{\sin(N_\psi/2)}{\sin(\psi/2)} \\E_T &= N E_0 = E_{\max} \\\psi &= \alpha + \beta d \cos 60^\circ = 0 \\\alpha + \frac{2\pi}{\lambda} \cdot \frac{\lambda}{4} \cdot \frac{1}{2} &= 0 \\\alpha &= -\frac{\pi}{4}\end{aligned}$$

Question 196: Antennas & Radiating Systems (GATE EC 2001)

MCQ

1 Mark

A medium wave radio transmitter operating at a wavelength of 492 m has a tower antenna of height 124m. What is the radiation resistance of the antenna?

- (A) 25Ω
- (B) **(Correct)** 36.5Ω
- (C) 50Ω
- (D) 73Ω

Correct Answer: (B)

Detailed Solution – Question 196

$\frac{\lambda}{4} = \frac{492}{4} = 123 \simeq L \therefore$ It is a quarter wave monopole antenna. So, $R_a = 36.5\Omega$

Question 197: Transmission Lines (GATE EC 2001)

MCQ

1 Mark

A uniform plane electromagnetic wave incident normally on a plane surface of a dielectric material is reflected with a VSWR of 3. What is the percentage of incident power that is reflected?

- (A) 10
- (B) **(Correct)** 25
- (C) 50
- (D) 75

Correct Answer: (B)

Detailed Solution – Question 197

$$\begin{aligned}s &= \frac{1 + |\Gamma|}{1 - |\Gamma|} \\ \Rightarrow 3 &= \frac{1 + |\Gamma|}{1 - |\Gamma|} \\ |\Gamma| &= 0.5 \\ \frac{P_r}{P_i} &= |\Gamma|^2 = 0.25\end{aligned}$$

$\therefore$ 25% of incident power is reflected.

Question 198: Uniform Plane Waves (GATE EC 2001)

MCQ

1 Mark

A material has conductivity of 10^{-2} mho/m and a relative permittivity of 4. The frequency at which the conduction current in the medium is equal to the displacement current is

- (A) **(Correct)** 45 MHz
- (B) 90 MHz
- (C) 450 MHz
- (D) 900 MHz

Correct Answer: (A)

Detailed Solution – Question 198

$$\begin{aligned}|\sigma E| &= |j\omega \epsilon E| \\ \sigma &= 2\pi f \epsilon_0 \epsilon_r \\ \Rightarrow f &= \frac{\sigma}{2\pi \times \epsilon_0 \epsilon_r} \\ &= \frac{9 \times 10^9 \times 2 \times 10^{-2}}{4} \\ f &= 45 \times 10^6 = 45\text{MHz}\end{aligned}$$

Question 199: Waveguides & Optical Fibers (GATE EC 2001)

MCQ

1 Mark

The dominant mode in a rectangular waveguide is TE_{10} , because this mode has

- (A) no attenuation
- (B) no cut-off
- (C) no magnetic field component
- (D) **(Correct)** the highest cut-off wavelength

Correct Answer: (D)

Detailed Solution – Question 199

Dominant mode in waveguide has lowest cut-off frequency and hence the highest cut-off wavelength

Question 200: Waveguides & Optical Fibers (GATE EC 2001)

MCQ

1 Mark

The phase velocity of waves propagating in a hollow metal waveguide is

- (A) **(Correct)** greater than the velocity of light in free space.
- (B) less than the velocity of light in free space.
- (C) equal to the velocity of light in free space.
- (D) equal to the group velocity.

Correct Answer: (A)

Detailed Solution – Question 200

$$V_p = \frac{c}{\cos \theta} = \frac{c}{\sqrt{1 - \left(\frac{f_c}{f}\right)^2}}$$

with $f > f_c$ $V_p > c$ always.

Question 201: Uniform Plane Waves (GATE EC 2001)

MCQ

1 Mark

If a plane electromagnetic wave satisfies the equation $\frac{\partial^2 E_x}{\partial x^2} = C^2 \frac{\partial^2 X}{\partial t^2}$ the wave propagates in the

- (A) x-direction
- (B) **(Correct)** z-direction
- (C) y-direction
- (D) xy plane at an angle of 45° between the x and z directions

Correct Answer: (B)

Detailed Solution – Question 201

The E field is a function of z and directed in X. i.e. harmonic function of z or travelling in z direction.

Question 202: Transmission Lines (GATE EC 2001)

MCQ

1 Mark

A transmission line is distortion-less if

- (A) $RL = \frac{1}{GC}$
- (B) $RL=GC$
- (C) **(Correct)** $LG=RC$
- (D) $RG=LC$

Correct Answer: (C)
